# 1. About this Open Learning Module

## Target Audience and Prerequisites

This Open Learning module is aimed at new recruits in Great Britain into the role of Signalling Technician on either the installation or maintenance sides. It can also be used to update or refresh the knowledge of experienced technicians who are undergoing re-certification of their competence.

Apart from a basic Secondary Education, an ability to understand moderately complex English language and the presentation of information in the form of diagrams, plus a reasonable curiosity and a willingness to learn, the only requirement is that you should have successfully completed the study of

- the associated Open Learning Module “Introduction to Railway Signalling”
- Signalling For Technicians Module 1 “Circuit Reading”

or have acquired equivalent basic knowledge about the purpose and implementation of railway signalling, the forms of diagrams and schematic drawings used and the various symbols and methods of labelling employed.

# Objectives

This module is intended to provide an introduction to and an appreciation of

- the requirements laid down by the Department of Transport for the operation of railway points
- the types of point layout in use
- the types of stock rail and switch rail in use
- the various methods of point operation
- the components and operation of combined electric point machines
- the components and operation of non-combined electric point machines
- the components and operation of electro-pneumatic point machines
- the components and operation of rail point clamp locks
- the need for and the application of facing point lock and point detection tests
- the need for insulation of points components and the methods of providing insulation
- the components and operation of back drives

## What you will require

Apart from this Open Learning module, the only other material you require is the Signalling Maintenance Specifications.

## The Results expected from this Module

When you have completed this Module and the assessment, and been awarded your certificate, you should

- have an appreciation of the various types of point configuration and their components
- be able to identify the various types of stock rails and switch rails
- be able to identify and understand the construction and operation of the following types of point mechanisms:

  - mechanical- the GEC HW 2000 combined point machine- the Westinghouse Style 63 combined point machine- the SGE Type HA non-combined point machine- the SGE Electro-Pneumatic Point Machine (AC Immune Version)- rail point clamp locks- mechanical and hydraulic back drives

- be able to identify and understand the construction and operation of various types of insulation used with point mechanisms
- be able to understand the purpose and application of

  - facing point lock tests- points detection tests

- be able to correctly use and understand the jargon commonly used in association with switches and points

## Jargon

In the main body of this module the officially recognised jargon descriptions are used and some commonly used variations are introduced. The Glossary of Terms (see Section 16) provides some of the other relevant variations which are used.

# A Safety Warning

This Open Learning Module is designed to inform you about railway points. However, please remember that, until you have undergone proper training and been certified as competent in Personal Track Safety (PTS) and you have been authorised by your Supervisor, you are forbidden to go onto any part of the railway that is not open to passengers, unless you are properly supervised and it is an official part of your training.

Study and remember what it says in your Rule Book and the Track Safety Handbook. There is no requirement for you to go onto the railway to successfully complete this course of study. Of course, your supervisor or appointed mentor may use their discretion and properly arrange for you to go onto the railway to look at practical examples of the points, associated equipment and layouts described in this manual and which you have not already had an opportunity to observe in the course of your work and other training.

# 2 Requirements laid down by the Department of Transport

## 2.1 Introduction

Points are used to divert trains from one line to another.

This Section will examine in detail the requirements for point operation as laid down by the Department of Transport.

At this stage do not expect to understand all the terms used, familiarise yourself with the requirements and bear them in mind when you continue your work through the module.

Before looking at the requirements, look closely at figure 1 which provides you with an example of a rail point layout.

*An example of a Point Layout*

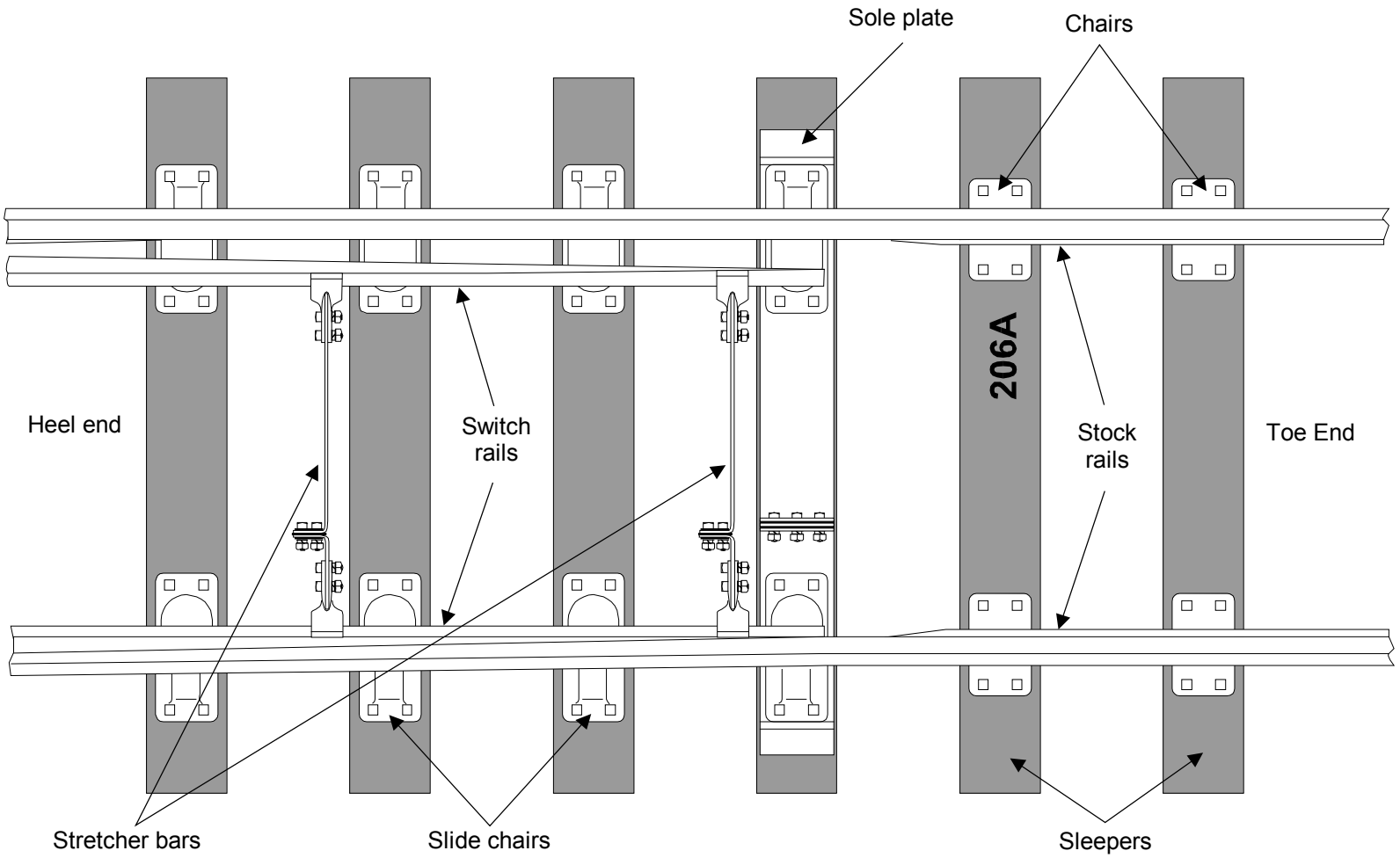



Figure 1

*Point Identification*

The point identification is fixed in the four foot next to the normally closed point switch rail (see figure 1) and contains the following information

- the point number
- the point end

An arrow may be provided to indicate the normally closed switch rail if there is likely to be confusion.

If there is more than one point end operated from one control, the same point number will be used, but the individual ends will be uniquely identified by a letter placed after the number, e.g. 206A, 206B, etc.

The relevant component parts and the operation of points and the methods of moving them will be examined in greater detail in later sections.

The normal side is the side on which the switch rail is normally closed. For testing purposes the lie of the points should be described as either the

- left hand switch rail closed
- right hand switch rail closed

This is determined by standing at the toe end of the points looking towards the heel end.

A short explanation follows. Each extract has been taken directly from the Department of Transport Requirements.

# 2.2 Department of Transport Requirements

Paragraph 20.

All points to have two or more stretcher bars.

## Stretcher Bars

The function of the stretcher bars is to hold both switch rails together. They ensure

- correct point opening
- the switch rails move together as one unit

Paragraph 21.

Facing points on passenger lines and all points regularly used in the facing direction by passenger trains to have:-

a) a bolt lock through a third stretcher bar, with its bolt either worked through a locking bar or controlled by track circuit

b) a stock rail gauge tie

c) apparatus to detect each switch is in its proper position and that points are bolted, before the relevant signals can be cleared

## Facing Point Third Stretcher Bar

This is also termed the lock stretcher bar, it

- is connected to both switch rails
- provides bolt locking notches

## Bolt Lock

The function of the bolt lock is to lock the switch rails to the stock rails by passing through a notch in the lock stretcher bar so that the points cannot move. This is important, as movement could be caused by vibration due to the passage of the train.

## Stock Rail Gauge Tie

This is also termed the sole plate. Its function is to tie both stock rails together in order to maintain their gauge and prevent road spread.

## Point Detection

The function of the point detection is to prove that the

- closed switch rail is fitting correctly to the stock rail
- open switch rail has the necessary clearance from the stock rail

In the case of facing points it also proves they are locked.

## Chairs

The chairs sit on and are fastened to the sleepers. The rail then sits in the chair and is held firmly in place by clips or wedges (keys)

## Slide Chairs

These are provided to hold the stock rails in place and support the moveable section of the switch rails, whilst allowing them to move smoothly from one side to the other as shown in figure 1.

## 2.3 Section Summary

This section has examined the requirements of the Department of Transport concerning points. Before moving on to Section Three complete Self-Assessment Questions 1 and 2.

### Self-Assessment Question 1

State the Department of Transport requirements concerning the number of stretcher bars to be used on points.

### Self-Assessment Question 2

In addition to the stretcher bars referred to in Question 1, what other Department of Transport requirements apply to facing points on passenger lines.

Now check your answers with those given in the Answers Section. Before moving on to Section Three make sure that you fully understand all the information given in Section Two. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 3 Types of Point Layout

## 3.1 Introduction

Point layouts vary depending on

- the number of lines involved
- the direction of travel
- speed of traffic
- the switching to be done

All points fall into one of two categories

- facing points
- trailing points

Each of these will now be examined in greater detail.

*Facing Points*

## Facing Points

Look carefully at figure 2 which shows an example of facing points

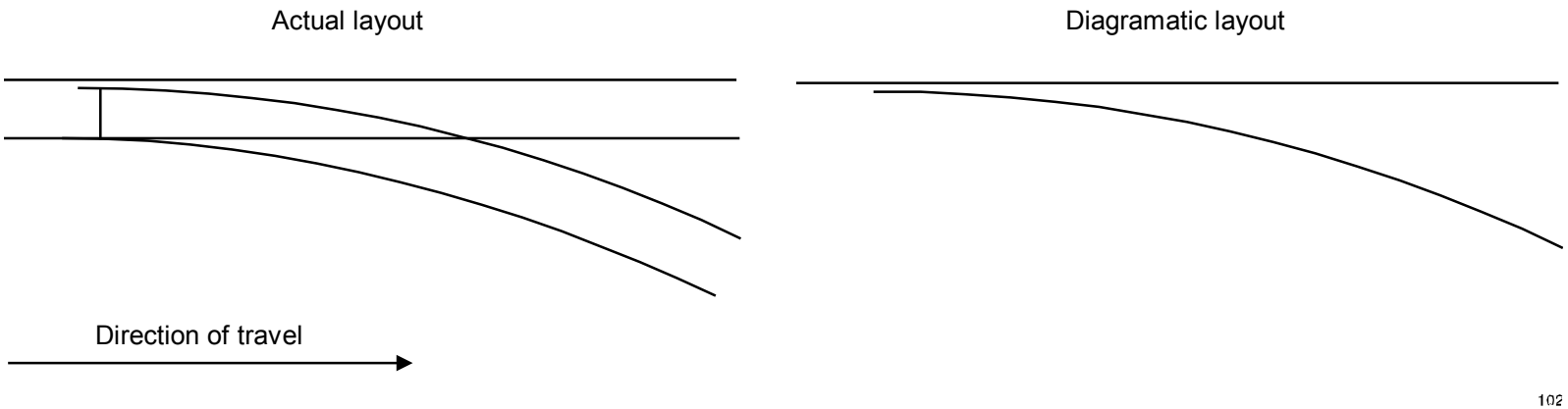



Figure 2

You can see from figure 2 that a train passing over these points can be directed onto one of two lines. The lines diverge (separate) at these points.

## Trailing Points

Look carefully at figure 3 which shows an example of trailing points.

*Trailing Points*

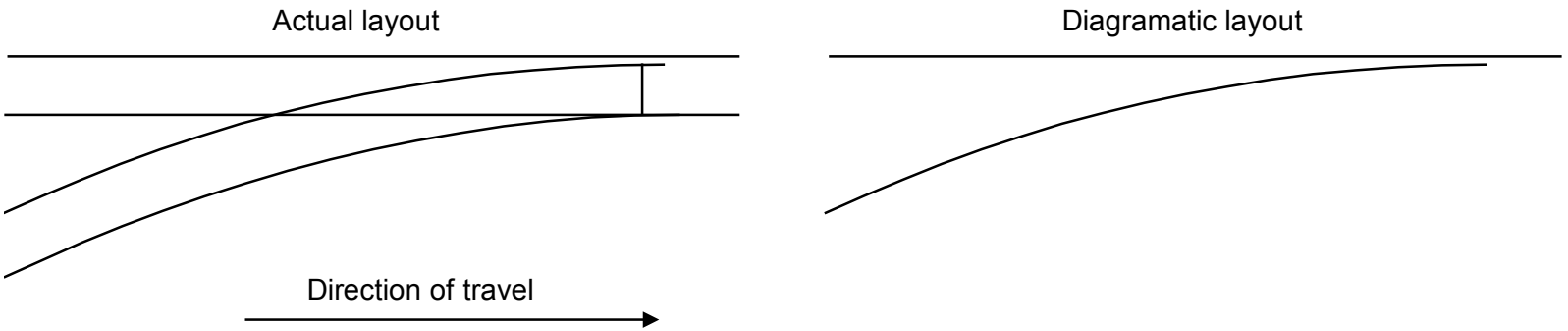



Figure 3

With the trailing points shown in figure 3 approaching trains will be directed onto the same line. This means that the lines converge (come together) at these points.

## Point Layouts

There is a variety of point layouts which include

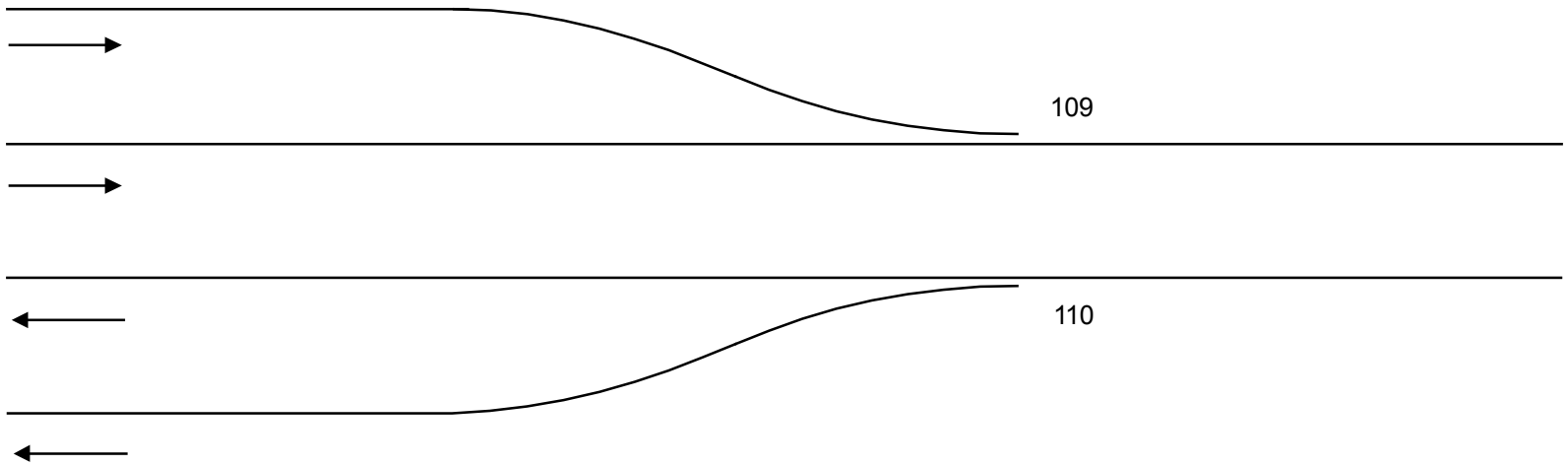
- single end
- double end
- multiple end

Each of these layouts will now be examined in greater detail.

## Single Ended Point Layout

A single ended point layout is a single set of points working independently of any other points. They will be identified by a unique number.

Figure 4 provides a diagrammatic example of this type of layout showing two single ended points.




Arrows indicate direction of travel

104

Figure 4

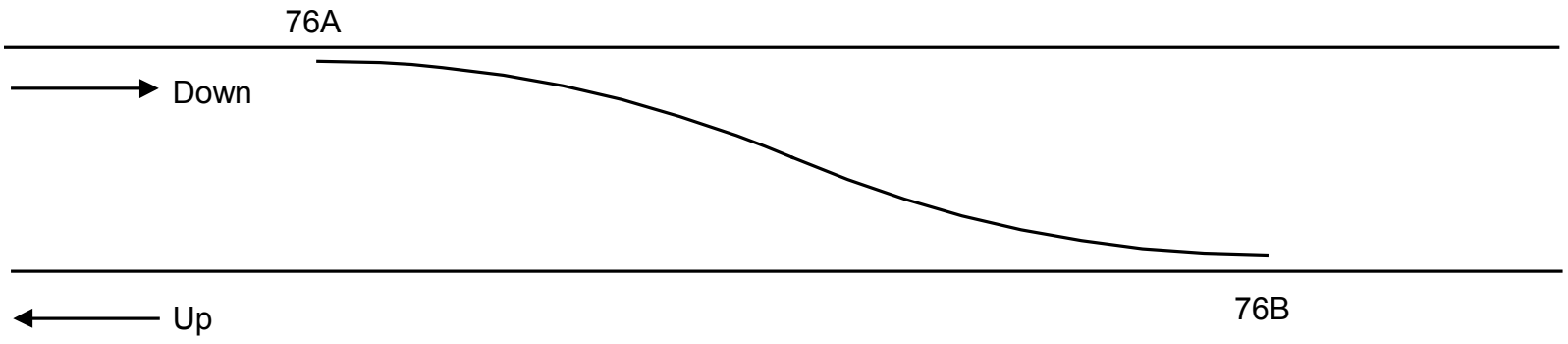
## Double Ended Point Layout

A double ended point layout is two ends of points working from one control. In addition to the point number, each end is given a letter to uniquely identify the point end. The A end of the points is either

- the end nearest the Up end of the line (current standard)
- the end nearest the controlling signal box (old standard)

Figure 5 provides a diagrammatic example of double ended point layout.

*Double Ended Point Layout*




10

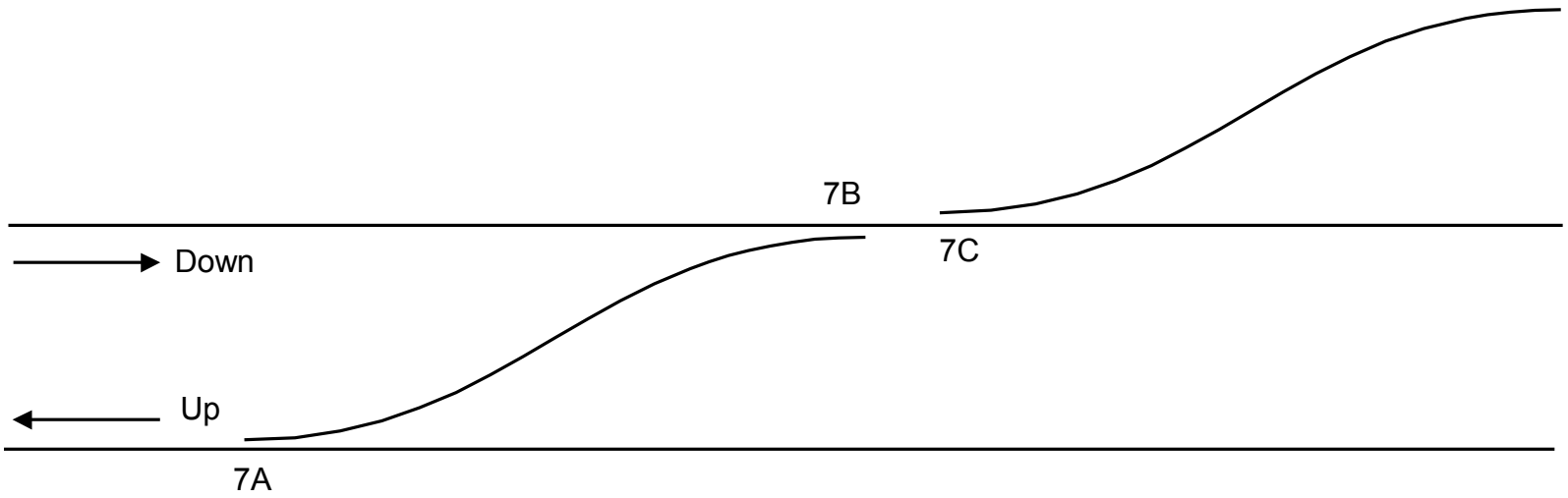
Figure 5

## Multiple Ended Point Layout

A multiple ended point layout is three or more point ends working from one control. Each end is given a letter to uniquely identify it. The use of multiple ended point layouts is now being limited to three ends.

Figure 6 provides a diagrammatic example of a multiple ended point layout. In this example three ends are shown.

*Multiple Ended Point Layout*




10

Figure 6

## 3.3 Point Configurations

You will now examine a number of different point configurations which may incorporate

- single end point layout
- double end point layout

The configurations to be examined are as follows

- fixed diamond crossing
- plain lead
- single slip
- double slip
- switch diamond
- swing nose crossing

In the figures which illustrate the point configurations, the chairs have been removed. This is to clarify the information given. Except for the fixed diamond crossing each configuration will be examined in terms of

- diagrammatic representation
- configuration
- detail at toe end

## Fixed Diamond Crossing

The term fixed diamond crossing is often used when references are made to switches and crossings. An example is shown in figure 7. Strictly speaking it does not belong to the category of points because no points are involved and no switching occurs.

It can be seen in figure 7 that a train can travel from A to D, D to A, B to C or C to B.

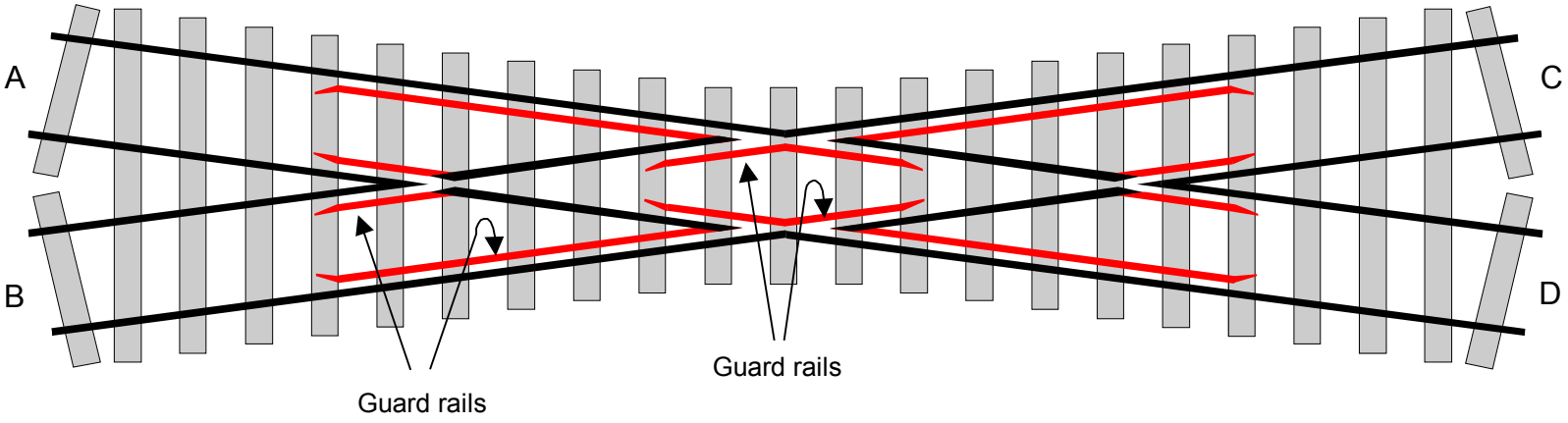



Figure 7

This configuration does not allow any switching, for example A to C, or B to D are not allowed.

Notice the guard rails. Their function is to keep the train wheels in line as they pass over the gaps.

In this diagram the guard rails are marked in red.

## Plain Lead

The plain lead configuration is the simplest form of points in use.

Figure 8 shows a diagrammatic representation and figure 9 shows the configuration.

*Plain Lead Diagrammatic*

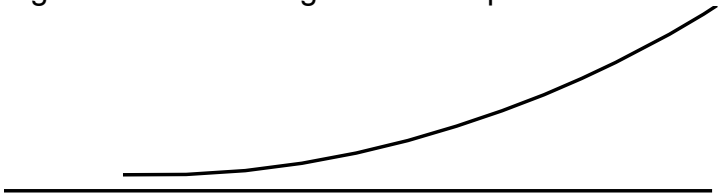


Figure 8

*Plain Lead Configuration*

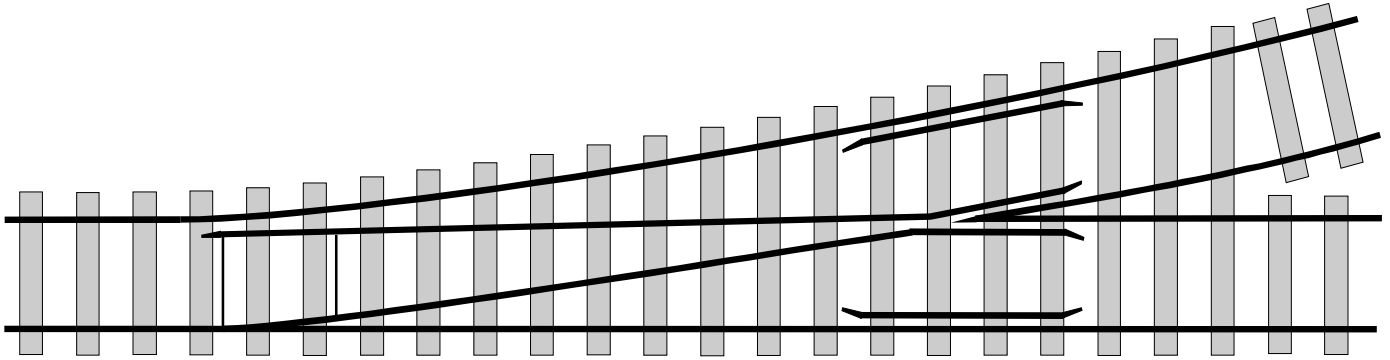


Figure 9

Figure 10 shows the detail at the toe end of the plain lead.

*Detail at the Toe End of Plain Lead*

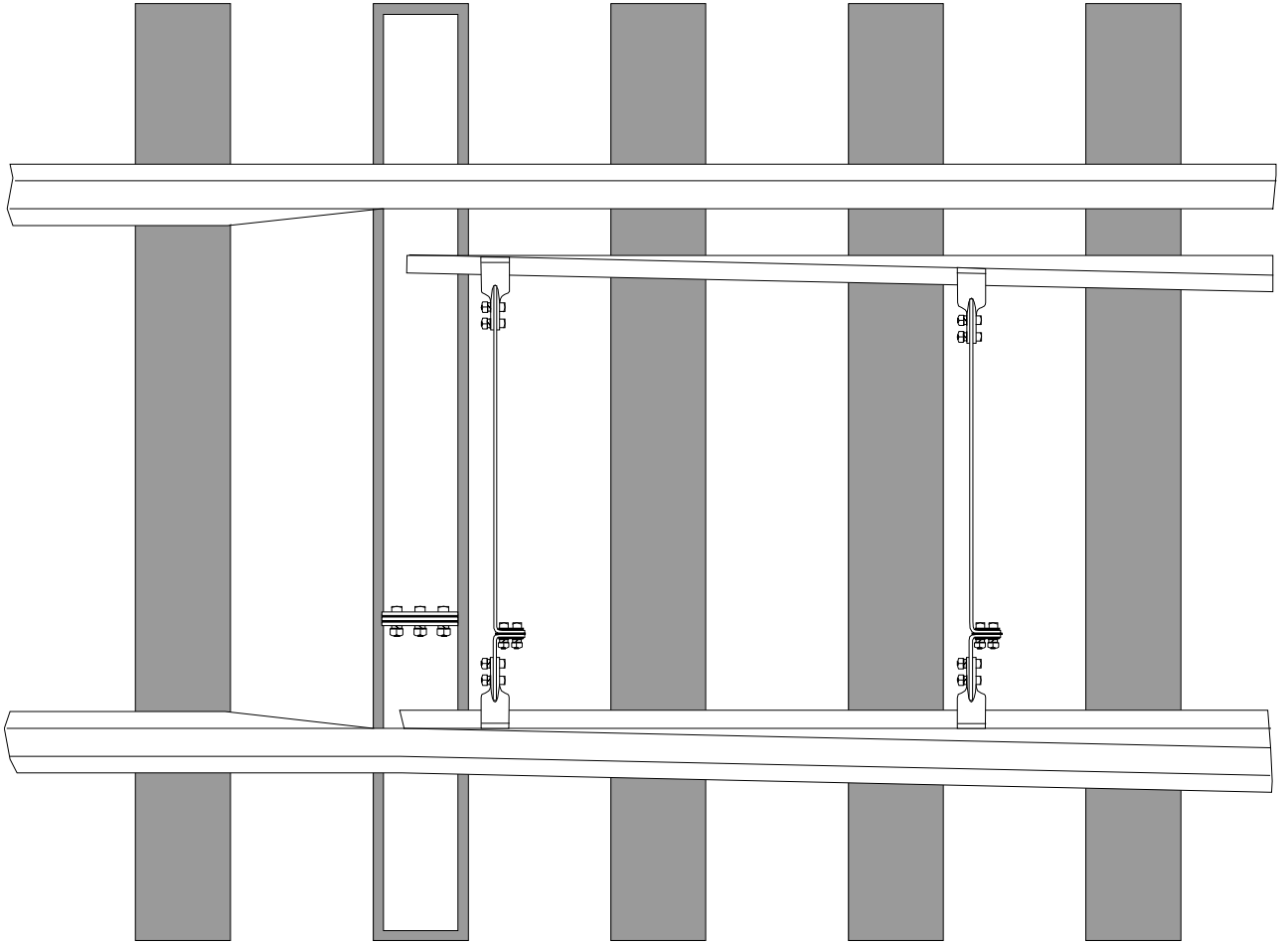


Figure 10

## Single Slip

Figure 11 shows the diagrammatic representation and figure 12 shows the configuration of a single slip, which exists as a modification of the fixed diamond crossing. Note that the letters on figure 11 are included to illustrate the passage of trains and are not part of the plan symbol.

*Single Slip Diagrammatic*

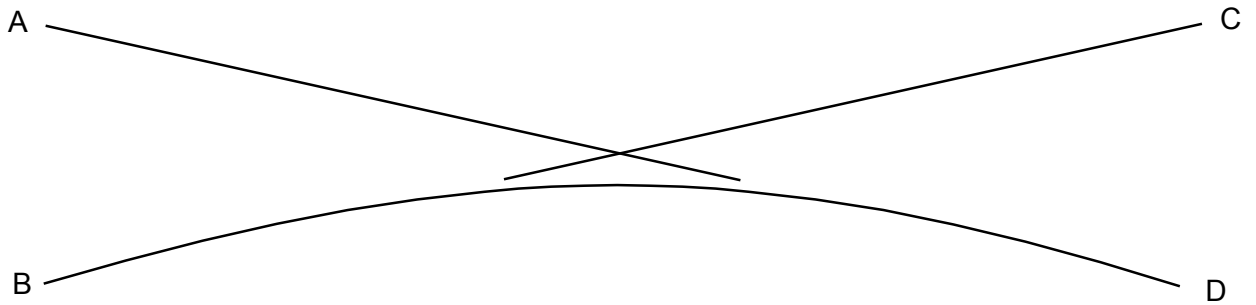



Figure 11

*Single Slip Configuration*

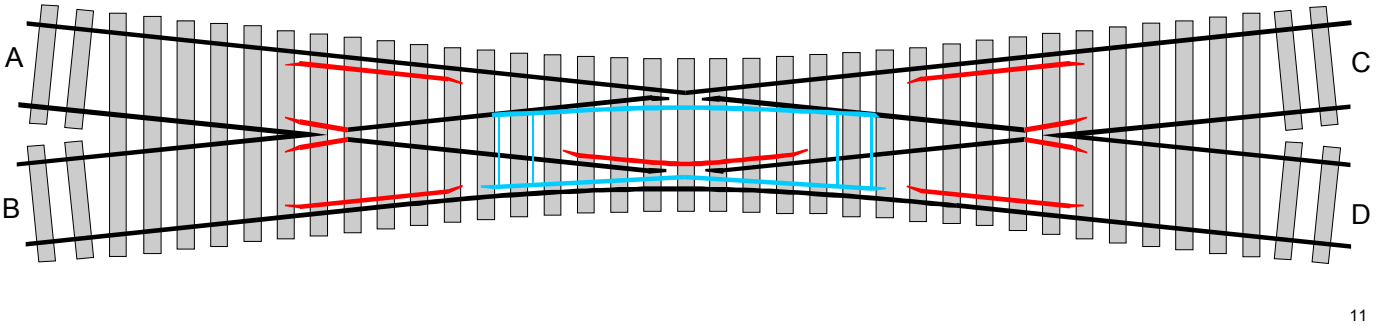



Figure 12

In addition to allowing trains to travel

A to D

D to A

B to C

C to B

the single slip also allows movements

B to D

D to B

Figure 13 shows the detail at the toe end of the left hand set of switches. A similar arrangement exists for the right hand switches.

.

*Detail at Toe End of Single Slip*

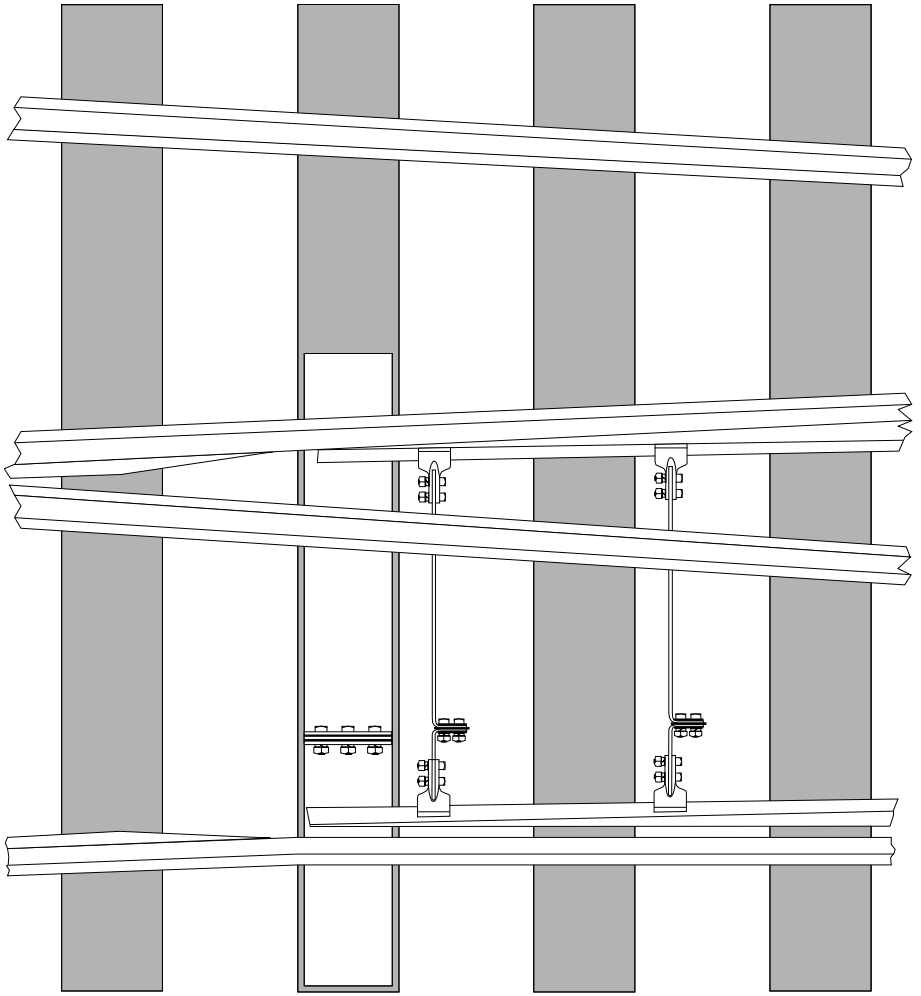


Figure 13

## Double Slip

The double slip configuration is a natural progression from the single slip configuration. Look carefully at figures 14 and 15

*Double Slip Diagrammatic*

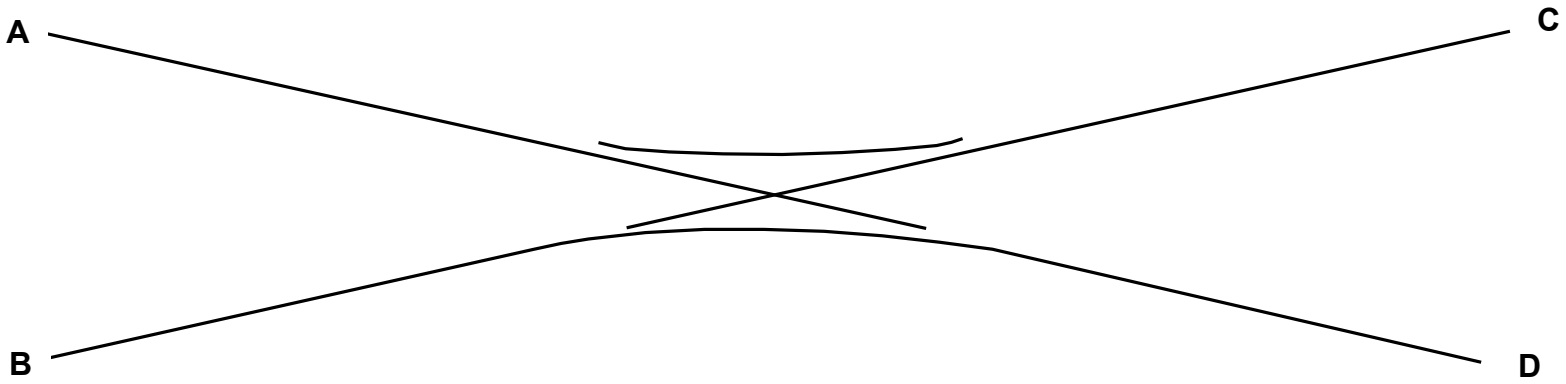



Figure 14

*Double Slip Configuration*

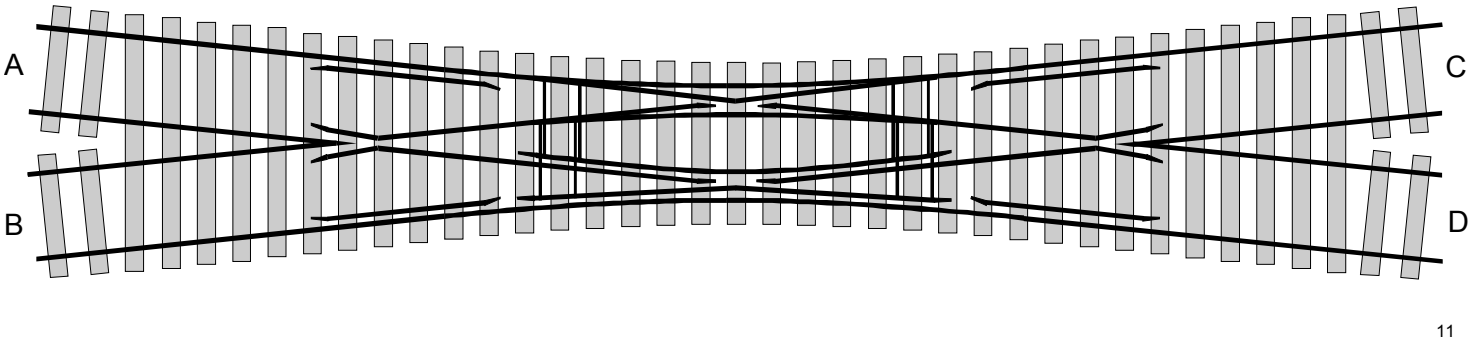



Figure 15

The double slip configuration is basically the same as the single slip configuration but it also allows passage of trains from

- A to C
- C to A

Figure 16 shows the detail at the toe end of the left hand set of switches. A similar arrangement will exist at the right hand set of switches.

*Detail at Toe Ends of Double Slip*

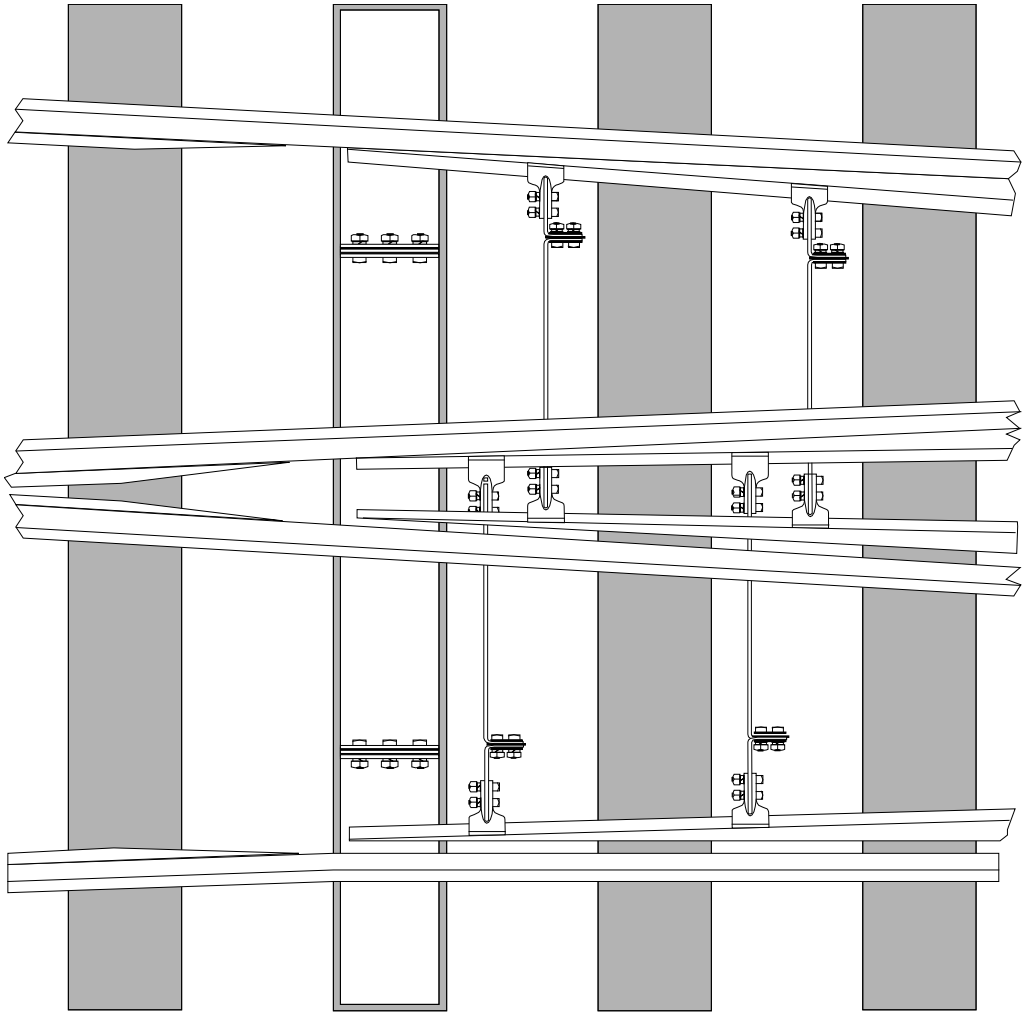


Figure 16

## Switch Diamond

In a fixed diamond crossing (illustrated in figure 7), where two lines cross, there are gaps to allow the wheel flanges to pass through the opposing rails. When high speed trains are passing over the crossings these gaps cause severe hammering to the wheels and to the rails at the edges of the gaps. To avoid this hammering the switch diamond layout was developed. This configuration uses two short points positioned toe to toe. Figure 17 shows the diagrammatic representation, figure 18 shows the configuration, and figure 19 shows the toe ends of a switch diamond.

*Switch Diamond Diagrammatic*

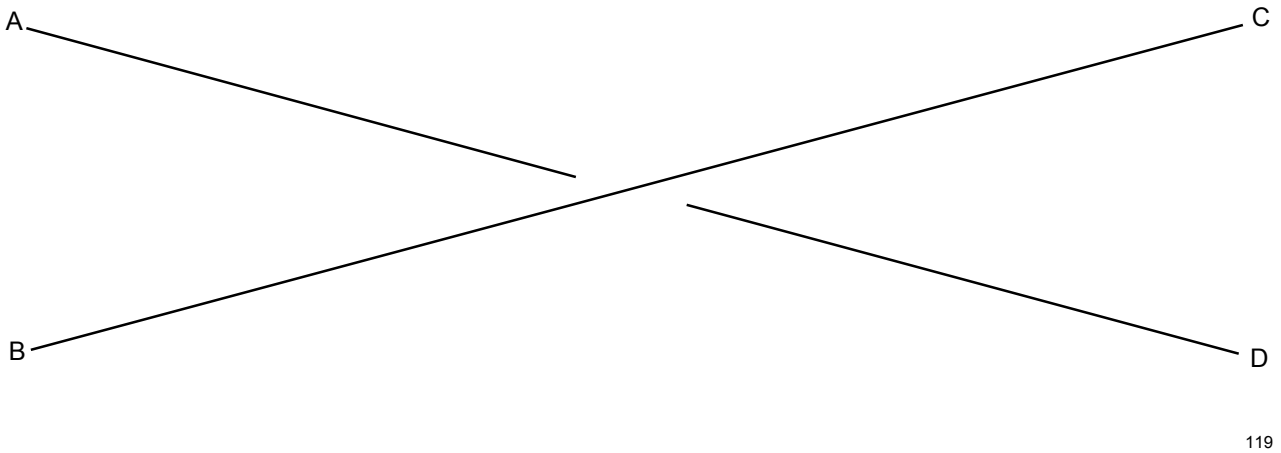



Figure 17

*Switch Diamond Configuration*

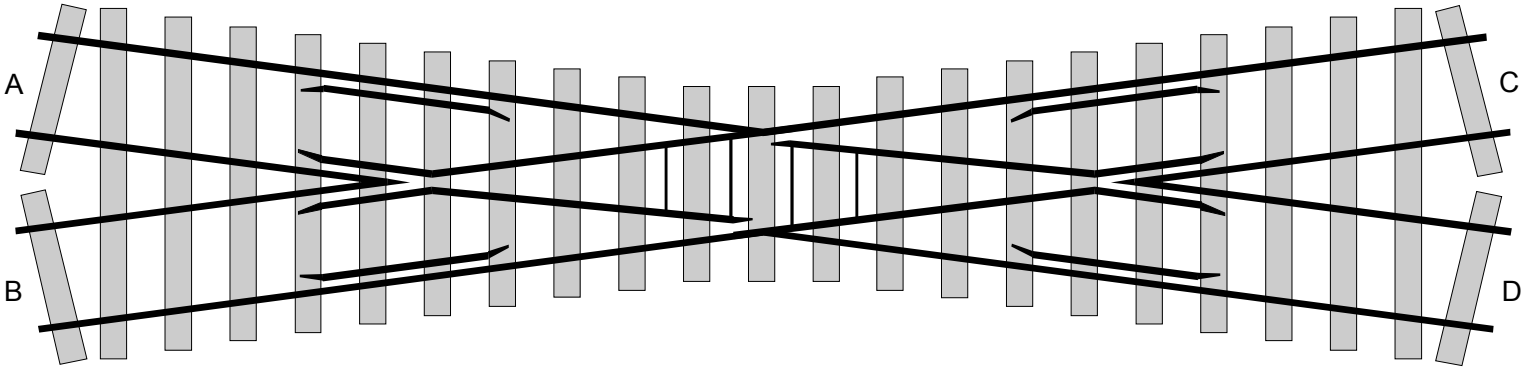



Figure 18

Look carefully at figures 18 and 19. You should be able to see that when the points are set as shown the gaps are closed for trains to pass from B to C or C to B. When the points are both reversed, the gaps are closed for trains to pass from A to D or D to A.

*Detail at Toe Ends of Switch Diamond*

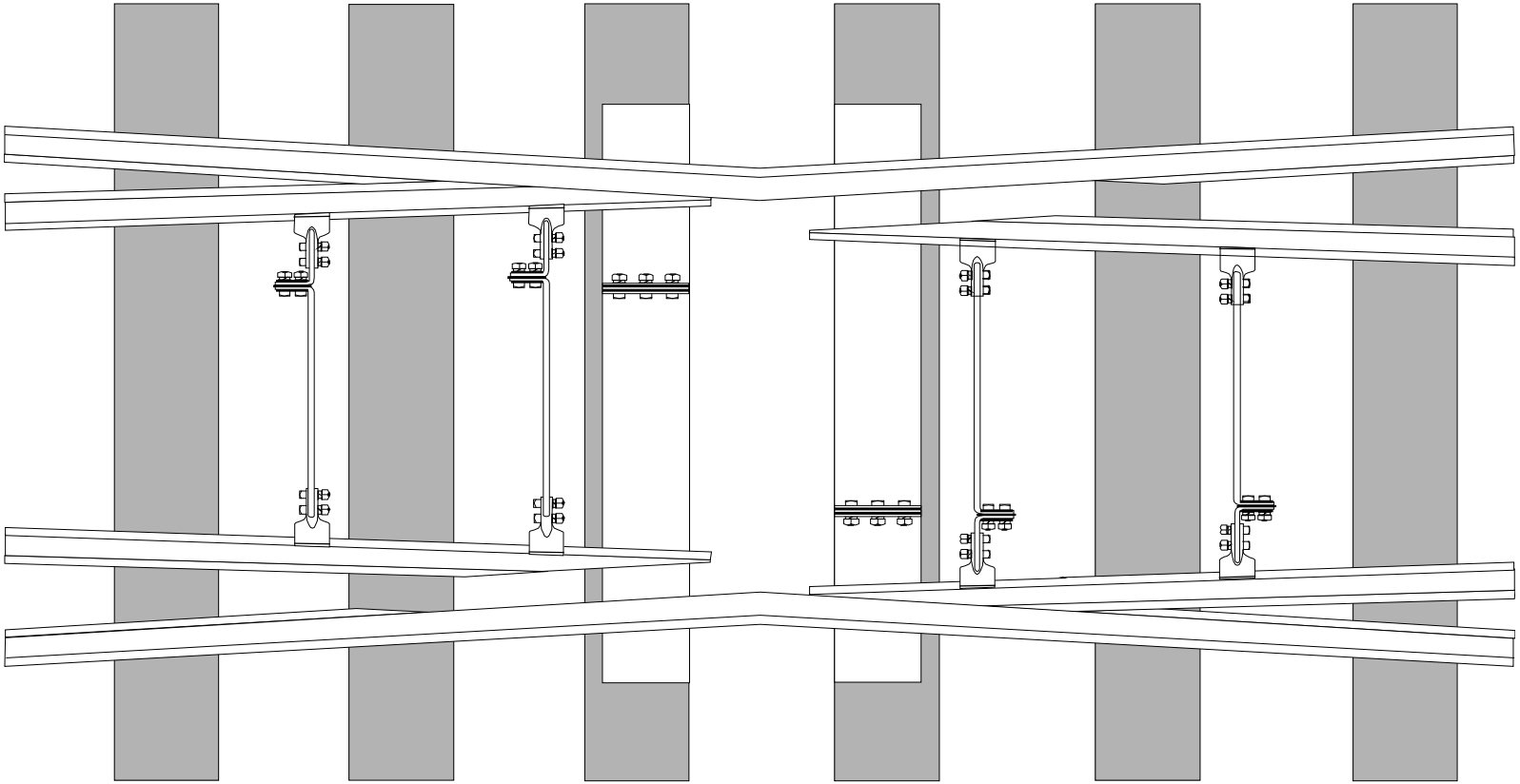


Figure 19

## Swing Nose Crossing

On all types of points and crossings (see figures 7, 9, 12, 15 and 18) rails cross and gaps have to be left for the wheel flanges to pass through. When the line speeds are 90 mph or more hammering of the wheels and rails at these gaps becomes a serious problem and has led to the development of the swing nose crossing. Figure 20 shows the diagrammatic representation of a swing nose crossing associated with a plain lead, figure 21 shows the configuration of a typical plain lead incorporating a swing nose crossing, and figure 22 shows the toe end of the switch nose.

*Plain Lead with Switch Nose Diagrammatic*

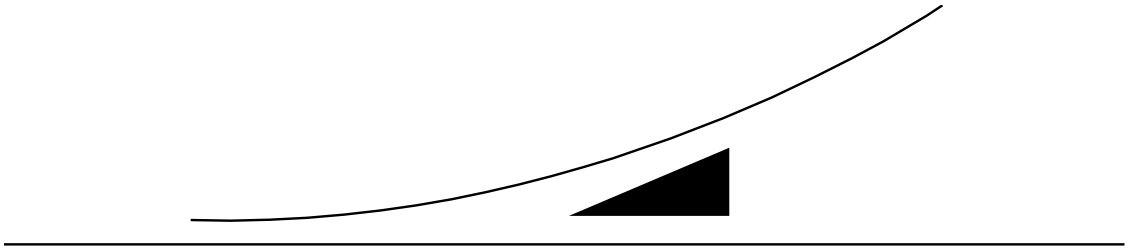
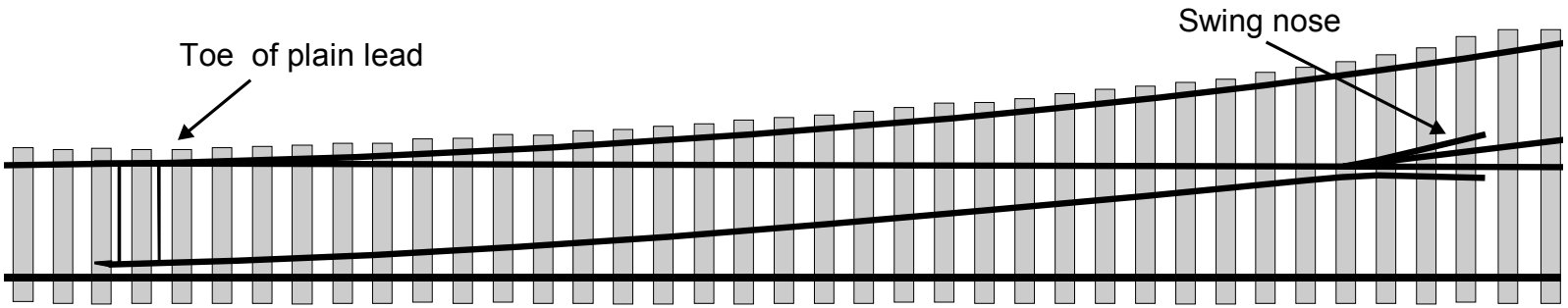


Figure 20

*Plain Lead with Switch Nose Configuration*




15

Figure 21

*Detail at Toe End of Switch Nose*

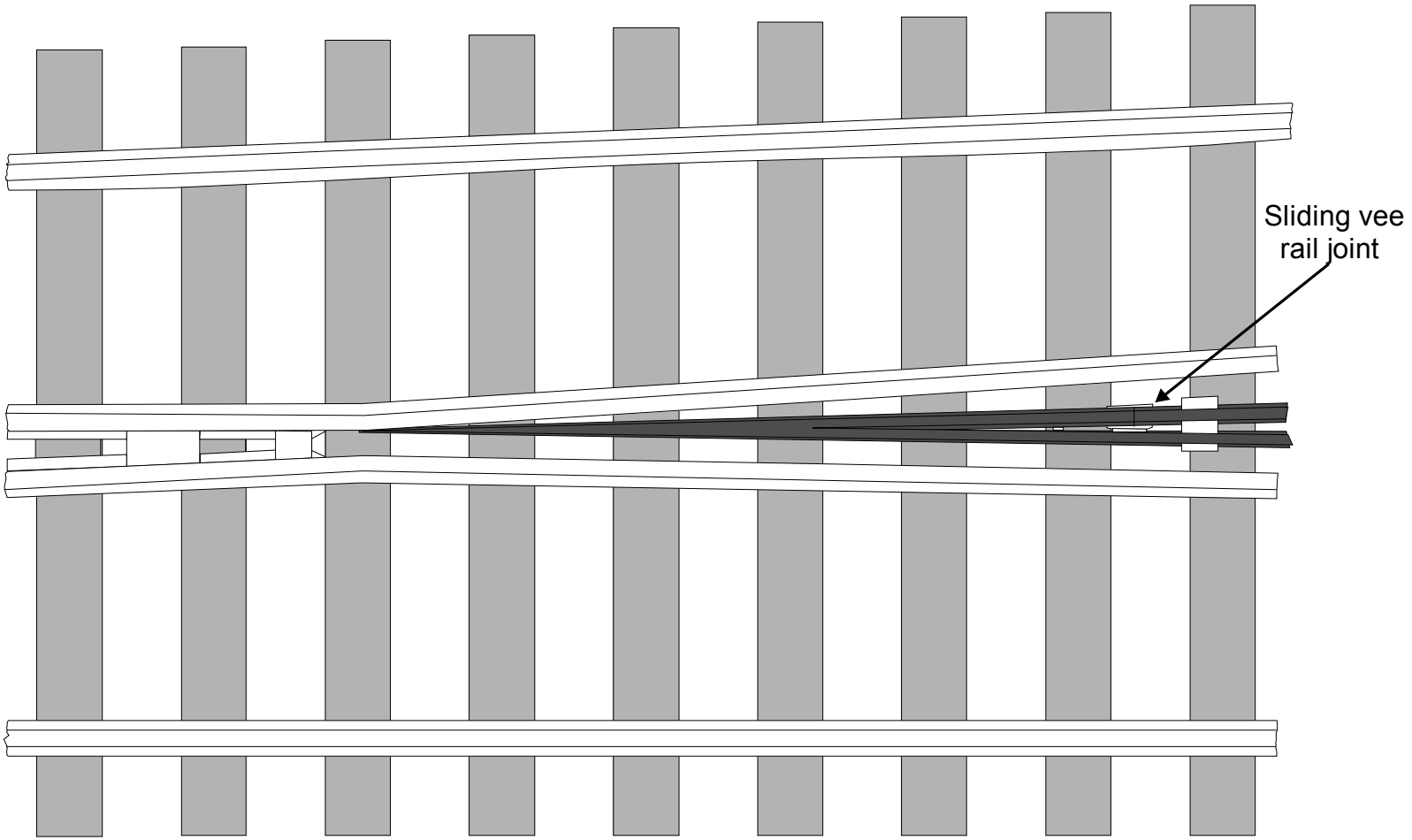



Figure 22

## 3.4 Point Configurations

Different point configurations are utilised depending on specific circumstances and requirements. Although a plain lead is always preferred, lack of space and complex layouts can require alternative point configurations to be used. Some examples of these configurations will now be looked at in greater detail.

Figure 23 provides an example of a point layout using plain leads.

*Point Layout using Plain Leads*

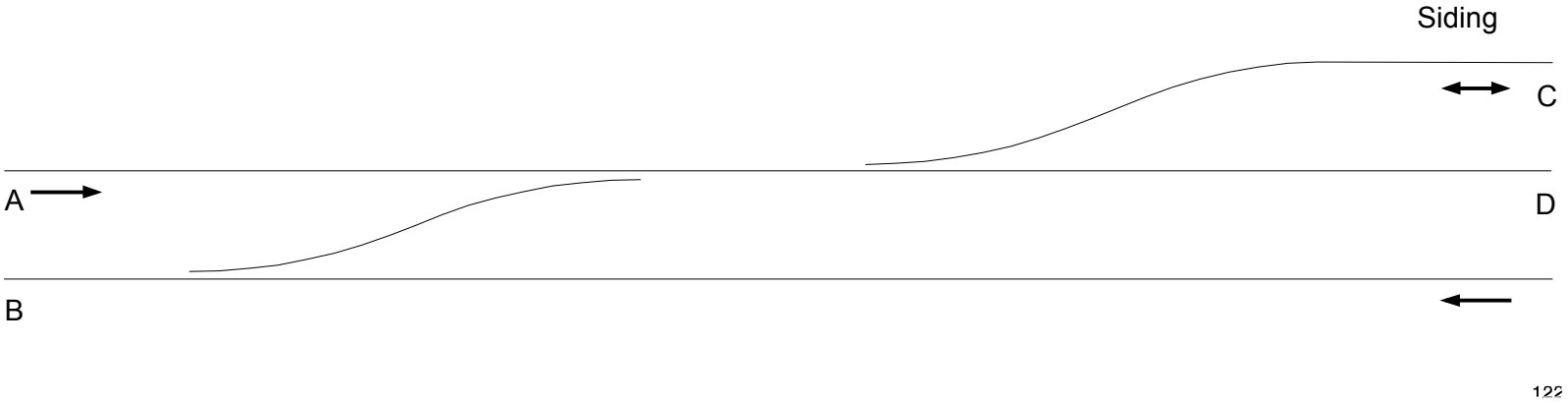
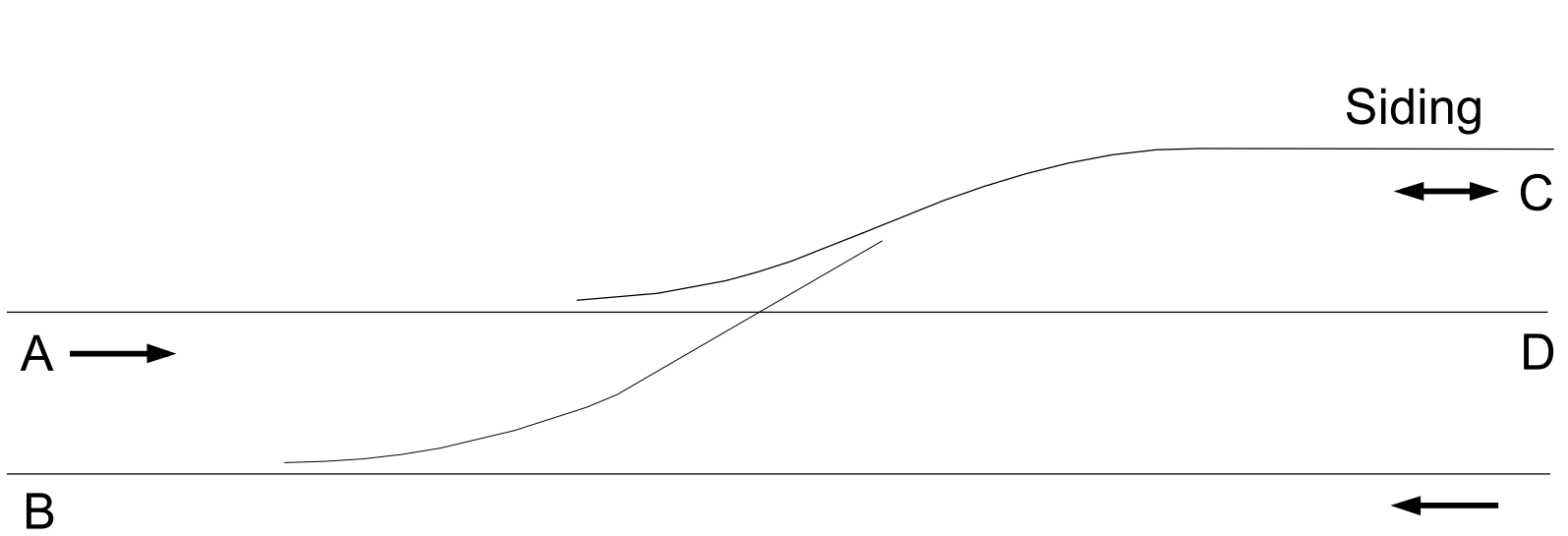



Figure 23

The same layout can be achieved in less space by the use of a single slip configuration as shown in figure 24.

*Point Layout*
*sing Single Slip*




123

Figure 23

In both layouts under normal operating conditions, a train can travel from A to D. It can also switch lines into the siding to C. A train can also travel out of the siding and be switched to A or B.

*Ladder Crossing Using Plain Leads*

Figure 25 provides a layout using plain leads. There are four tracks and three double ended sets of points. It can be seen that with this layout a train can travel from line A to E and also switch lines to F, G or H and vice versa.

From H a train can travel to D and also switch lines to C, B or A. This is known as a ladder crossing. In this layout it requires a great deal of space.

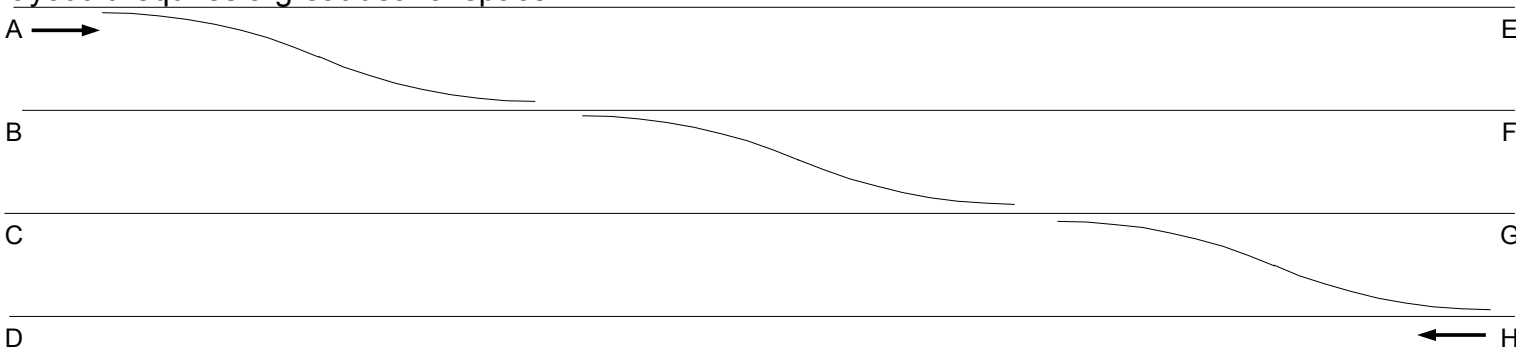



Figure 25

Where space is limited double slip configurations may be used, as in figure 26.

*Ladder Crossing using Double Slips*

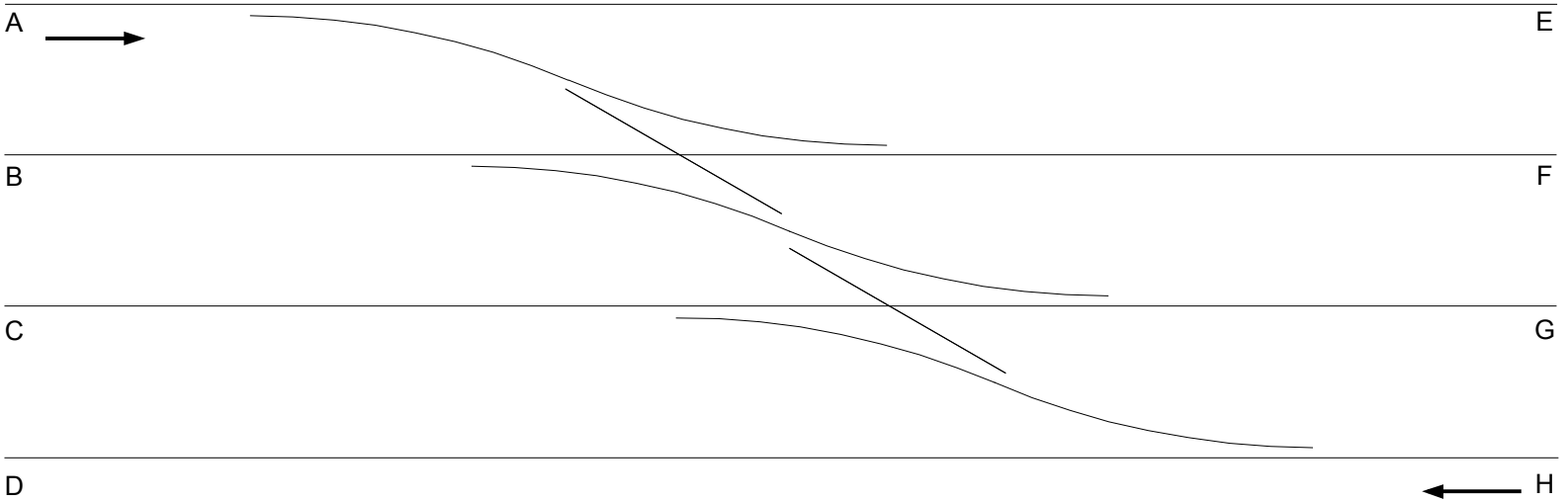



Figure 26

There are an infinite number of point layouts which can be achieved by using a combination of the point configurations examined in this section.

It will be useful at this stage to find a scheme plan for an area familiar to you or your place of work and examine it to see if you can recognise the different point configurations.

## 3.5 Section Summary

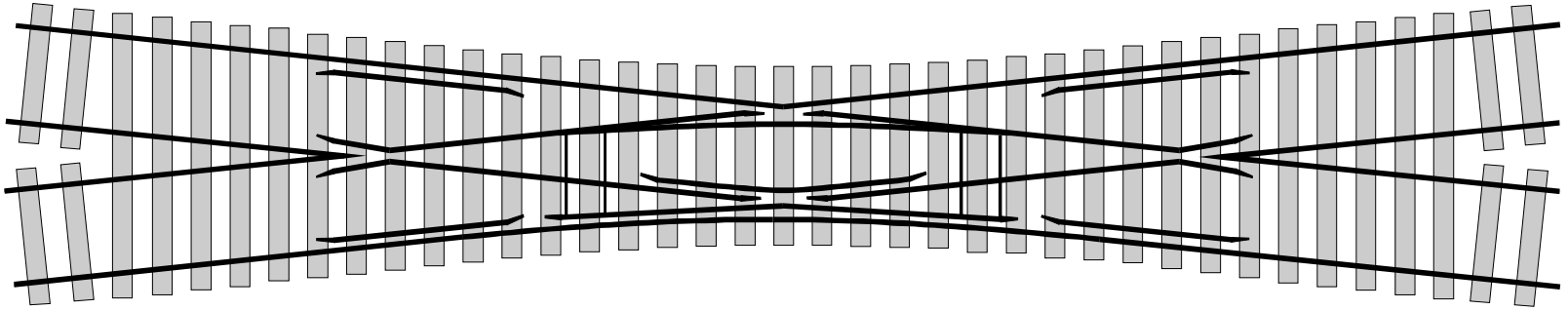
This section has looked at

- point categories
- point layouts
- point configurations

Now complete Self-Assessment Question 3 before proceeding to Section Four.

## Self-Assessment Question 3

Correctly identify the point configuration shown in figure 27.



126

Figure 27

Now check your answer with the one given in the Answers Section. Before moving on to Section Four make sure that you fully understand all the information given in Section Three. If there are any areas which you do not understand, re-read the relevant parts of the text.

This page has been left blank intentionally.

# 4 Stock Rails and Switch Rails

## 4.1 Introduction

There are two basic two types of rail used, namely

- flat bottom
- bullhead

A pair of points normally has two stock rails which are fixed and two switch rails which are the moving part of the point layout.

Each of these will now be examined in greater detail.

## 4.2 Stock Rail

There are a number of variants of stock rail in use, these include

- flat bottom inclined
- flat bottom vertical
- bullhead

## 4.3 Switch Rail

Switch rails are the moving parts of the point layout and may be

- closed up to the stock rail on the left hand side and away from the stock rail on the right hand side in which case the points are set for one direction
- away from the stock rail on the left hand side and closed up to the stock rail on the right hand side in which case the points are set for the opposite direction

The types of switch blade available include

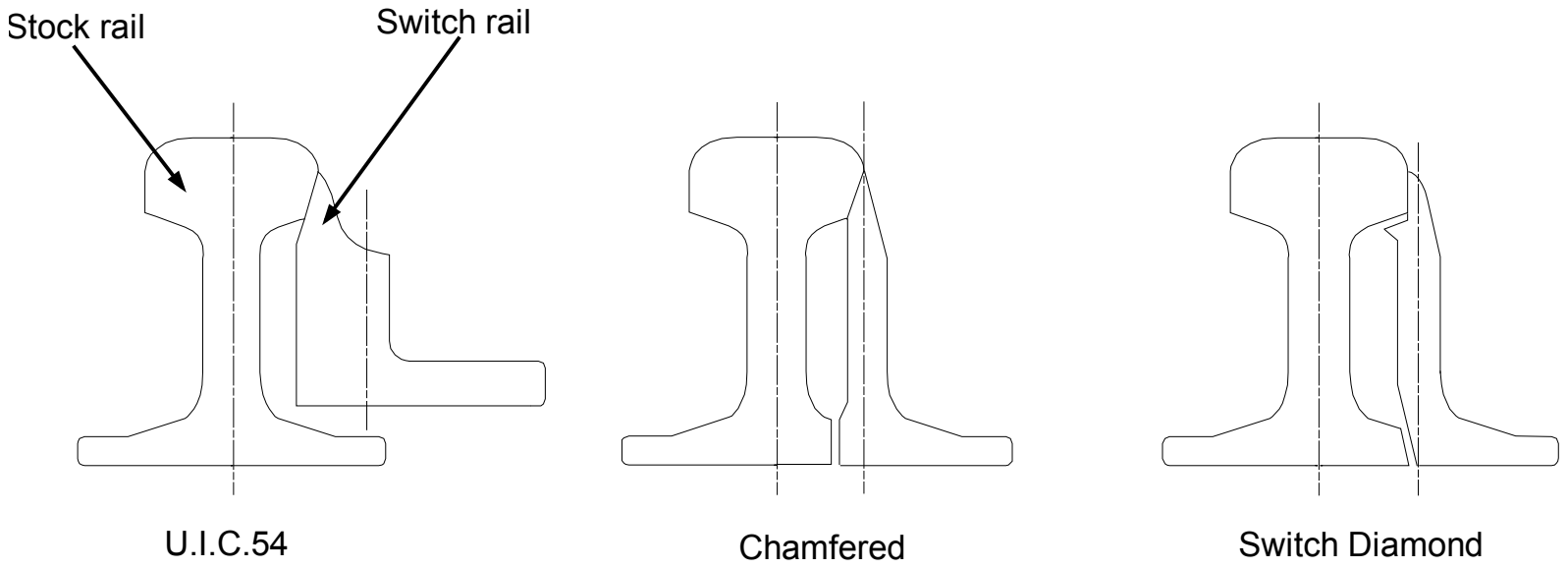
- undercut
- straight cut
- chamfered
- switch diamond
- U.I.C. 54

Each of these will now be examined in greater detail.

## Flat Bottom Vertical

Figure 28 shows the stock and switch rail configurations for the flat bottom vertical which sits upright on its chair.

*Examples of Flat Bottom Vertical Switch and Stock Rails*




127

Figure 28

Modern installations invariably utilise a flat bottom vertical type.

# Flat Bottom Inclined

Figure 29 shows the stock and switch rail configurations for a flat bottom inclined track.

*Examples of Flat Bottom Inclined Switch and Stock Rails*

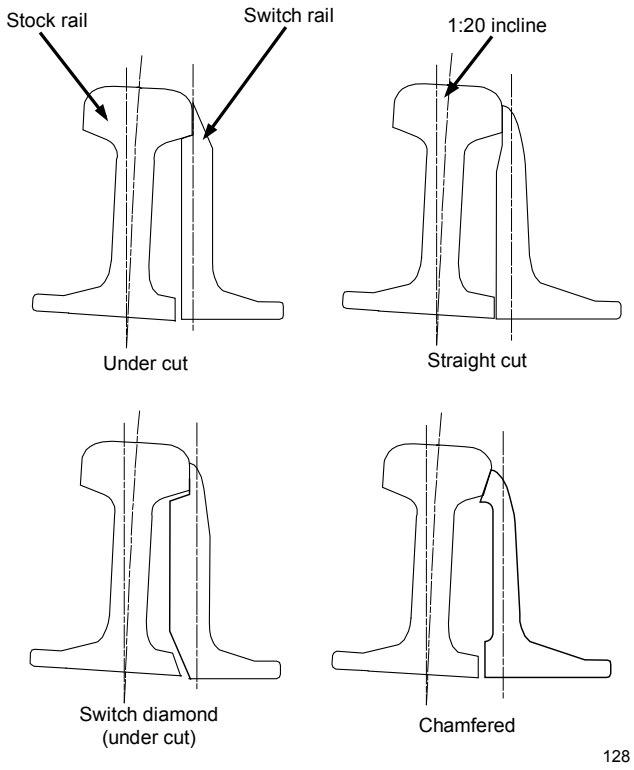



Figure 29

The flat bottom inclined type, as the name suggests is inclined inwards towards the centre of the track. It should be noted that although the stock rail is inclined, the switch blade remains vertical.

*Examples of Bullhead Switch and Stock Rails*

# Bullhead

Figure 30 shows the stock and switch blade configurations for the bullhead type.

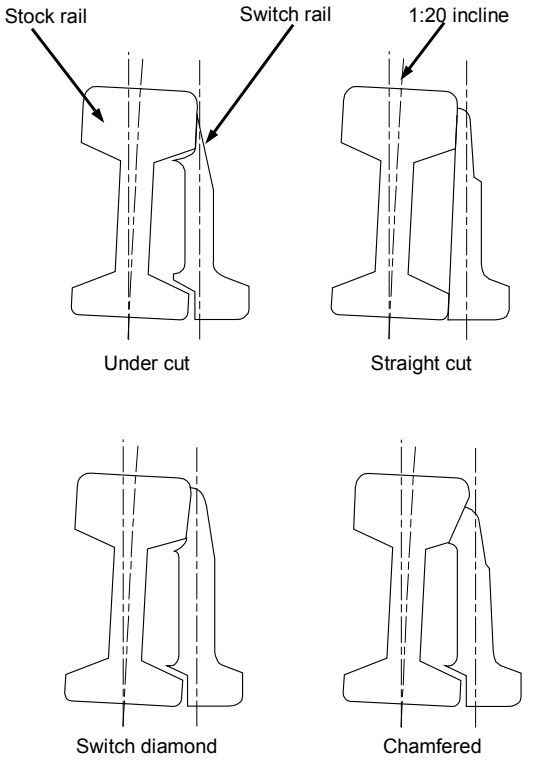



Figure 30

The bullhead rail is similar to flat bottom inclined in the fact that the stock rail is inclined inwards towards the centre of the track and the switch rail remains vertical.

At this stage it will be useful for you to examine different rail types on a section of track at your place of work. See if you can recognise the different types. Check your findings with a colleague or your mentor.

## 4.4 Section Summary

This section has looked at

- stock rails
- switch rails

Now complete Self-Assessment Question 4 before proceeding to Section Five.

## Self-Assessment Question 4

Correctly identify the following types of stock and switch rail configuration shown in figure 31.

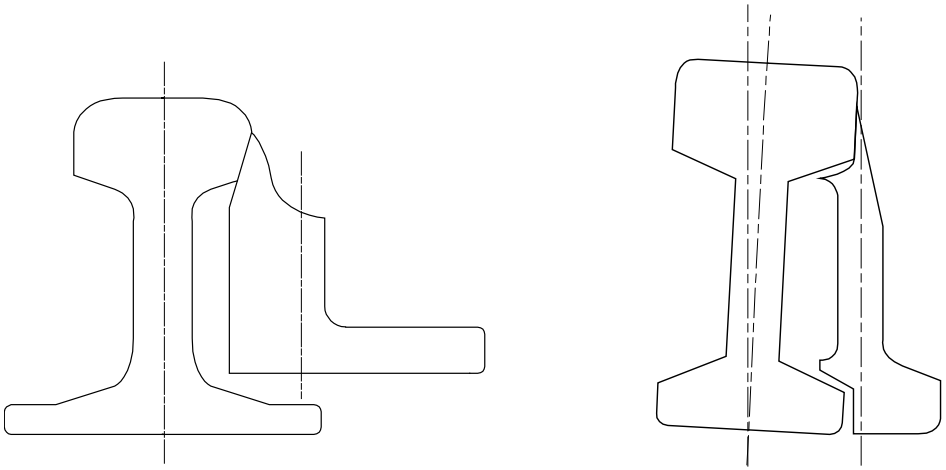


Figure 31

Now check your answer with the one given in the Answers Section. Before moving on to Section Five make sure that you fully understand all the information given in Section Four. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 5 Methods of Point Operation

## 5.1 Introduction

There are several types of point operation

- mechanical
- electrical
- electro-pneumatic
- hydraulic

These will now be examined in greater detail.

## 5.2 Mechanical

Mechanical points are the simplest of point mechanisms. The Signaller physically operates a large lever in the signal box which in turn pushes or pulls rodding and cranks. Finally the rodding drives the switch blades. Where a facing point lock is provided its lock plunger is operated by a separate lever and rodding.

Consideration has to be given to the effect of temperature change on the rodding. This is termed compensation and describes how allowance is made for metal

- expansion
- contraction

This effect could cause the switch rail to be either

- pulled hard against the stock rail
- pushed away from the stock rail

Cranks and compensators are employed to cancel out the effects of expansion and contraction. The principle is that every rodding run must include equal amounts of pushing and pulling, but a more detailed explanation is beyond the scope of this module.

Figure 32 illustrates a mechanical point layout. The rodding and cranks have been drawn schematically in order to aid clarity and any compensation arrangements have been omitted.

*Mechanical Points Layout*

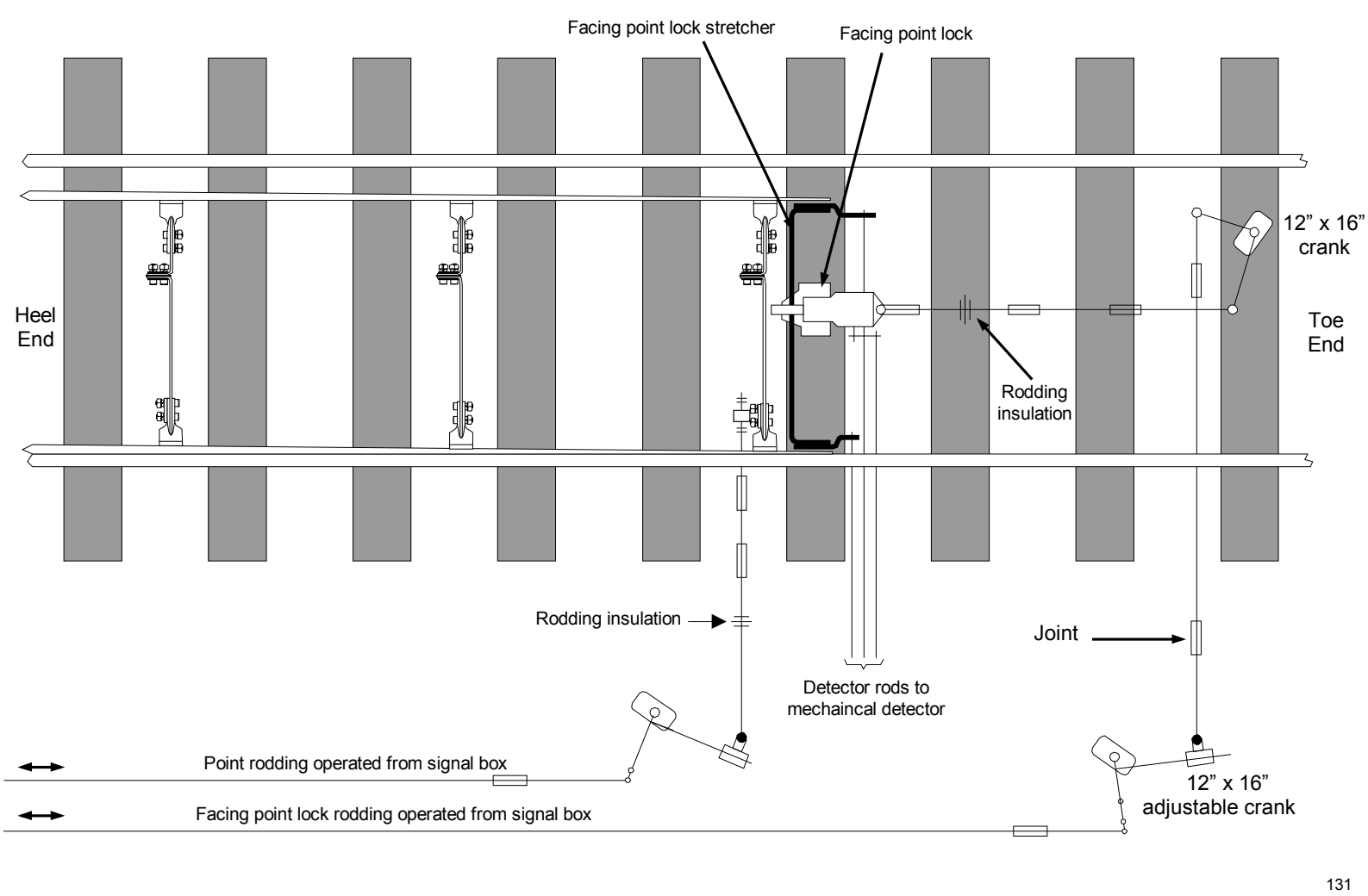



Figure 32

The facing point lock as shown in figure 33 illustrates in greater detail the toe end of a set of mechanical points detailing the

- lock plunger
- facing point lock stretcher
- near side detector rod
- far side detector rod
- the facing point lock detector rod

The near side and far side detector rods are used to ensure the points are fitting up to the stock rail in both positions and the facing point lock detector rod is used to ensure the points are locked. From these three rods the points can be detected (checked) to ensure that they are in the correct position and locked before a signal can be cleared for train movement over them.

*Arrangement of Mechanical Facing Point Lock*

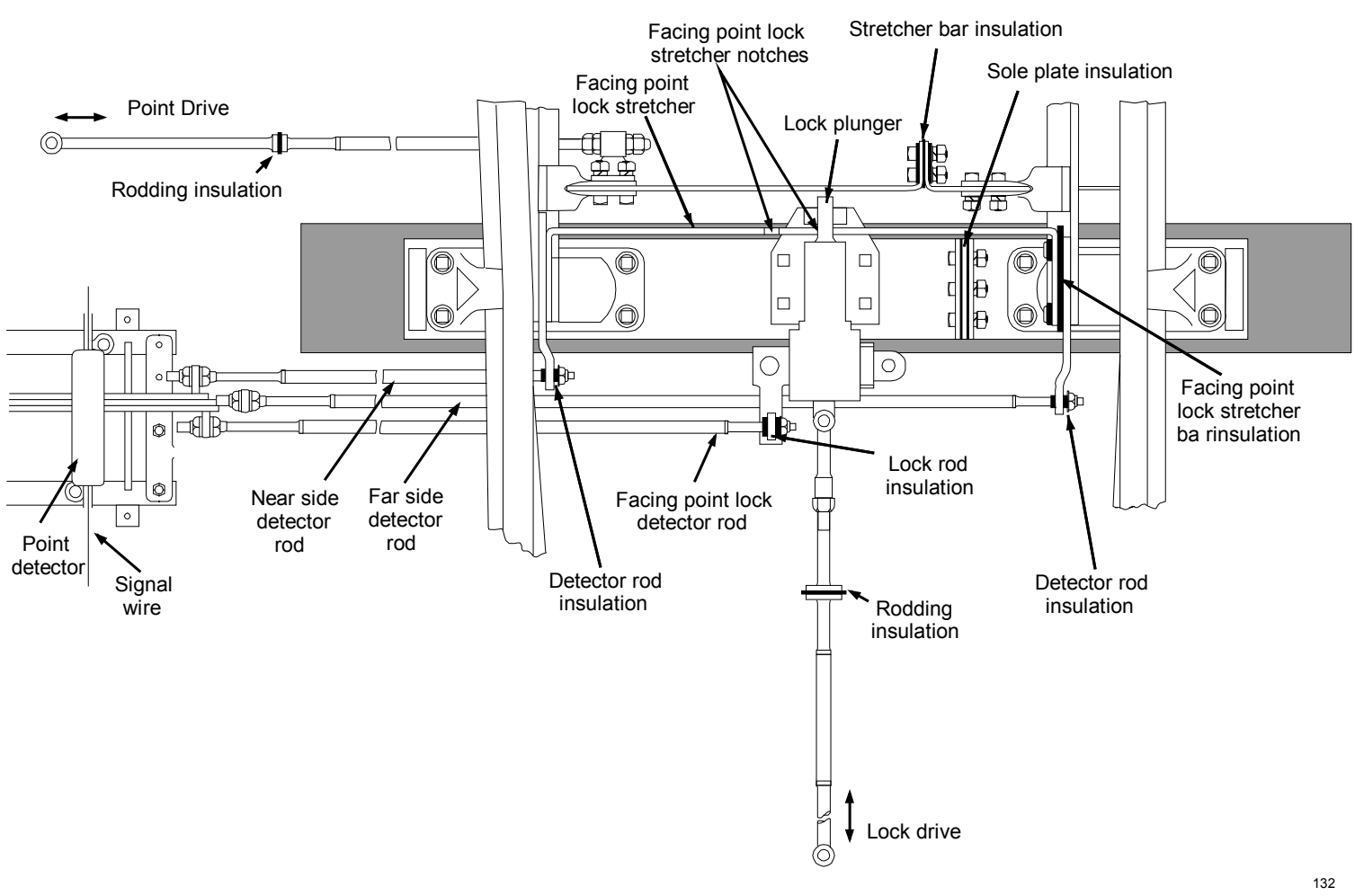



Figure 33

In figure 33 the facing point lock has a lock plunger operated by the Signaller pushing/pulling a lever which in turn moves the rodding to operate the lock plunger.

The lock plunger enters a notch cut in the facing point lock stretcher so that the points will only lock when the switch rail is fitting up against the stock rail in either normal or reverse positions.

There are two notches cut in the facing point lock stretcher bar one for the normal position and one for the reverse position.

Also shown in figure 33 are the insulations required for the track circuit to operate correctly. Insulation will be looked at in more detail in a later section.

## 5.3 Electrical

Two types of electric point machine are used. Both use an electric motor to move the points.

The combined machine incorporates the point detection and lock.

The non-combined machine does not incorporate point detection or locking. They have to be provided separately.

## 5.4 Electro-Pneumatic

Electro-pneumatic points are powered by compressed air which drives a piston to move the points. The compressed air supply is delivered from a local reservoir which is maintained by an independent compressor. The points movement is controlled by electrically operated air valves which regulate the air supply to the piston. The machine incorporates point detection and locking.

## 5.5 Hydraulic

Hydraulic operation of points is performed by hydraulic fluid moving a piston. The points movement is controlled by switching the electrical supply to an electric motor which drives the hydraulic compressor. The mechanism incorporates electrical point detection.

## 5.6 Practical Examples

A number of different point machines are described in greater detail in sections which follow.

## 5.7 Section Summary

This section has examined a number of methods of point operation.

Now complete Self-Assessment Question 5 before proceeding to Section Six.

## Self-Assessment Question 5

Table 1 which follows lists the various methods of point operation and identifies various options covering control and point detection. Tick the appropriate boxes to identify the correct options in each case.

| Method of Point Operation | Points moved manually | Points moved by electric motor | Points moved by compressed air supply | Points moved by hydraulic fluid | Points locked by separate lock | Points locked by a lock in the machine | Mechanical Point detection | Electrical Point detection |
|---|---|---|---|---|---|---|---|---|
| Mechanical | | | | | | | | |
| Combined Electrical | | | | | | | | |
| Non-combined Electrical | | | | | | | | |
| Electro-pneumatic | | | | | | | | |
| Hydraulic | | | | | | | | |

Now check your answer with the one given in the Answers Section. Before moving on to Section Six make sure that you fully understand all the information given in Section Five. If there are any areas which you do not understand, re-read the relevant parts of the text.

This page has been left blank intentionally.

# 6 Combined Point Machines

## 6.1 Introduction

As was previously stated the function of a point machine is to move a pair of switch rails so a train can be switched from one line to another.

Electrically operated point machines fall into two basic categories

- combined point machines
- non-combined point machines

Non-combined point machines will be examined in Section Seven.

In a combined point machine the

- locking
- detection

take place within the machine itself.

There are numerous combined point machines in use but two of the most common versions in use on British railways are the

- GEC HW series point machines
- Westinghouse Style 63 point machine

Although these machines are quite different they have been designed to be interchangeable at the point fittings.

An example of the point fittings will be examined later in this section.

The component parts and operation of these machines will now be examined in greater detail.

## 6.2 The GEC HW 2000 Point Machine

Look carefully at figure 34 which illustrates a GEC HW 2000 point machine. This machine is very similar to the GEC HW 1000 point machine. The only major difference between them is that the HW 1000 has a split field motor with a friction clutch and the HW 2000 has a permanent magnet motor with an electro-magnetic clutch. The differences between these motors and clutches will become apparent later. Use of the HW 2000 is generally confined to AC traction areas.

*GEC HW 2000 Points Machine*

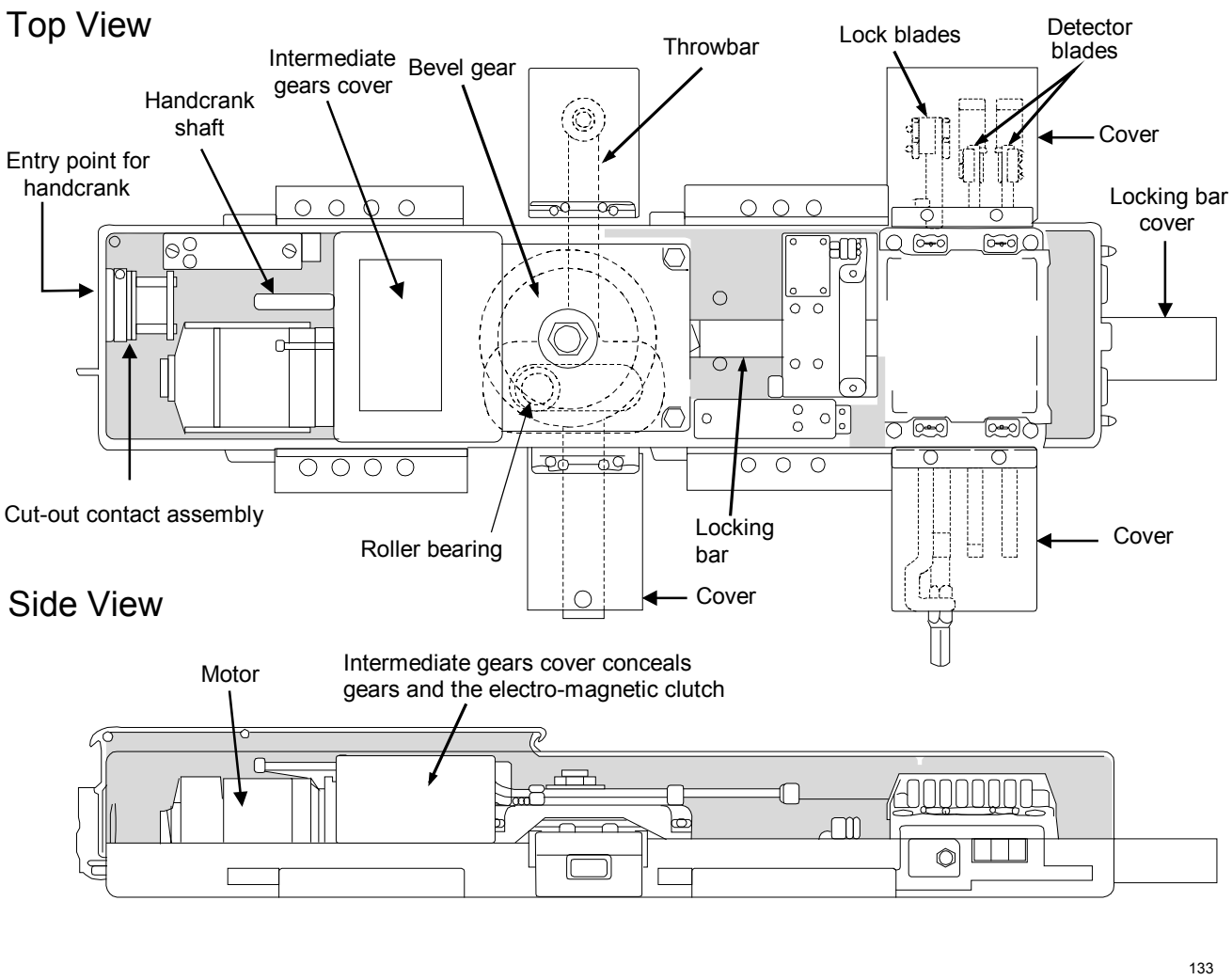



Figure 34

The basic components which can be seen in figure 34 are as follows

## Motor

The DC electric motor provides the power by which the points are operated.

## Clutch

The clutch is of the electromagnetic type and is energised with the motor. It is adjusted to slip by means of a variable resistor in the event of the

- points being unable to move
- points being unable to complete their travel

This ensures that the motor does not burn out if it is overloaded.

When

- the point switch blades have completed their travel
- the locking bar has almost completed its travel
- the motor and the clutch are de-energised.

Disengaging the clutch ensures that the gear mechanism and locking bar complete their travel under their own momentum. This reduces the stress and wear in the point machine.

### Intermediate Gears

The intermediate gears transmit the rotation of the motor to the bevel gear via the bevel pinion. This action will be illustrated later in this section.

### Bevel Gear

The bevel gear wheel carries a roller bearing. This roller bearing is used in converting the rotation of the motor into a linear motion which will move the points.

## 6.3 Operation of the GEC HW 2000 Point Machine

In order to understand the operation of the GEC HW 2000 point machine it is necessary to be aware of the additional components which are illustrated later in figure 35.

### Escapement

The escapement slot in conjunction with the roller bearing converts the rotation of the motor into a linear motion of the throw bar to move the points.

### Locking Bar

The locking bar is connected to the bevel gear via a connecting link. As the bevel gear rotates, the connecting link pushes the locking bar which in turn pushes the lock dog out of the notch in the lock blades. This action unlocks the points.

## Lock Blades

The lock blades are connected via a lock rod to the lock stretcher bar. The lock blades have two notches through which the lock dog locks the points either

- normal
- reverse

When the points have completed their travel and the switch rail is correctly fitted to the stock rail in either

- normal position
- reverse position

the lock dog on the locking bar will enter the appropriate notch in the lock blades. This will

- prove that the switch rails are in position
- prevent the closed switch rail from opening during the passage of a train

When the machine starts its next movement, the lock dog disengages from the lock notch in the lock blades.

## Detector Blades

The detector blades are used to detect the

- closed switch rail is fitting correctly to the stock rail
- the open switch rail has a necessary clearance from the opposite stock rail

The detector blades are connected to the switch rails via the detector rods.

The points are fully detected when the appropriate switch rail is closed, the other switch rail is open and the lock dog is fully engaged in the notch in the lock blades.

Look carefully at figure 35 and locate the relevant components as the operation of the machine is described step by step.

*GEC HW 2000 Point Machine, movement of Throw Bar and Locking Bar*

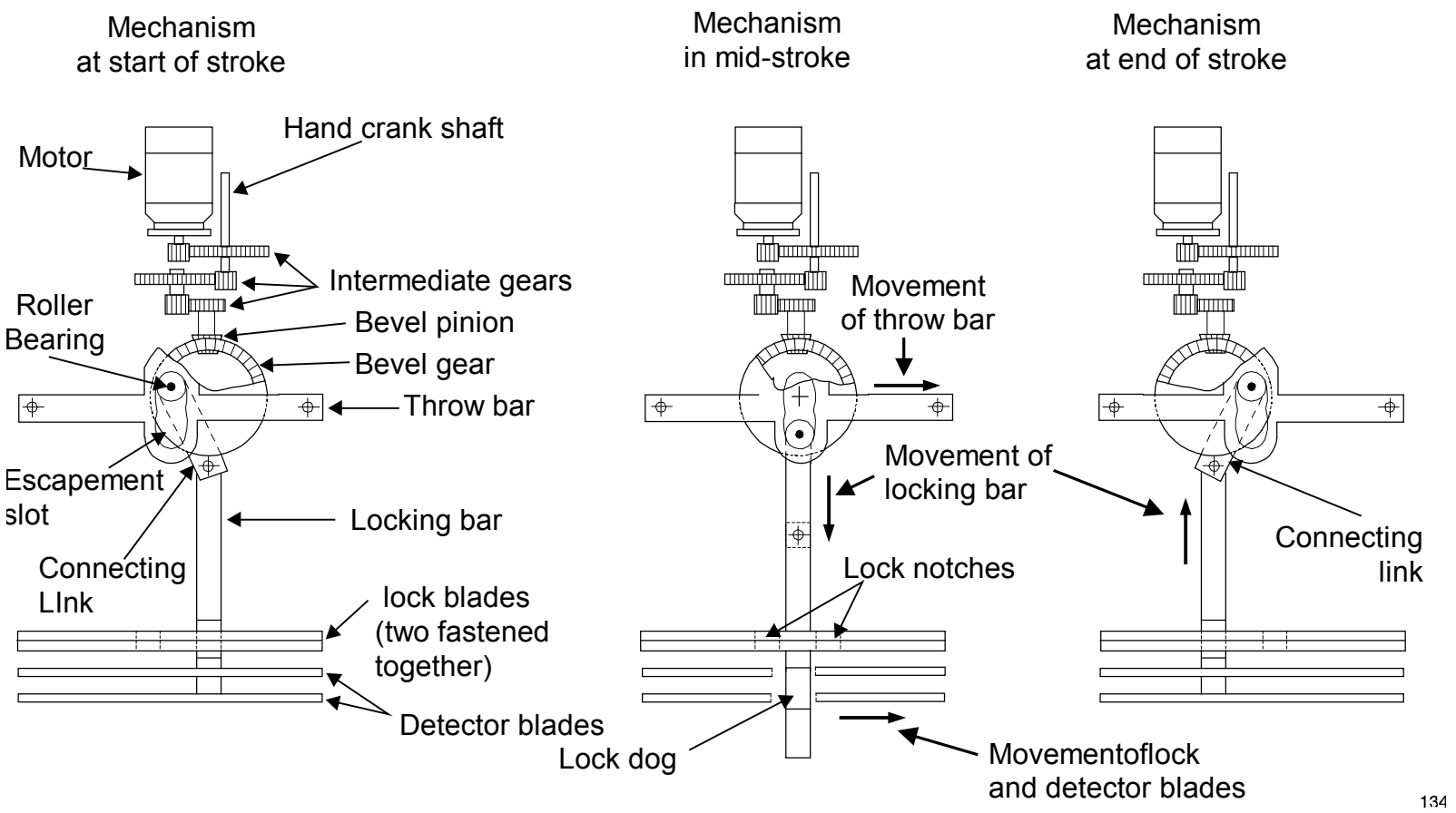



Figure 35

For the purpose of explaining the operation the terms horizontal and vertical are used in relation to figure 35. In reality the movement is

- longitudinal
- lateral

*Stage 1*

When the motor rotates it

- drives the bevel wheel via the intermediate and main gears
- the roller bearing which is connected to the bevel wheel starts to move down along the escapement slot

At this stage the roller bearing is

- not moving horizontally therefore the throwbar is not moving and the switch rails remain held in place
- moving vertically which causes the lock dog to be moved out of the notch in the lock blades.

*Stage 2*

During this stage the roller bearing

- approaches the end of the escapement slot
- starts to move horizontally which moves the throw bar horizontally thus moving the point switch rails via the drive rod

The point switch rails complete their travel and the appropriate one fits to the stock rail.

*Stage 3*

The points now need to be locked. The

- roller bearing begins to move vertically up the escapement slot
- connecting link moves the locking bar upwards which moves the lock dog into the second notch in the lock blades

The operation is now complete.

Figure 36 shows the point fittings which are used with a combined point machine for a plain lead with the point machine mounted on the right hand side. The first two timbers at the toe end of the points and the sole plate are extended to accommodate the combined point machine.

Also shown in figure 36 are the insulations required for the track circuit to operate correctly. Insulation methods are examined in greater detail in Section 11.

*Point fittings for a plain lead driven by a combined point machine*

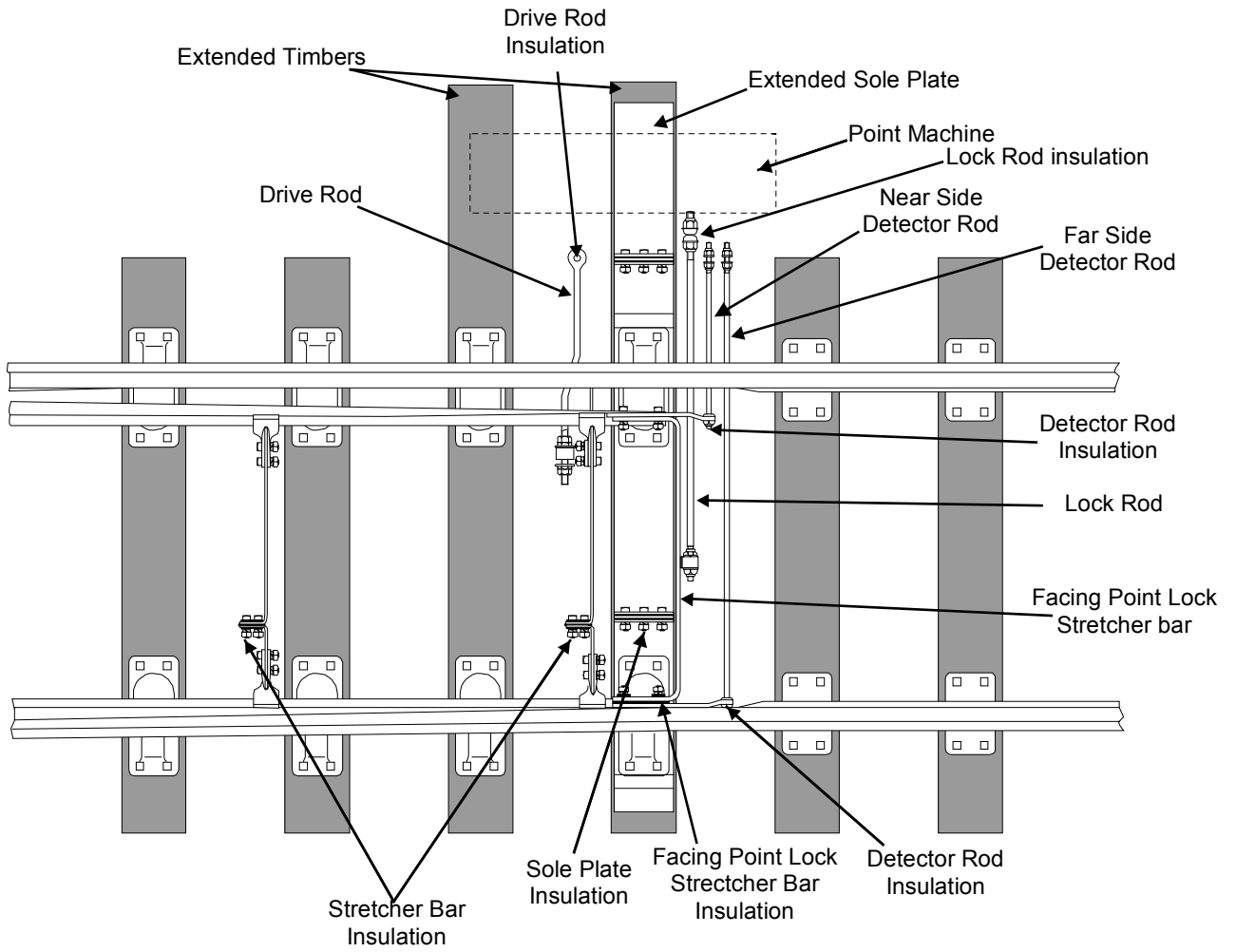



Figure 36

## *6.4* The Westinghouse Style 63 Point Machine

The main components of this type of point machine will now be examined. It is worthwhile locating these components on figure 37.

### Motor

The DC electric motor provides the power by which the points are operated.

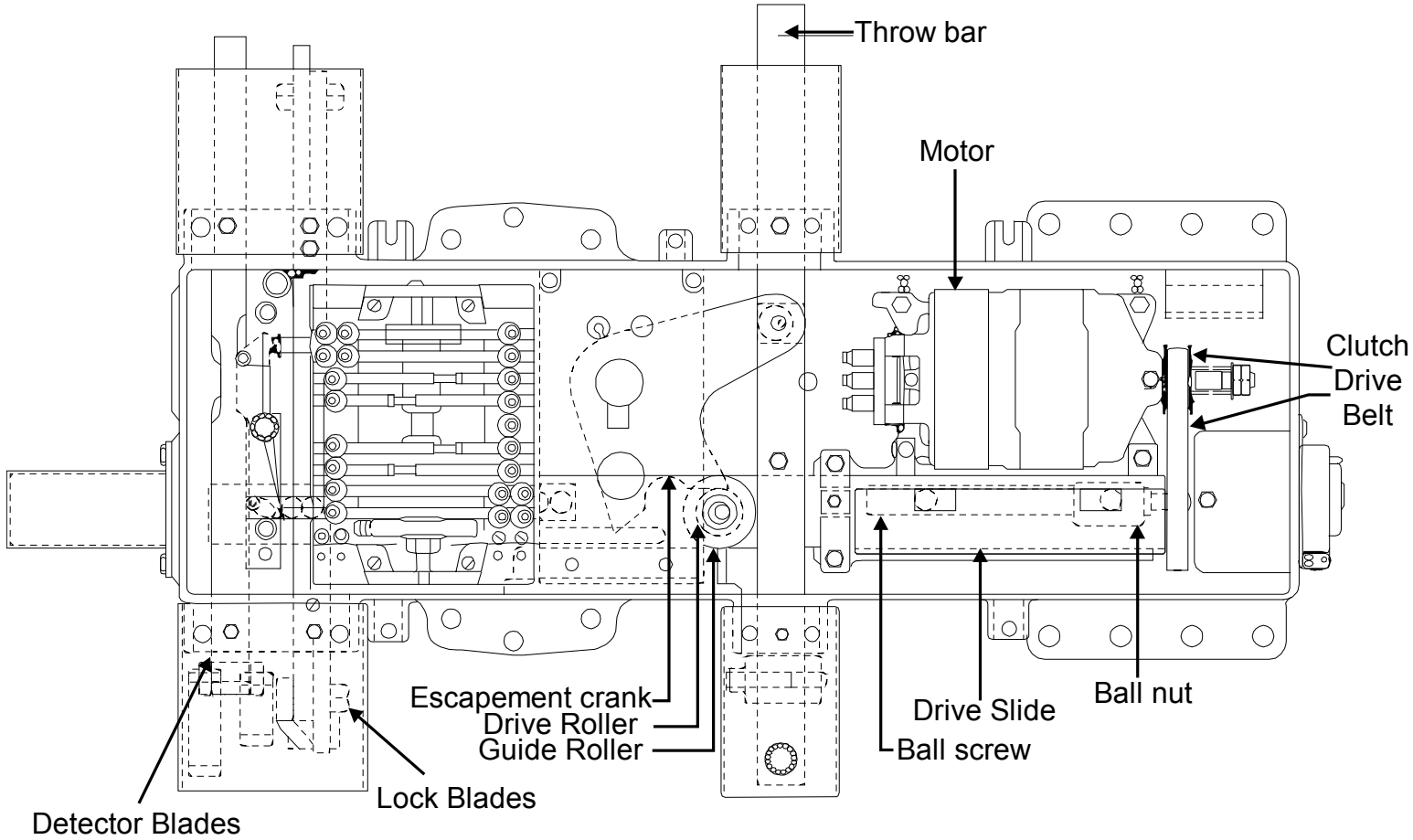
### Clutch

The clutch is a dry plate friction type and can be adjusted by means of a pressure spring to slip in the event of the points being unable to complete their travel. The slipping action of the clutch protects the motor from burning out.

### Escapement Crank

The escapement crank in conjunction with the drive slide produces the movement of the throw bar which moves the points.

*Westinghouse Style 63 Point Machine*




136

Figure 37

## Throw Bar

The throw bar is connected to the escapement crank to produce the movement of the points switch rails.

## Lock Blades

The lock blades are connected via a lock rod to the lock stretcher bar. It has two notches through which the lock dogs will lock the points

- normal
- reverse

When the points have completed their travel and the switch rail is correctly fitted to the stock rail in either

- normal position
- reverse position

then the lock dog on the drive slide will enter the appropriate notch in the lock blades. This will

- prove that the switch rails are in position
- prevent the closed switch rail from opening during the passage of a train

## Detector Blades

The detector blades are used to detect

- the closed switch rail is fitting correctly to the stock rail
- the opposite switch rail has the necessary clearance from its adjacent stock rail

They are connected to the switch rails via the detector rods.

The points are fully detected when the appropriate switch rail is closed, the other switch rail is open and the appropriate lock dog is fully engaged in the corresponding notch in the lock blades.

# 6.5 Operation of the Westinghouse Style 63 Point Machine

Look carefully at figure 38 and locate the relevant components as the operation of the machine is described step by step. This illustration shows the movement of the throw bar from left to right.

*Operation of Escapement and Lock on Westinghouse Style 63 Point Machine*

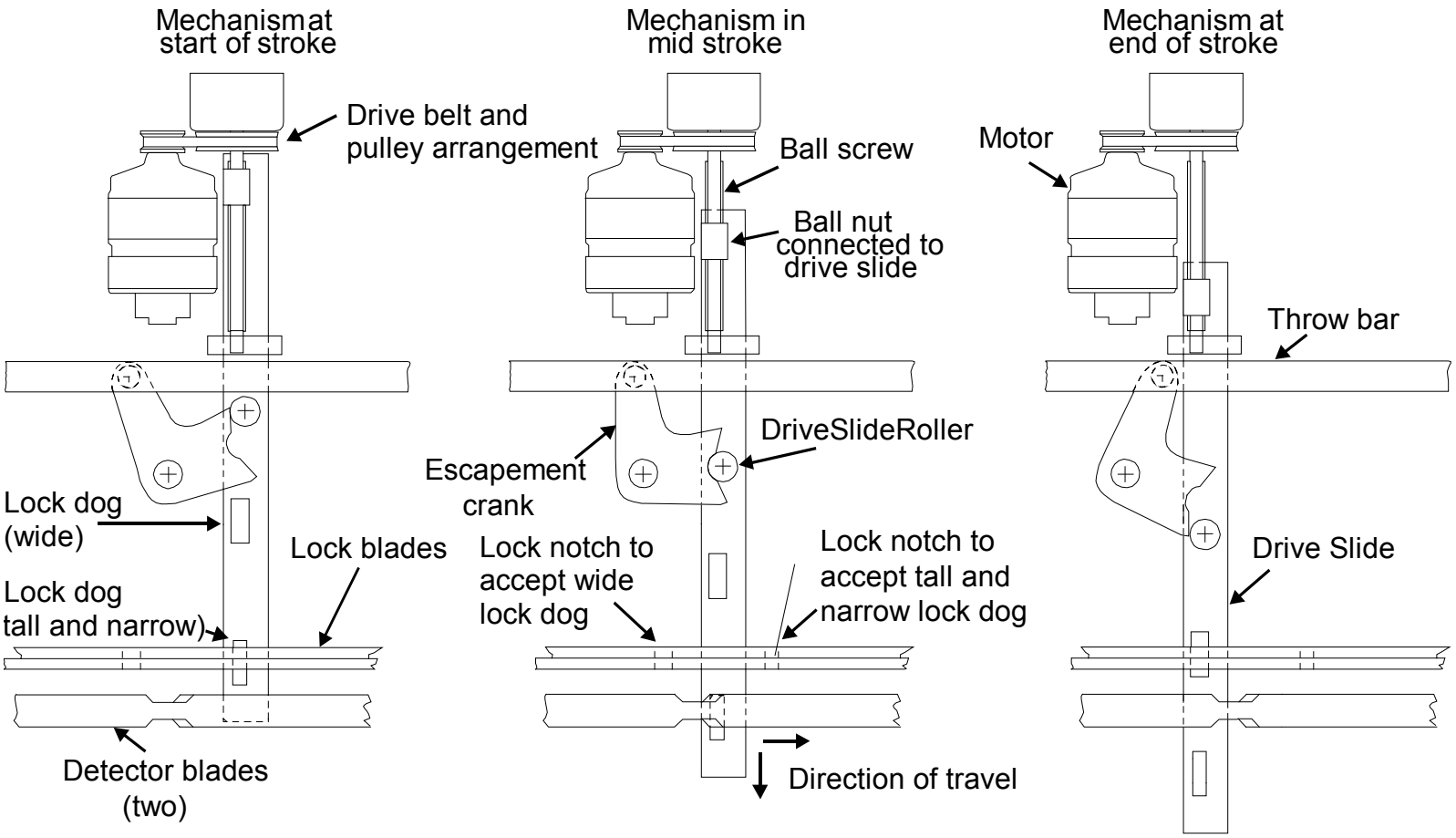



Figure 38

## To Unlock

The motor rotates

- this drives the ballscrew via a belt and pulley arrangement
- as the ballscrew turns, it moves a ballnut which is connected to the drive slide
- the movement of the drive slide removes the lock dog from the notch in the lock blades

The points are unlocked and therefore the switch rails are free to move to the opposite lie.

## To Drive

After the unlocking sequence the drive slide roller moves along the escapement crank and

- the drive slide roller engages the escapement crank, which moves the throw bar
- the throw bar moves the switch blades over to the opposite lie

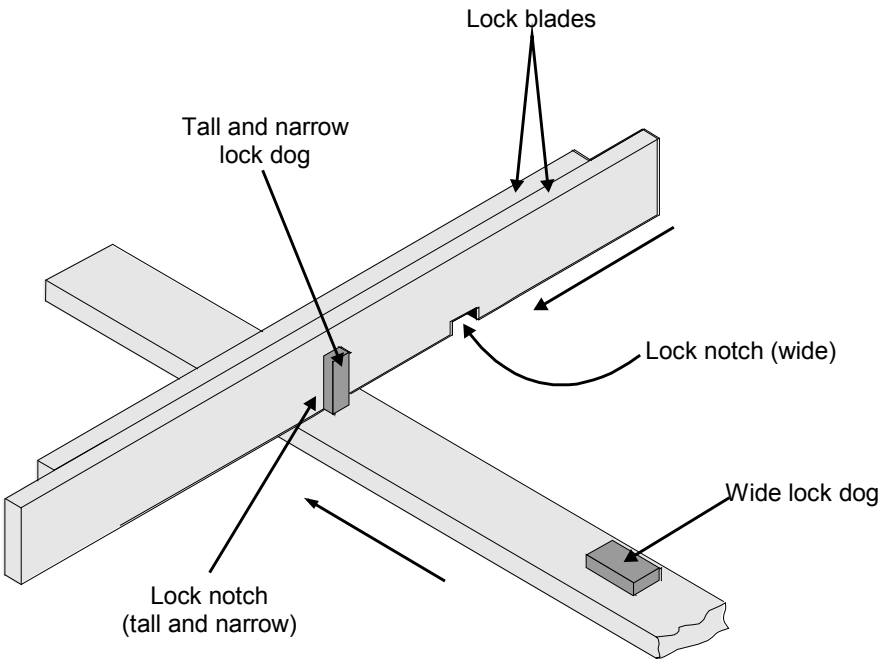
## To Lock

As the drive slide approaches the end of its travel

- the drive slide roller disengages from the escapement crank
- the drive slide moves the lock dog into the opposite notch in the lock rod

The locking mechanism is illustrated in figure 39.

*Locking mechanism of Westinghouse Style 63 point machine*




138

Figure 39

## Snubbing

When the drive slide is nearing the end of its travel it operates electrical contacts which

disconnect the electrical supply to the motor

create a short circuit across the motor windings which causes the motor to act as a brake

The resulting effect is called “snubbing” and brings the mechanism to a controlled stop.

## 6.6 Motors

The motors used in combined point machines can be one of two types

- split field (also known as a series field)
- permanent magnet

The split field motor has the following advantages over the permanent magnet motor

- it is smaller
- it provides a much higher starting torque
- it is much cheaper

The permanent magnet motor has one major advantage over the split field motor. It is immune to induced AC currents. Consequently it is normal practice to use the permanent magnet motors in AC traction areas and the split field motors elsewhere.

## 6.7 Section Summary

This section has looked at

- GEC HW 2000 combined point machine
- Westinghouse Style 63 combined point machine

Now complete Self-Assessment Question 6 before proceeding to Section Seven.

## Self-Assessment Question 6

State the two most common types of combined electrical point machine in use on British railways and the purpose of the clutch mechanism on those machines.

Now check your answer with the one given in the Answers Section. Before moving on to Section Seven make sure that you fully understand all the information given in Section Six. If there are any areas which you do not understand, re-read the relevant parts of the text.

This page has been left blank intentionally.

# 7 Non-Combined Point Machines

## 7.1 Introduction

A non-combined point machine does not contain an integral lock or detector. It provides two separate drives

- one operating movement of the switch rails
- the other operating a bolt lock located in the four foot

Detection is provided by a separate unit.

The SGE Type HA point machine is typical.

## 7.2 The SGE Type HA Non-Combined Point Machine

This machine is illustrated in figure 40. The main components are as follows

### Motor

The DC electric motor supplies the power by which the points are operated.

*View of an SGE Type HA point machine with the cover removed*

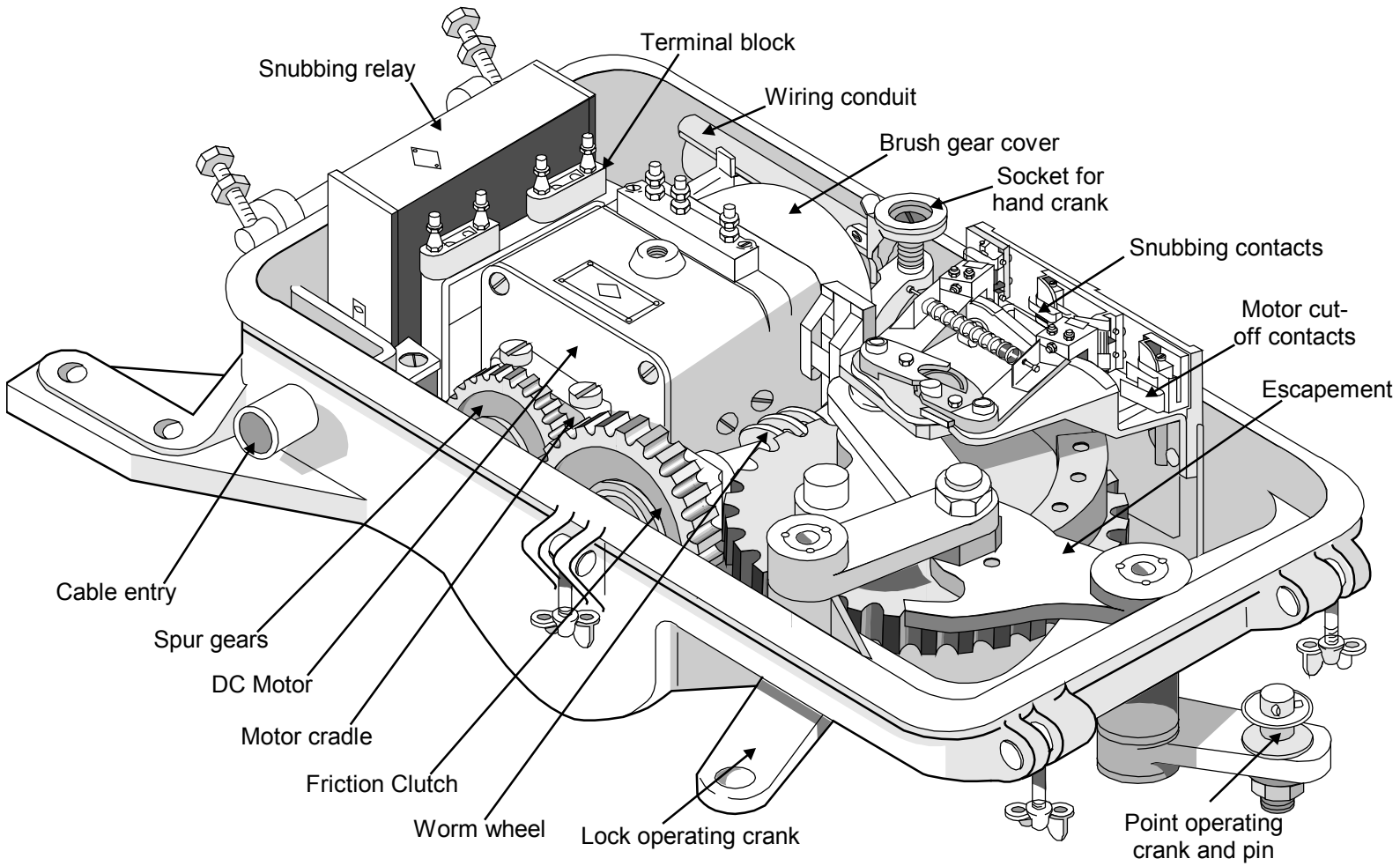



Figure 40

## Gears

The gears transmit the motor rotation through a clutch to a worm wheel to move the points.

## Clutch

The clutch is a dry plate friction type and can be adjusted to slip by means of a pressure spring in the event of the points being unable to complete their travel.

The slipping action of the clutch protects the motor from burning out.

## Escapement

The escapement is connected to the worm wheel and produces the thrust to move the points and the lock operating crank.

## Point Operating Crank

The point operating crank is connected via rods or cranks to the switch rails.

## Lock Operating Crank

This is connected via rods and cranks to a remote lock.

## 7.3 Operation of the SGE Type HA Non-Combined Point Machine

This machine was designed for installation on point layouts using the British Railways standard facing point lock. On trailing points a version is used which has the lock operating crank omitted.

The rotation of the motor is transmitted through the gears and clutch to the worm wheel. As the worm wheel turns it engages the escapement and provides the necessary thrust to the point operating crank and the lock operating crank.

Figure 41 shows the layout typically used on a facing point. For clarity the chairs and the rod insulation have been omitted.

*Typical layout employed with an SGE Type HA point machine*

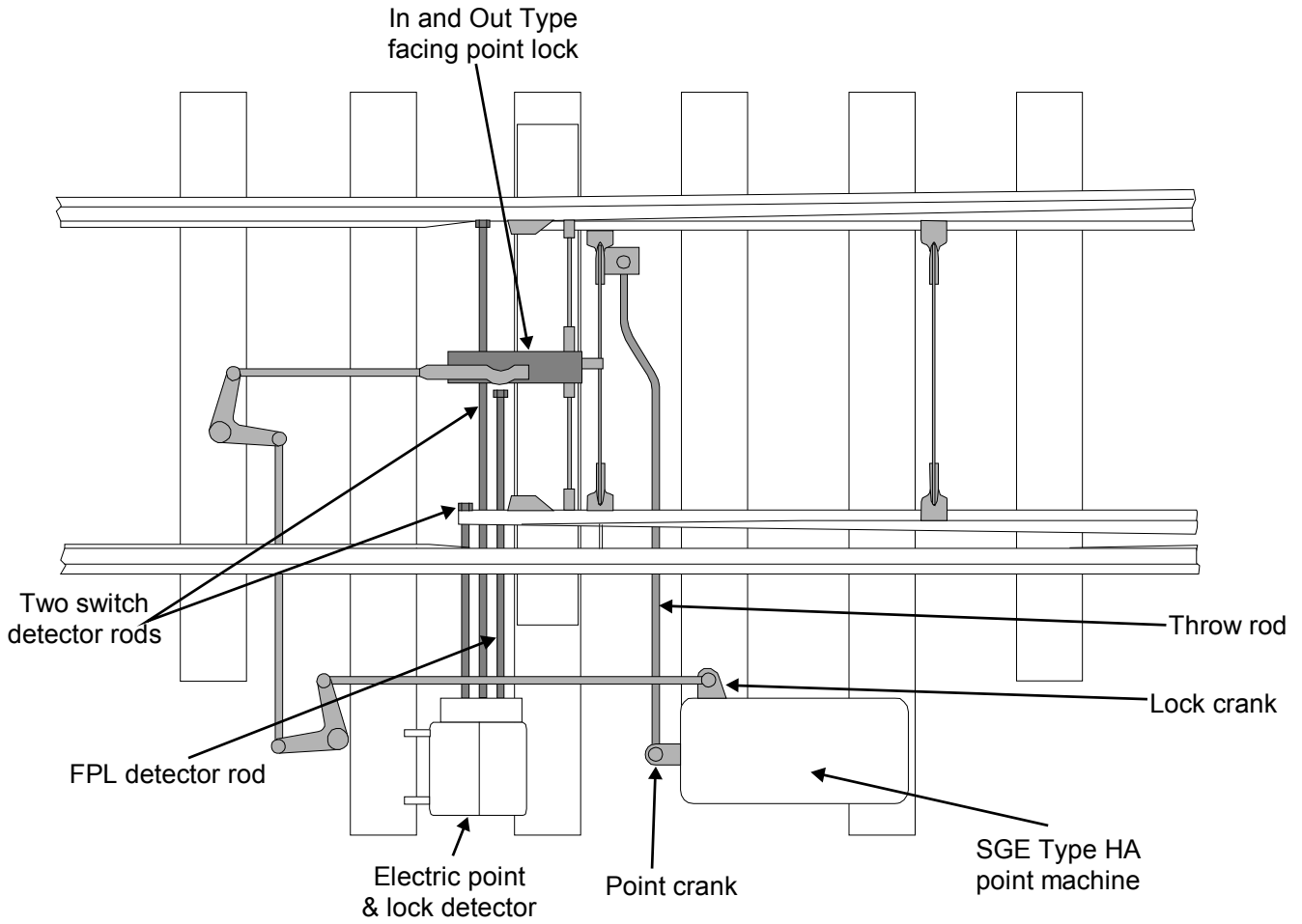



Figure 41

## 7.7 Section Summary

This section has looked at the SGE Type HA non-combined point machine in terms of its

- components
- operation

Now complete Self-Assessment Question 7 before proceeding to Section Eight.

## Self-Assessment Question 7

State the principal differences between the non-combined and the combined electrically operated point machines.

Now check your answer with the one in the Answers Section. Before moving on to Section Eight make sure that you fully understand all the information given in Section Seven. If there are any areas which you do not understand, re-read the relevant parts of the text.

This page has been left blank intentionally.

# 8 Electro-Pneumatic Point Machine

## 8.1 Introduction

The distinctive features of the electro-pneumatic point machines are the power to move the point switch rails is provided by compressed air and the control is provided via electro-magnetic needle valves. The AC immune version of the SGE Electro-Pneumatic Point Machine is typical of these types of machine and is described below. There are other versions of the SGE machine which are functionally very similar, and Westinghouse machines which have detailed design differences but employ the same principles.

## 8.2 Components of the SGE Electro-Pneumatic Point Machine (AC Immune)

A typical general arrangement of this machine is shown in figure 42. Not illustrated is the compressed air supply to the air control valve unit. This is supplied from a locally controlled compressed air reservoir which usually serves all of the EP points in the locality and delivers the air at 50 psi. Also not shown is the metal cover which is normally fitted to protect the escapement mechanism and is secured by a single captive screw. The covers on the air control valve unit and the point detector are provided with padlocks to secure them against unauthorised access and the entry of foreign matter.

*General Arrangement of SGE EP Point Machine (AC Immune)*

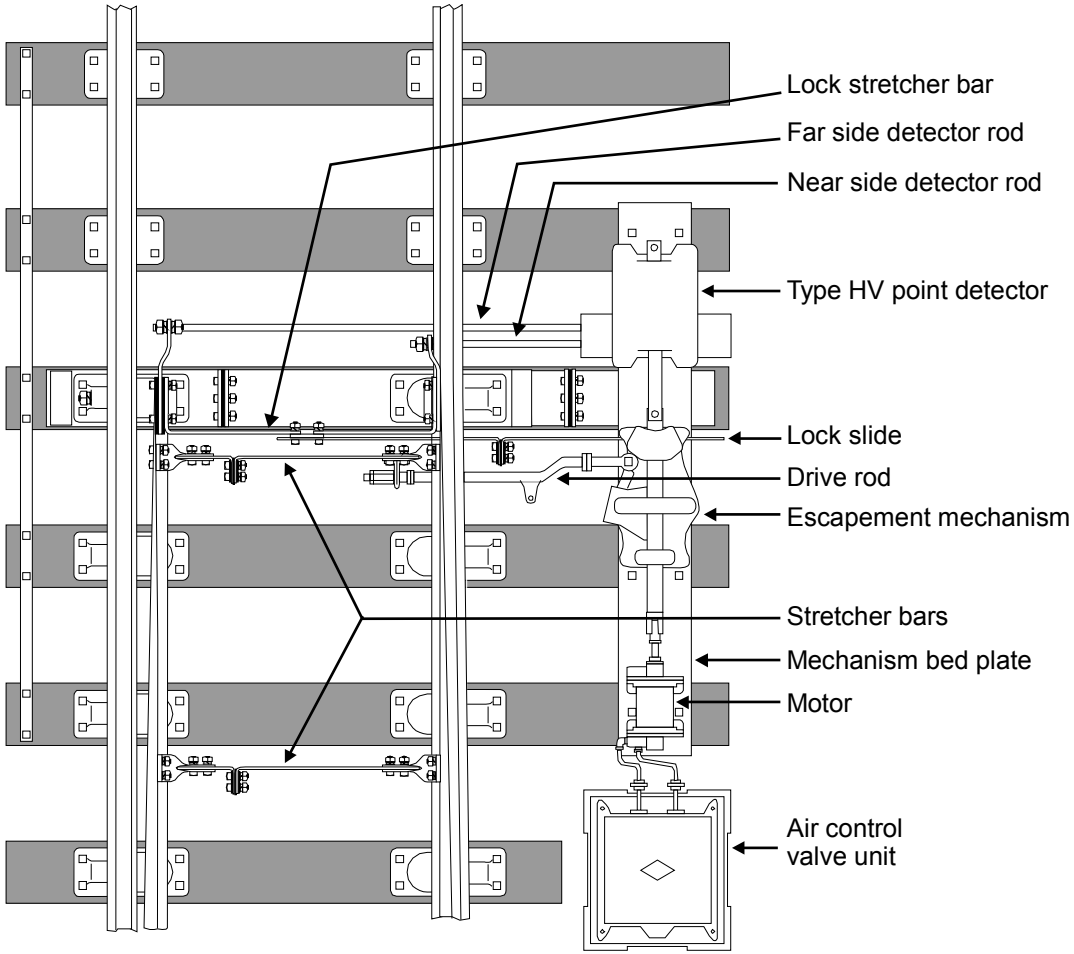



Figure 42

Note the four parts of the mechanism

- the motor
- the air control valve unit
- the escapement mechanism
- the point detector

## The Motor

A cut away view of the motor is shown in figure 43. It consists of a simple piston in a cylinder. Air is applied to one side of the piston to move the points to their normal position and to the other side of the piston to move them reverse. The piston has a stroke of 8 inches and normally has a diameter of 5 inches. However, for heavy duty such as double slips, a motor with a 6 inch diameter piston can be used.

*Cut away illustration of SGE EP Motor*

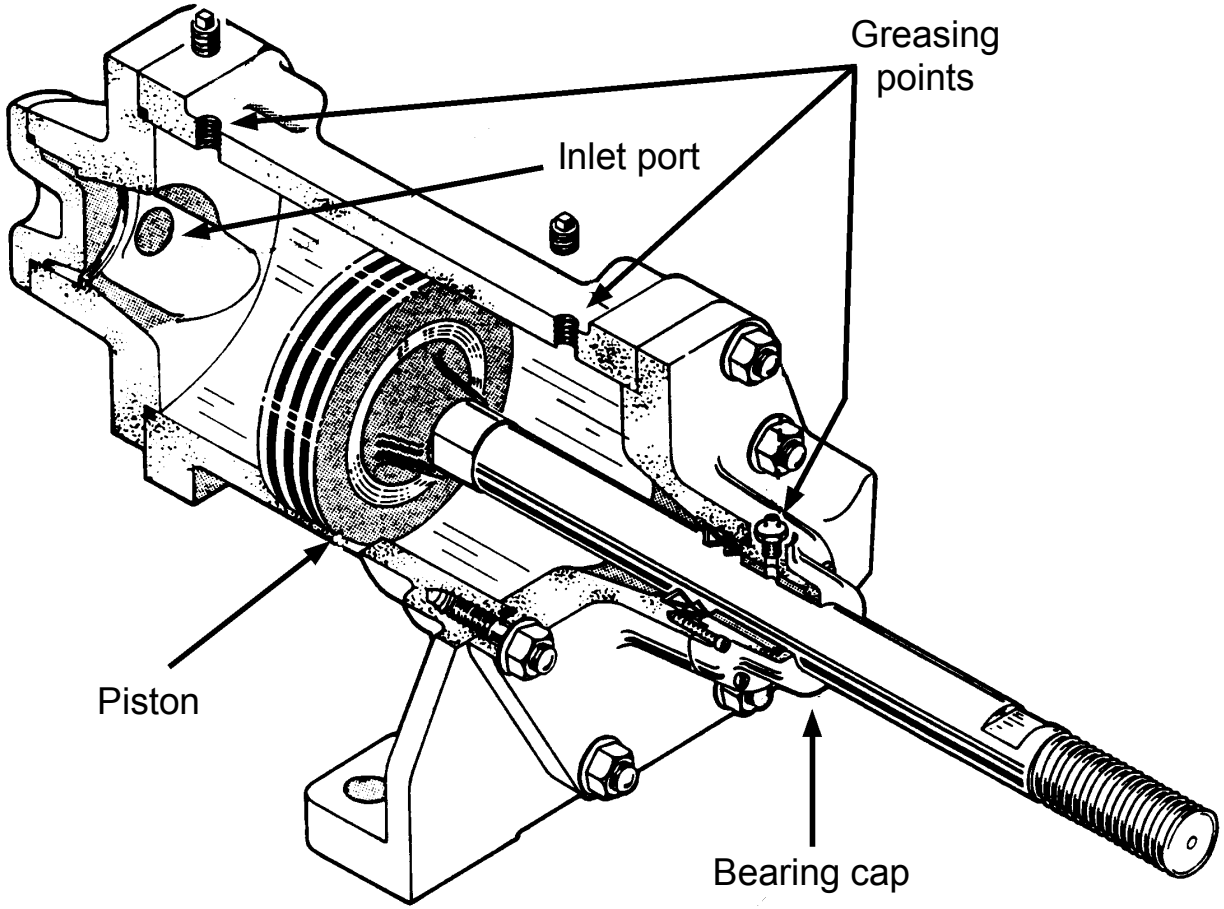



Figure 43

## The Air Control Valve Unit

The function of this unit is to deliver compressed air to the appropriate side of the motor piston when a points movement is required and to vent the other side of the piston to the atmosphere. When the full travel of the piston is completed the compressed air in the motor cylinder is vented to take the stress off the mechanism and leave it ready to make the next movement. The control is provided by three electro-magnetic needle valves. They are the

- lock valve
- normal valve
- reverse valve

All three valves have the same construction which is illustrated in figure 44.

*Construction of needle valve*

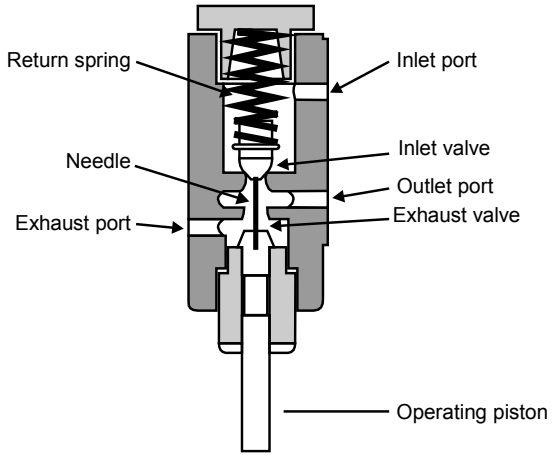



Figure 44

The needle valve operating piston is moved by the armature of a relay which resembles and has the characteristics of a BR930 series AC immune relay. The relay coil can be rated for operation by 50V DC or 24V DC. When the relay is de-energised the operating piston is pushed down by the return spring and the exhaust valve is open to give an air flow path from the outlet port to the exhaust port. When the relay is energised the operating piston is pushed up so that the exhaust valve is closed and the inlet valve is opened to give an airflow path from the inlet port to the outlet port.

On their own the needle valves provide too small an air path to operate the motor so a motor control valve is put between them and the motor. A simplified illustration of the arrangement is shown in figure 45.

*Simplified illustration of the Motor Control Valve*

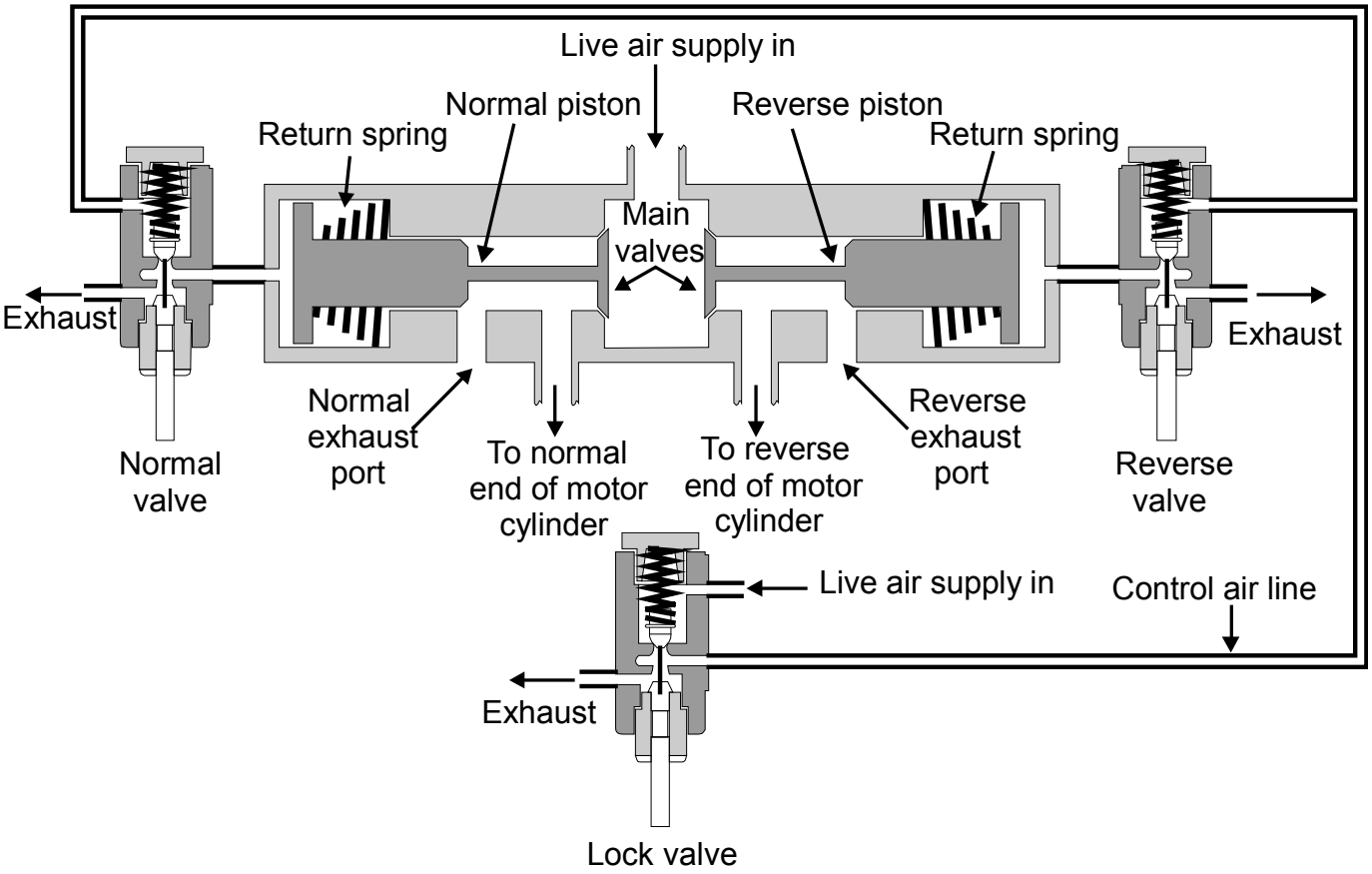



Figure 45

Note: In practice the arrangement of the pistons and the main valves is more complex, in order to ensure that each exhaust port is closed before the corresponding main valve is opened and the main valve closes before the exhaust port opens.

## Operation of the Motor Control Valve

*Motor Stationary*

When the EP points are not required to move the lock valve is de-energised. Irrespective of the state of the air supply and the state of the normal and reverse valves the control air line is open to the exhaust and no air pressure is applied to either the normal piston or the reverse piston. Both of these pistons are driven outwards by their return springs so that

- both main valves are closed
- both pistons are leaving open the respective exhaust ports
- the normal end and the reverse end of the motor cylinder are both open to the exhaust

*Motor moving points normal*

To move the points to their normal position, the normal valve is energised and the reverse valve is de-energised. (Electrical interlocking ensures that both valves cannot be energised simultaneously.) Then the lock valve is energised. This allows the live air supply to drive the normal piston to the right, which closes the normal exhaust port and opens the main valve to admit the live air supply to the normal end of the motor cylinder. When the motor has driven the points to their normal position and the points are detected normal the lock valve is de-energised, which connects the control air line to the exhaust, allows the normal piston to return to its outward position and removes the air pressure from the motor.

*Motor moving points reverse*

To move the points to their reverse position, the reverse valve is energised and the normal valve is de-energised. Then the lock valve is energised. This allows the live air supply to drive the reverse piston to the left, which closes the reverse exhaust port and opens the main valve to admit the live air supply to the reverse end of the motor cylinder. When the motor has driven the points to their reverse position and the points are detected reverse the lock valve is de-energised, which (as for the normal position) connects the control air line to the exhaust, allows the reverse piston to return to its outward position and removes the air pressure from the motor.

***Motor regulator***

To ensure that only just sufficient pressure is applied to the motor, in order to keep to a minimum the stress and wear on the mechanism, a screw regulator valve (not illustrated) is used to restrict the flow of the air supply to the main valves. This regulator is adjusted on site as part of the initial commissioning and the subsequent routine maintenance.

## The Escapement Mechanism

This is illustrated in figure 46. It has three functions

- to convert the linear movement produced by the motor into the required movement of the throw bar
- to provide the locking function on the lock slide. One of two lock dogs on the drive slide engages in a notch on the lock slide. During the first part of drive slide movement the dog leaves the notch to allow the points to move. During the last part of the drive slide movement the second dog engages in a second notch on the lock slide to lock the points in their other position.
- to transfer the motor piston movement to the point detector

*Escapement Mechanism for SGE EP Point Machine*

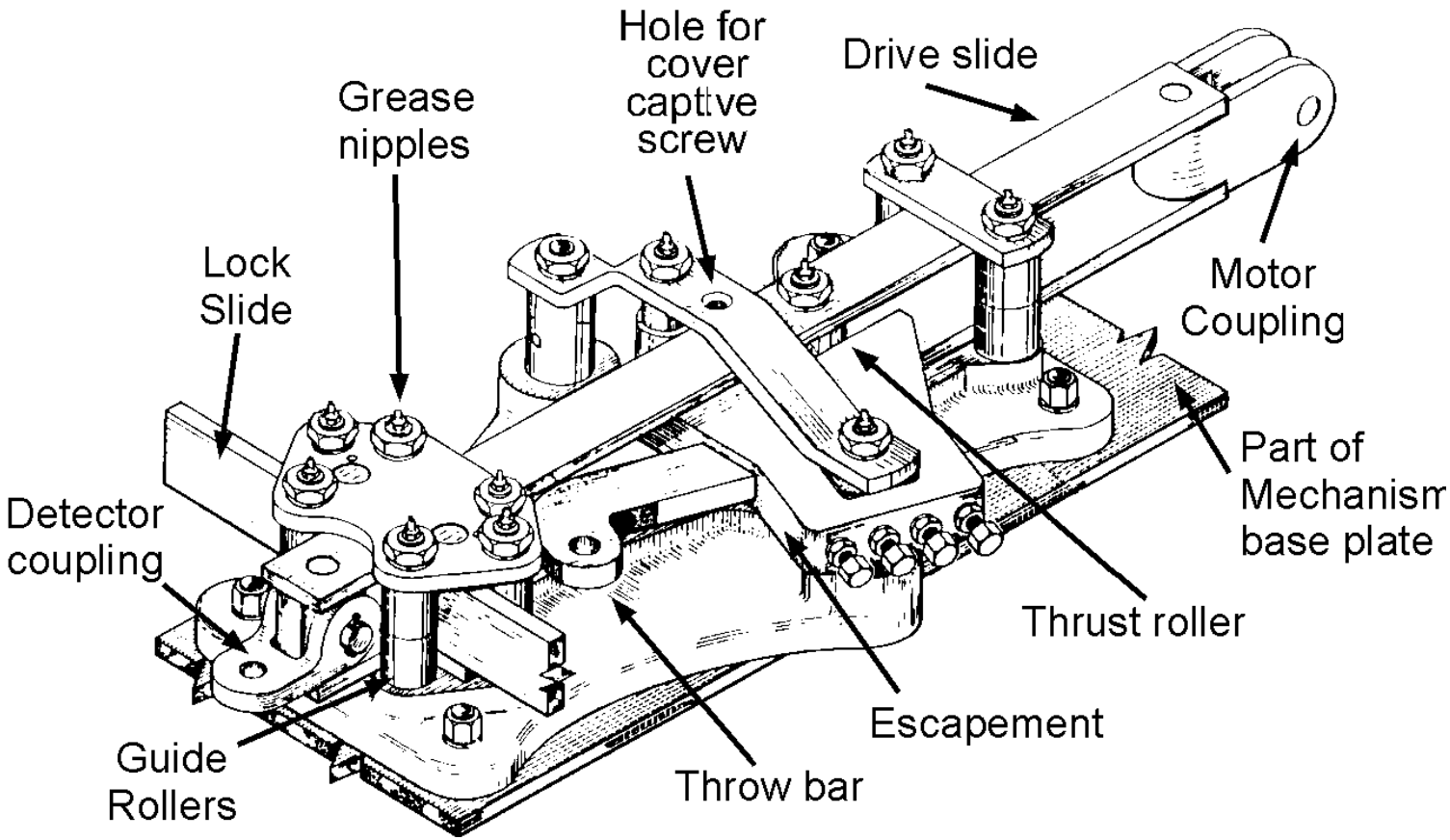


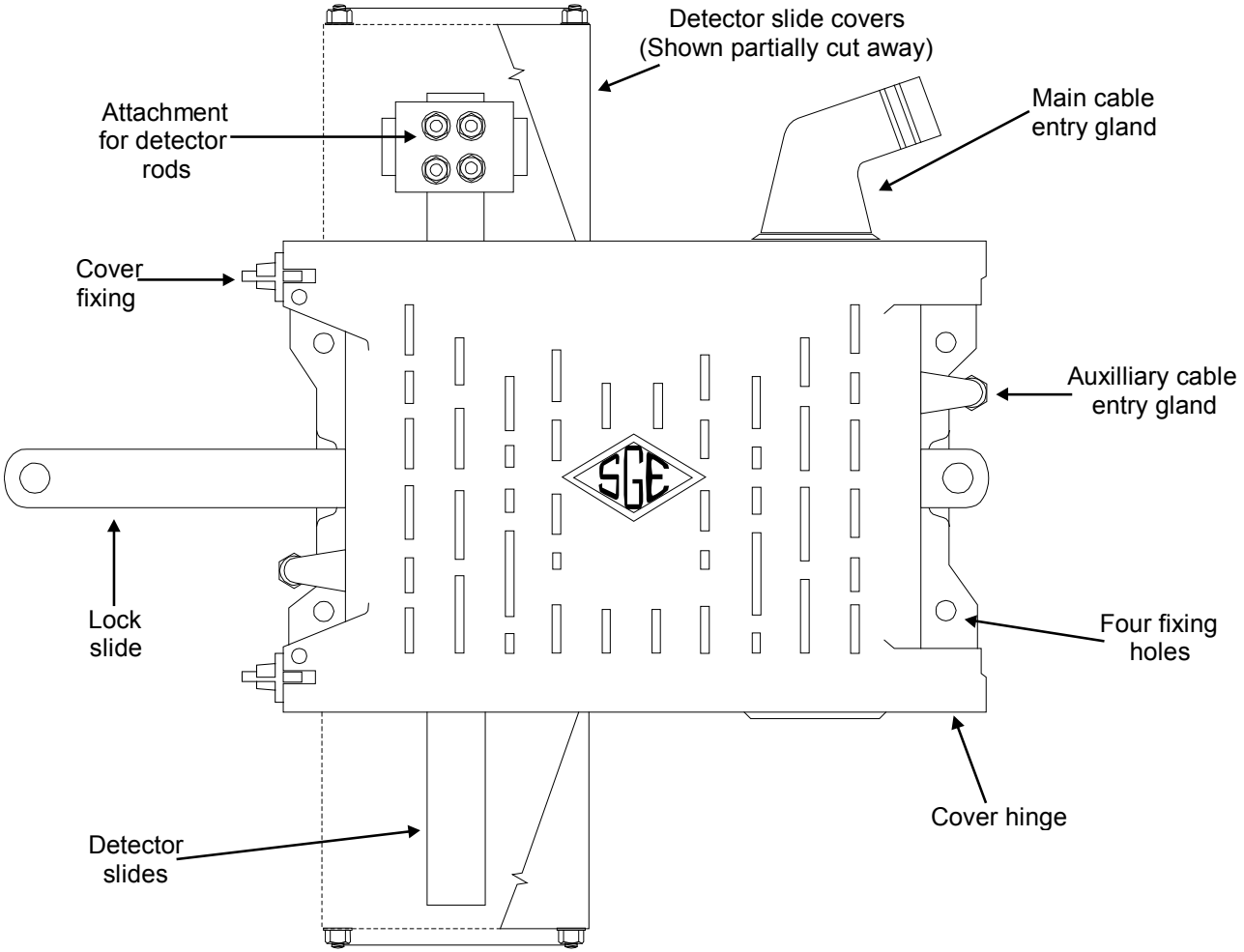

Figure 46

## SGE Type HV Point Detector

This is illustrated in figure 47 with its cover attached. The function of the point detector is to verify that both switch rails have moved to their correct positions when the lock slide (driven by the drive slide of the escapement mechanism) has completed its travel, thus indicating that the points are correctly operated and locked either normal or reverse.

The cover fixings include provision for padlocking to prevent unauthorised entry. The attachments for the detector rods and the main cable entry gland can be fitted on either side to suit different point layouts.

*External View of the SGE Type HV Point Detector*




15

Figure 47

# 8.3 Manual Operation of the SGE EP Points

Special provision is made on these EP points for manual operation. There are two alternatives

- power operation
- wholly manual operation

## Power Operation

The electrical connections to the needle valve operating coils are made via a plug connector which is normally concealed by the air control valve unit cover. When this plug connector is withdrawn it disables the external control of the needle valves and reveals two manual control slots. Inserting a trident shaped special tool into one of these slots allows manual operation of the valves. Provided the air supply is still available the points can be moved either normal or reverse.

## Wholly Manual Operation

Provided the plug connector is removed, or the air supply is off or disconnected, the points can be moved by placing a suitable lever in the coupling between the motor and the escapement drive slide and levering against the row of holes provided in the mechanism bed plate. It is very dangerous to do this if the normal power operation is not disabled, otherwise the motor may become energised as soon as the point detector recognises that the points are no longer fully made and locked.

## 8.4 Section Summary

This section has looked at the AC immune version of the SGE Electro-Pneumatic Point Machine as typical of EP point machines in general. It has covered the following main components

- the motor
- the air control valve unit
- the escapement mechanism
- the point detector

and their functions. In addition the arrangements for manual operation were described and explained.

Now complete Self-Assessment Questions 8 and 9 before proceeding to Section 9.

## Self-Assessment Question 8

List the main component parts of an SGE EP Point Machine

## Self-Assessment Question 9

In your own words, briefly describe the arrangements for manual operation of an SGE EP Point Machine. In what circumstances is it dangerous to apply wholly manual operation?

Now check your answers with those in the Answers Section. Before moving on to Section 9 make sure that you fully understand all of the information given in Section 8. If there are any areas you do not understand, re-read the relevant parts of the text.

# 9 Rail Point Clamp Locks

## 9.1 Introduction

Rail point clamp locks produce the same outcome as the electrically-operated point machines, but the operation of the clamp locks is achieved by different principles.

Advantages of rail point clamp locks include

- they are easily transportable in small units
- relatively low cost
- no need for extended timbers
- ability to cope with road spread up to a certain amount without failure
- ability to cope with switch creep

The disadvantage of rail point clamp locks is that if track movement occurs under a passing train, components wear out quickly.

*Road Spread*

Road spread occurs when the gauge of the line begins to exceed the standard width, usually caused by continual heavy usage.

*Switch creep*

Switch creep is movement of the toe of the switch rail in either direction away from its original position, usually caused by temperature variations.

*Clamp lock point arrangement*

Figure 48 provides an example of a rail point clamp lock fitted to UIC 54 rail. When fitted to flat bottom or bullhead rail a different centre thrust bracket is used and other detail differences apply.

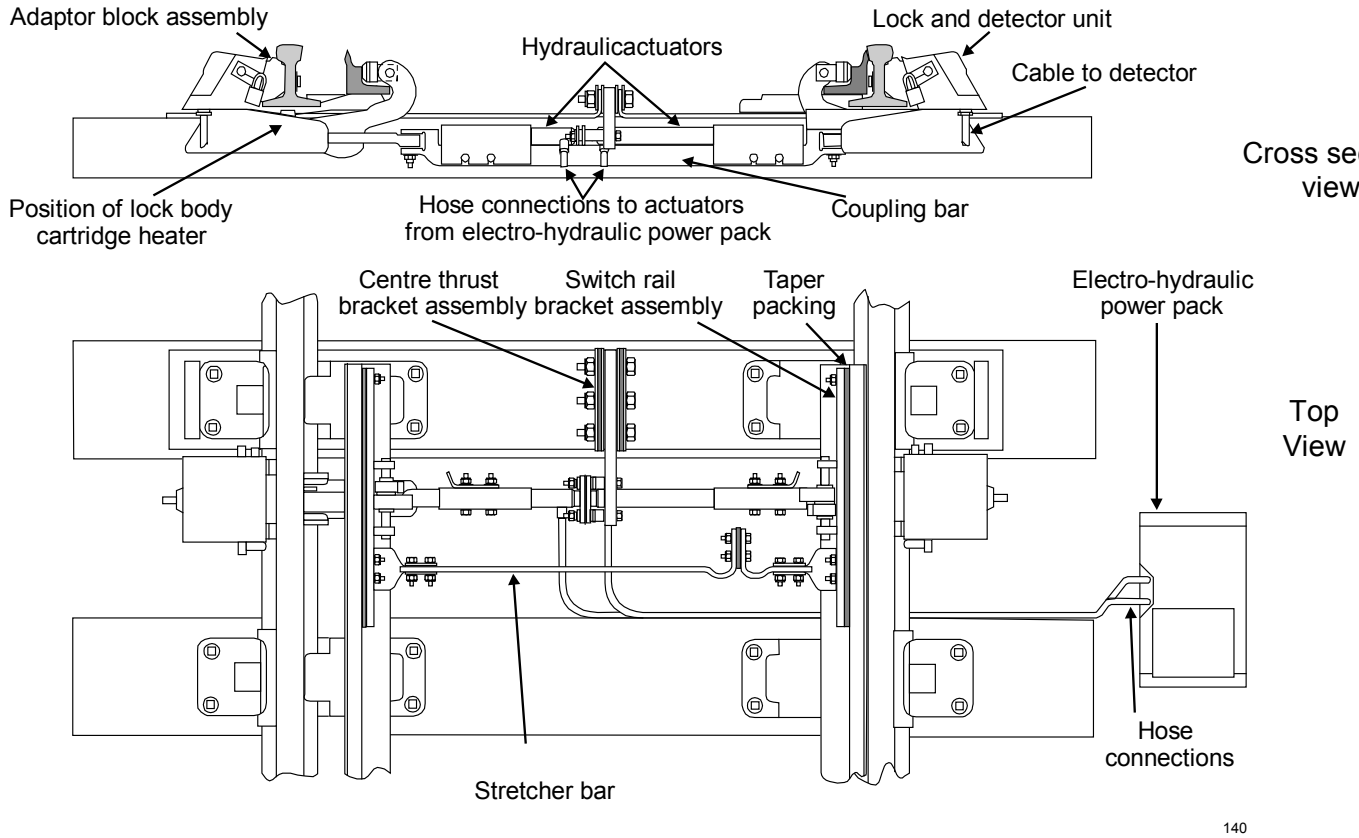



Figure 48

## 9.2 Components of the Rail Clamp Lock

The essential components of the rail clamp lock assembly are

- two lock and detector mechanisms
- an insulated coupling bar which links the two mechanisms. (this is just visible in figure 48)
- two single acting hydraulic actuators
- a centre mounted thrust bracket to which the fixed ends of the actuators are connected. The thrust bracket is attached to the insulated sole plate
- an electro-hydraulic power pack, which is energised whenever a switch movement is required

Each clamp lock comprises

- a lock and detector body attached to the stock rail. This incorporates a drive lock slide
- a bracket attached to the switch rail. This carries the lock arm and the point detection blade

## 9.3 Operation of the Rail Point Clamp Lock

Figures 49 to 53 show the sequence of operation of the clamp lock.

In figure 49 the left hand switch rail is closed and locked against its stock rail.

*Left hand switch closed and locked*

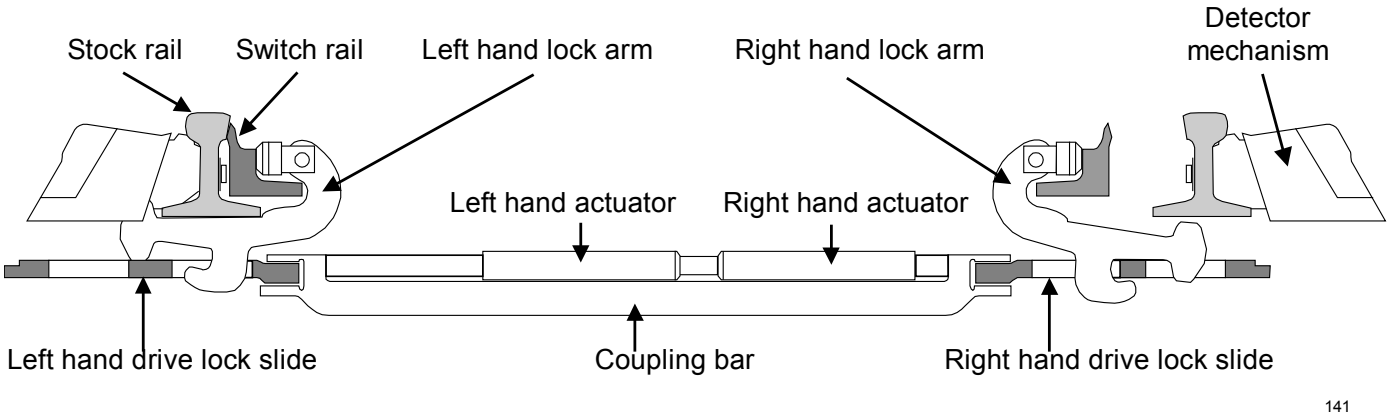



Figure 49

The left hand actuator

- has pushed the left hand drive lock slide through the left hand lock and detector mechanism
- is fully extended

The left hand drive lock slide is supporting the left hand lock arm in the fully raised (locked) position.

The left hand lock arm is holding the switch rail against the left hand stock rail.

When the points are thrown by the operator, hydraulic fluid is pumped from the power pack to the right hand actuator via a solenoid operated valve within the power pack. Standing pressure in the left hand actuator is released by a second valve, which allows the hydraulic fluid in the left hand actuator to return to the reservoir in the power pack.

On being pressurised the right hand actuator

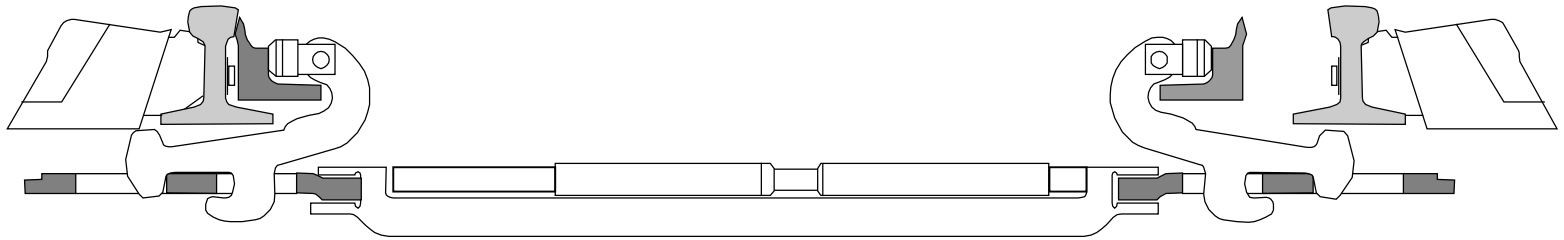
- starts to extend
- pushes the right hand drive lock slide into the right hand body
- pulls the left hand drive lock slide out of its body and causes the left hand actuator to retract

After the actuators and drive lock slides have moved approximately 40 mm to the right, the central bar of the left hand drive lock slide passes from under the left hand lock arm allowing it to drop by a combination of

- gravity
- the mechanical force being created when the lock arm starts to be pushed to the right

This movement unlocks the left hand clamp lock. Now look at figure 50.

*Left hand switch closed and unlocked*



142

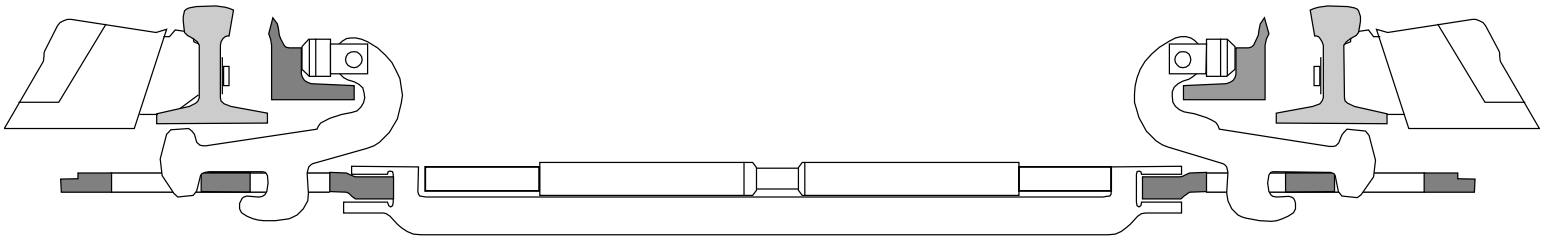
Figure 50

In figure 50 the left hand switch rail is

- free to move
- the central bar of the right hand drive lock slide is lodged against the right hand lock arm pushing it to the right and causing it to rise
- the right hand lock arm runs along the underside of the stock rail as the switch rails are moved to the right

Now look at figure 51.

*Mid travel*



143

Figure 51

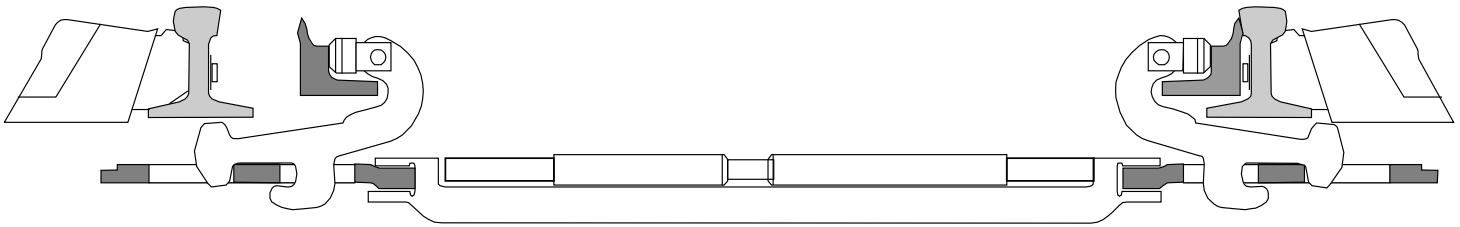
In figure 51 it can be seen that as the right hand actuator continues to extend, it

- pushes the right hand lock slide
- pushes the right hand lock arm to the right
- forces the right hand switch rail to the right until it closes against the right hand stock rail.

the stretcher bar pulls the left hand switch open.

Now look at figure 52.

*Right hand switch closed but not locked*



144

Figure 52

At this stage it is important to note that as the right hand switch rail is forced to the right it is the permanent way stretcher bar which forces the left hand switch rail to follow. The coupling bar does not contribute directly to this movement, but it does cause the left hand actuator to retract and moves the left hand lock slide to the position which holds the left hand lock arm in its lowered position.

When the right hand switch rail is pressed against the right hand stock rail the right hand lock arm cannot move any more to the right, but the right hand actuator continues to extend and forces the right hand lock arm into the position where it locks the switch rail against the stock rail and operates the microswitch, which indicates the point switch is fully closed and locked. In this position

- the right hand actuator reaches the end of its allowed travel
- the central bar of the right hand lock slide prevents the lock arm from dropping

Look at figure 53 which illustrates this last position.

*Right hand switch closed and locked*

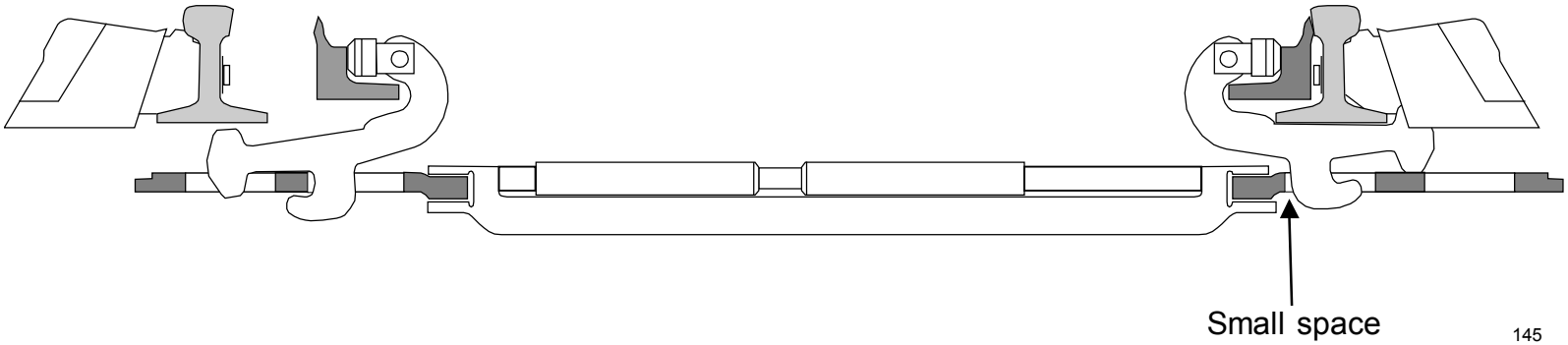



Figure 53

Note that the small space indicated is left between the arm of the lock slide and the lock arm to ensure that the lock arm is not under horizontal pressure from the lock slide after it is locked by the central bar of the lock slide.

## 9.4 Section Summary

This section has looked at rail point clamp locks in terms of

- component parts
- operation

Now complete Self-Assessment Questions 10 and 11 before proceeding to Section 10.

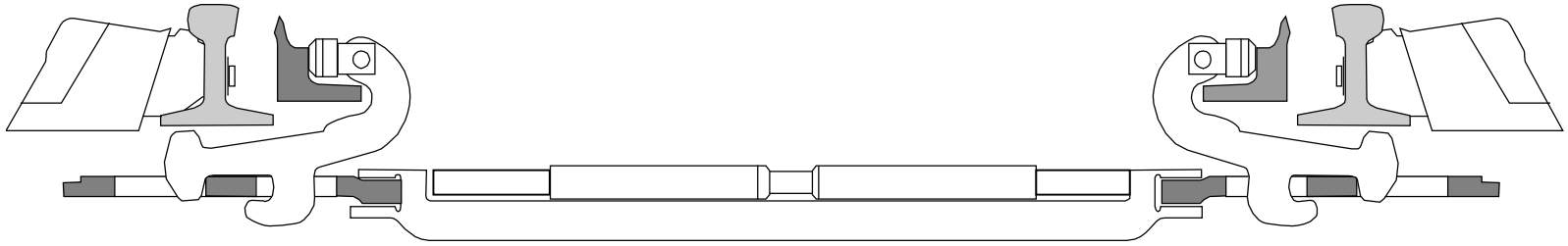
## Self-Assessment Question 10

State two advantages of the rail clamp lock.

## Self-Assessment Question 11

Figure 54 shows an illustration of the clamp lock mechanism at its mid travel point as the left hand switch is opening and the right hand switch is closing. In your own words describe the sequence of events as the operation is completed and the right hand switch rail is locked. Include in your explanation how the left hand switch rail is forced to its open position and the role of the coupling bar.

*Clamp lock points in mid-travel*



165

Figure 54

Now check your answers with those in the Answers Section. Before moving on to Section 9 make sure that you fully understand all of the information given in Section 8. If there are any areas you do not understand, re-read the relevant parts of the text.

This page has been left blank intentionally.

# 10 Facing Point Lock and Detection Tests

## 10.1 Introduction

In previous sections you looked at the operation of point machines and will have realised that the correct operation of the points cannot be assumed. Before any traffic is allowed over points the following must be correctly detected

- the closed switch is actually closed
- the open switch is fully open
- correct locking of facing points

In this section the function of the following tests will be explained

- facing point lock test
- points detection test

## 10.2 Facing Point Lock Test

The facing point lock test serves a double purpose, it serves as both a

- safety test
- reliability test

Both aspects of the test will now be examined in greater detail

Firstly the safety test ensures

- the points cannot become locked if the switch rail is standing off from the stock rail by 3.5 mm or more at a point in line with the stock rail bolt in the first slide chair
- when the points are locked there is no unsafe gap between the stock and switch rail which the flange of a wheel could enter and cause an accident

Secondly the reliability test ensures there is an adequate clearance within the locking mechanism by providing that the points can be locked when there is a 1.5 mm gap between the switch rail and the stock rail.

This allows

- a working tolerance
- prevents wear
- ensures the lock cannot become jammed

This test is performed by using two gauges, namely

- 3.5 mm thickness
- 1.5 mm thickness

Service A in Signalling Maintenance Specification (SMS) PA21 explains how this test should be carried out for electric and pneumatic point machines and Service A in SMS PA23 covers mechanical points.

For clamp locks see Signalling Maintenance Specification PB11 Service A.

## 10.3 Points Detection Test

The points detection test also serves a double purpose as both a

- safety test
- reliability test

For all types of points except the rail clamp point lock the facing point lock is first disabled and two tests are applied.

- to ensure that switch detection is not obtained when the switch rail is stood off from the stock rail by 5 mm or more
- to ensure that switch detection is obtained when the switch rail is stood off from the stock rail by 3.5 mm

The detailed procedure is given by Signalling Maintenance Specification (SMS) PA31 and uses gauges of thickness 5 mm and 3.5 mm.

For rail clamp point locks the facing point lock is separate from the detection mechanism and does not need to be maladjusted to carry out a detection test. It is performed by putting gauges between detector mechanisms. The details are given in SMS PA11 (Service C and Appendix B) and use gauges of thickness 4 mm, 2.5 mm, 2 mm and 1.5 mm.

## 10.4 Section Summary

This section has looked at the facing point lock and detection tests.

Now complete Self-Assessment Questions 12 and 13 before proceeding to Section 11.

Before attempting these questions make sure you have access to a copy of the Signalling Maintenance Specifications.

## Self-Assessment Question 12

State the first step of Service A in Signalling Maintenance Specification PA21.

## Self-Assessment Question 13

State the final step of the procedure for detection testing as laid down in the Signalling Maintenance Specifications.

Now check your answers with the ones in the Answers Section. Before moving on to Section Eleven make sure that you fully understand all the information given in Section Ten. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 11 Insulation Methods

## 11.1 Introduction

The installation of point fittings on a line means both rails are in danger of being electrically connected together through the mechanical linkages. This would prevent correct operation of track circuits and consequently it is necessary to insulate relevant parts of the point layout.

## 11.2 Details of Insulation Methods

All point layouts require the following items to be insulated

- sole plate
- four foot stretcher bars

Figure 55 shows how the sole plate is divided and the two parts are insulated. Figure 56 shows how a similar arrangement is used on stretcher bars.

*Sole plate insulation details.*

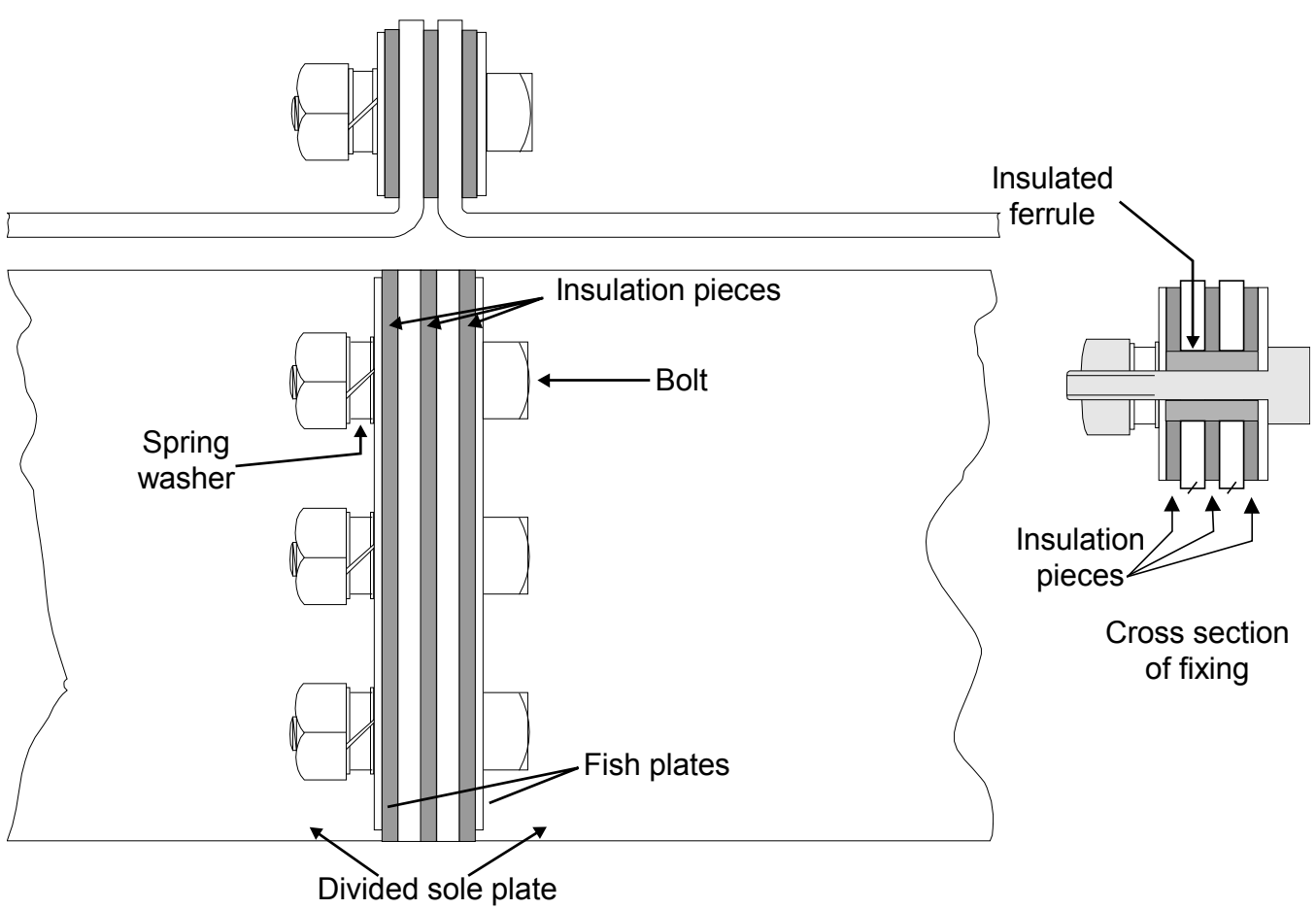



Figure 55

*Four foot stretcher bar Insulation details.*

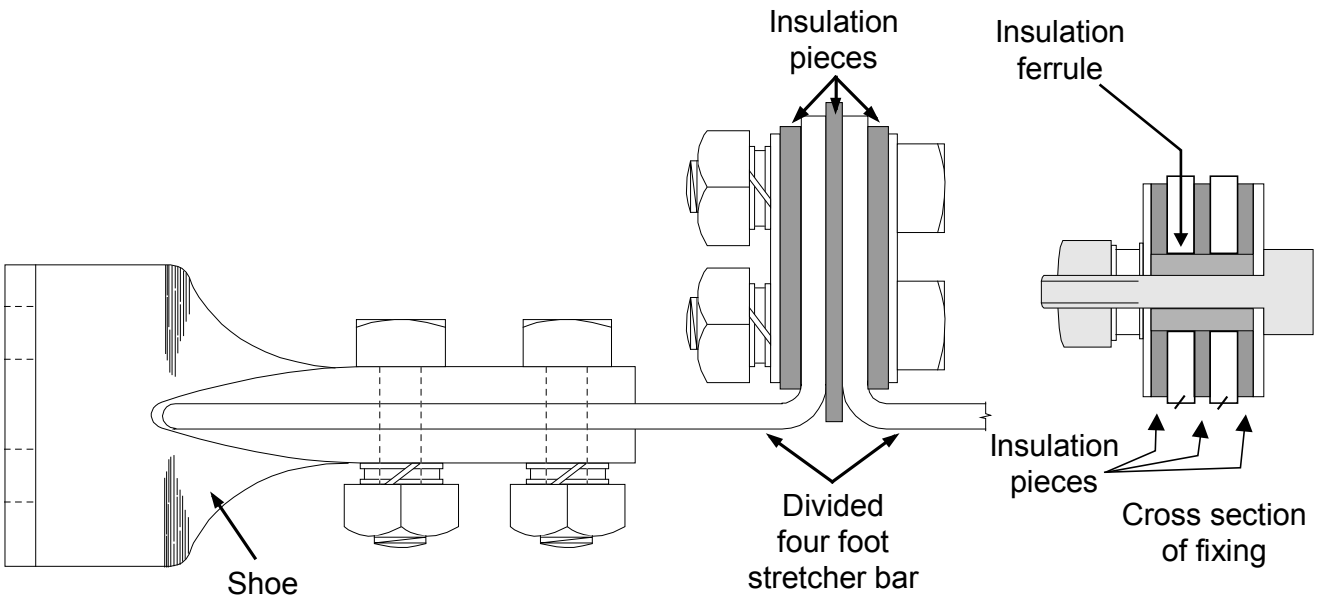



Figure 56

In the case of mechanical points the following additional fixings require insulation

- facing point lock stretcher bar
- lock rod
- detector rods
- point rodding
- facing point lock rodding

At this point refer back to figures 32 and 33 to re-familiarise yourself with the areas which are insulated on the mechanical point layout. These insulations will now be examined in greater detail.

## Facing Point Lock Stretcher Bar

Figure 57 shows the insulation arrangements used at both ends of the point lock stretcher bar on a flat bottom chamfered switch. The arrangements used on other rail types are very similar.

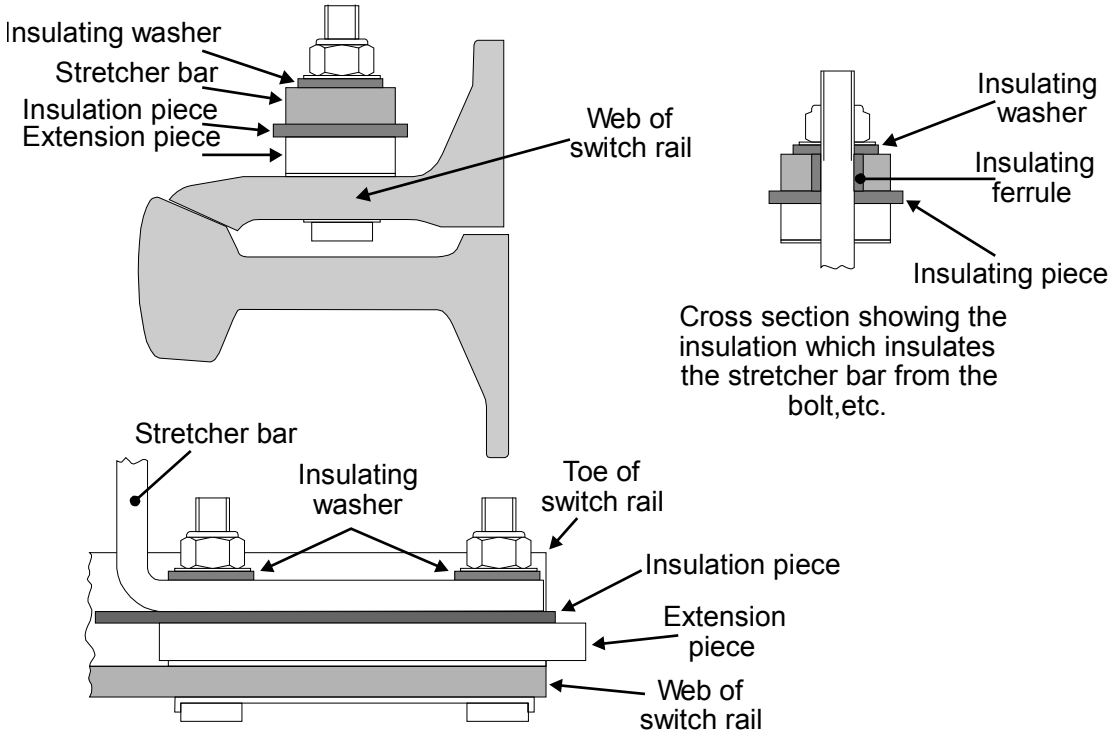



Figure 57

## Lock Rod

*Insulation on Lock Rod*

Figure 58 shows the insulation used for the lock rod. The insulated bushes fit into the mating lug on the point machine.

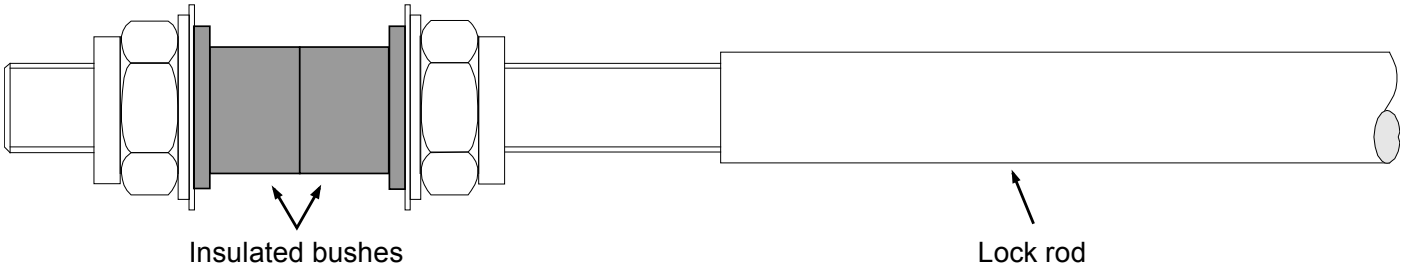



Figure 58

## Detector Rods

*Insulation on Detector Rod*

Figure 59 shows the insulation used on the detector rods. The insulated bushes fit into the hole in the end of the extension piece attached to the toe end of the switch rail.

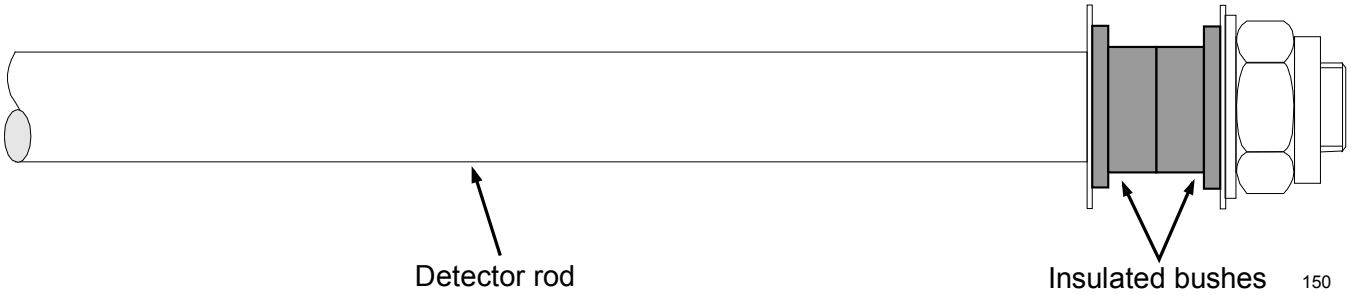



**Figure 59**

## Point Rodding and Facing Point Lock Rodding

*Insulation of Rodding Joints*

Figure 60 shows the insulation used for the joints on point rodding and facing point lock rodding.

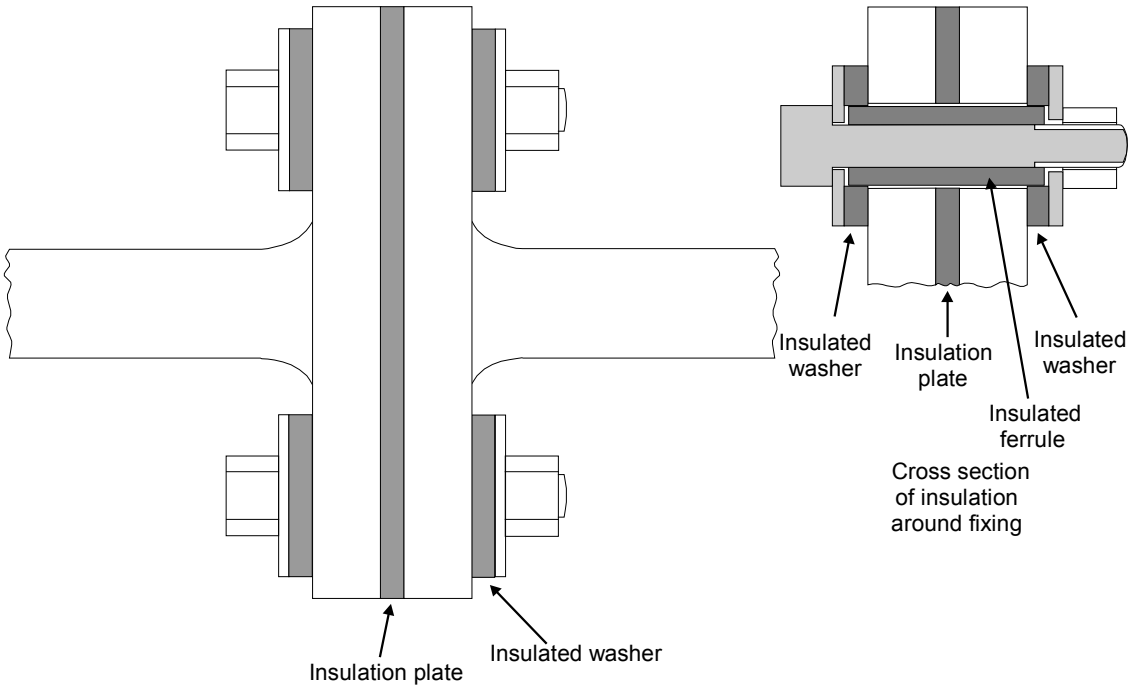
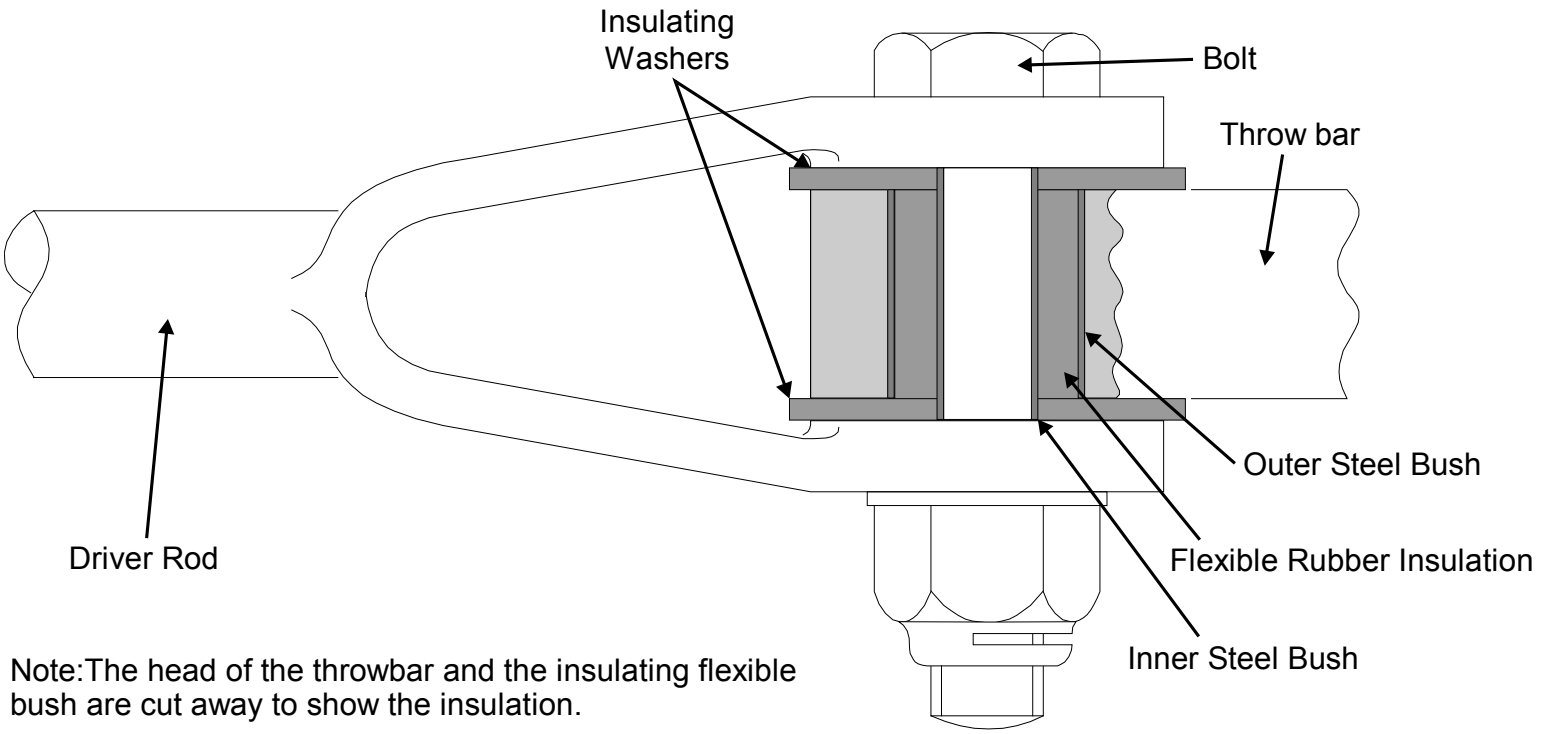



Figure 60

## Insulation of Power Worked Point Layouts

In most cases the insulation methods employed on power worked points are the same as those already described for mechanically operated points. The only significant addition is the insulation between the point machine throw bar and the point drive rod. An example is illustrated in figure 61.

*Insulation bush between the throw bar and the drive rod of power worked points*




Note:The head of the throwbar and the insulating flexible bush are cut away to show the insulation.

152

Figure 61

## 11.3 Section Summary

This section has looked at insulation methods used on point mechanisms.

Now complete Self-Assessment Question 14 before proceeding to Section Eleven.

## Self-Assessment Question 14

State the parts of a mechanical point layout that require insulation.

Now check your answer with the one in the Answers Section. Before moving on to Section Twelve make sure that you fully understand all the information given in Section Eleven. If there are any areas which you do not understand, re-read the relevant parts of the text.

## Self-Assessment Question 13

State the final step of the procedure for detection testing as laid down in the Signalling Maintenance Specifications.

Now check your answers with the ones in the Answers Section. Before moving on to Section Eleven make sure that you fully understand all the information given in Section Ten. If there are any areas which you do not understand, re-read the relevant parts of the text.

This page is intentionally left Blank

# 12 Back Drives

## 12.1 Introduction

On the points mechanisms you have been shown so far the drive connections have been on the stretcher bar and therefore near to the toe end of the switch rails. When the planed length of the switch rails is fairly short there is no problem that the planed section does not move correctly relative to the toe, so that the whole of the planed section is either in contact with the stock rail or fully away from it to enable the wheel flange to easily move between them. It is on the higher speed turn-outs, where the planed lengths become more extended, that problems can arise.

To overcome those problems the back drive was introduced. There are two types of back drive

- mechanical
- hydraulic

Each of these will now be examined in greater detail.

## 12.2 Mechanically Operated Back Drives

Figure 62 illustrates a mechanically operated back drive.

*An example of a mechanically operated back drive*

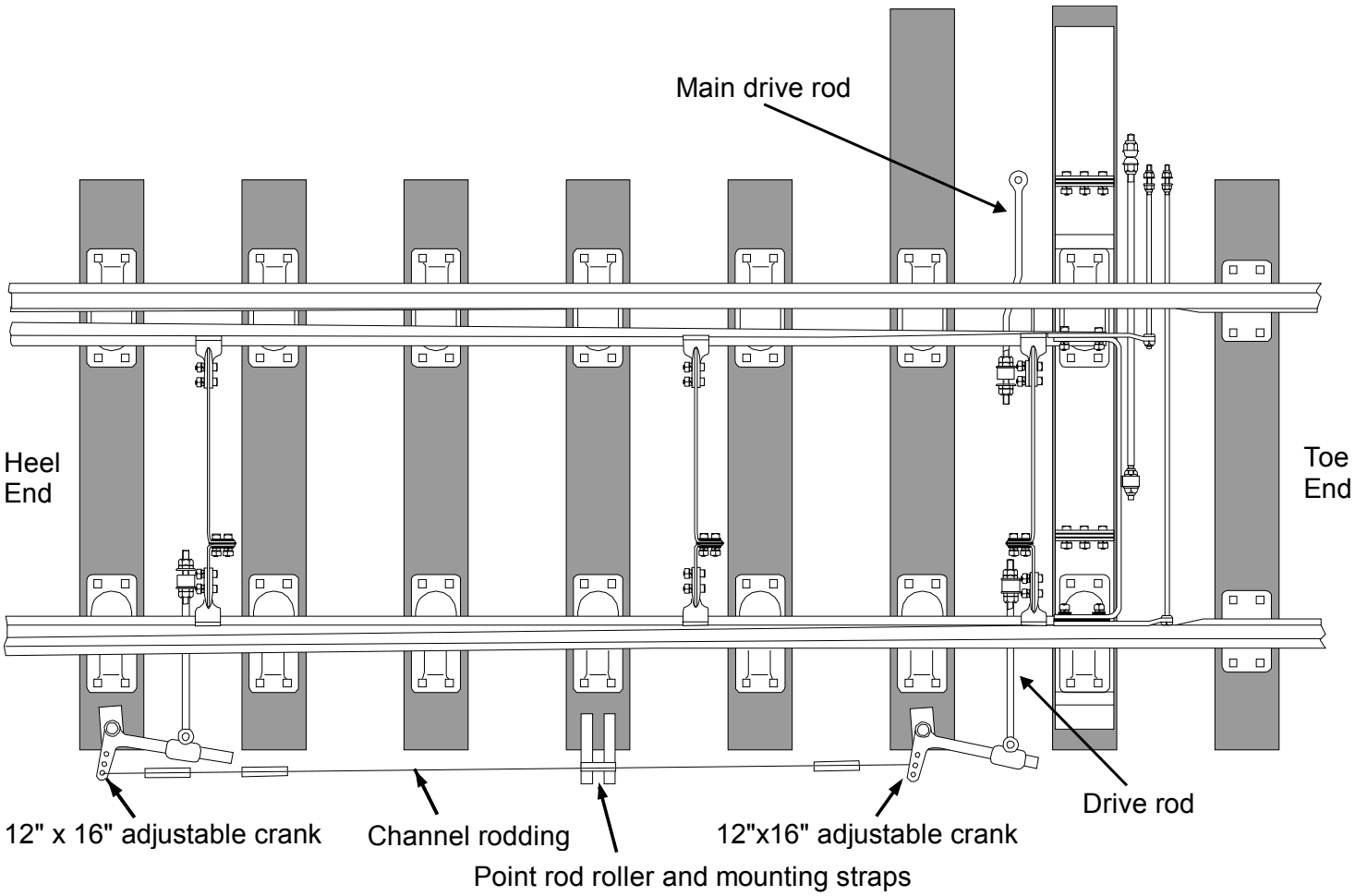



Figure 62

Look carefully at figure 62. In this back drive both the toe and heel of the switch rail are moved by the operation of the point machine

- the toe is moved by means of the main drive rod
- the heel is moved by means of the interconnecting rods and cranks

Figure 62 shows three stretcher bars. When the switches are even longer more stretcher bars are provided to maintain the gauge through the switch. On the longest switches up to 6 stretcher bars are used. If a 5th stretcher bar is employed an extra (intermediate) back drive is provided on the 3rd stretcher bar, but if a 6th stretcher bar is used the intermediate back drive is placed on the 4th stretcher bar.

## 12.3 Hydraulically Operated Back Drives

On high speed switches hydraulically operated back drives are used in conjunction with clamp locks, where

- the toe is moved by means of a clamp lock
- the back drives are moved by means of additional hydraulic actuators

Figure 63 shows the arrangement of a hydraulically operated high speed switch with hydraulic back drives. Look carefully at figure 63 and observe in this example that in addition to the clamp lock mechanism there are

- 9 stretcher bars
- 4 back drives each comprising 2 actuators driven from the same electro-hydraulic power pack as the clamp locks
- 2 supplementary detectors operating on the 5th and 7th stretcher bars.

Figure 64 shows the arrangement of an individual hydraulic back drive actuator with the combined thrust bracket and base plate.

*Arrangement of a high speed switch with hydraulic back drives*

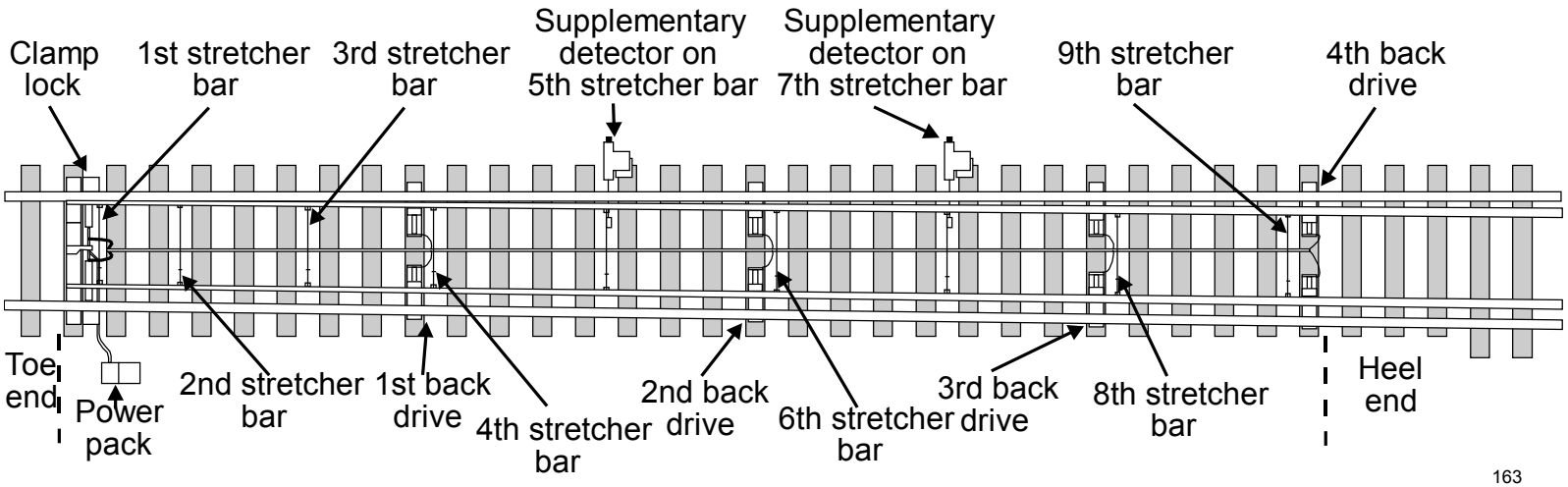



Figure 63

*Arrangement of individual hydraulic back drive actuator and combined thrust bracket/base plate*

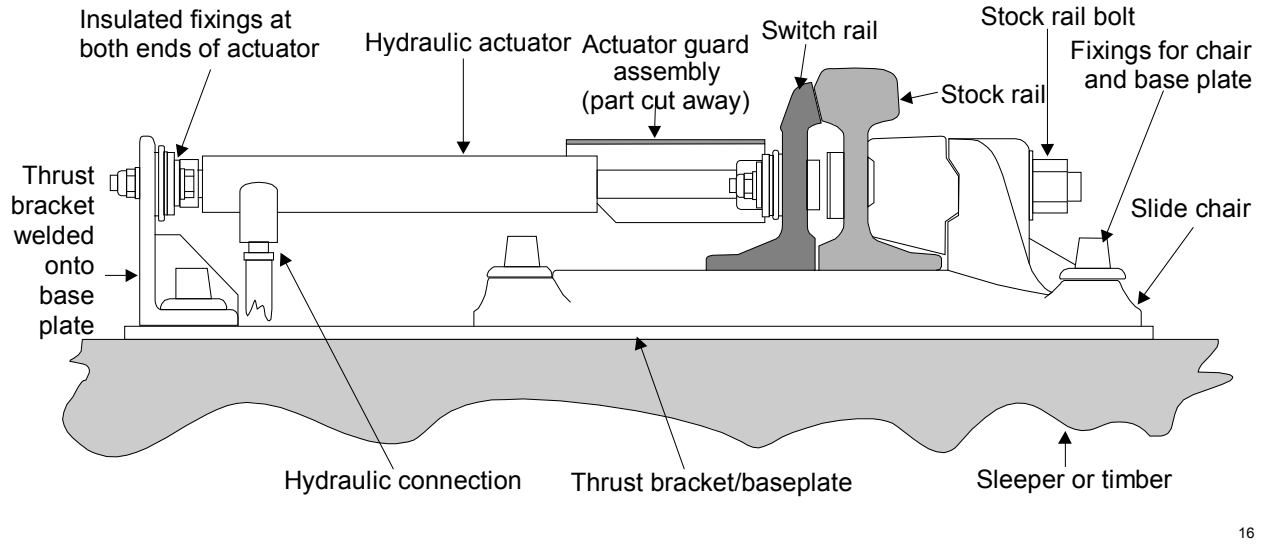



**Figure 64**

## 12.4 Section Summary

This section has looked at back drives.

Now complete Self-Assessment Questions 15 and 16.

### Self-Assessment Question 15

State the two types of back drive.

### Self-Assessment Question 16

In your own words state the purpose of back drives.

Now check your answers with those in the Answers Section. Before moving on to the Module Summary make sure that you fully understand all the information given in Section Twelve. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 13 Module Summary

This Module has looked at

- the DOT requirements
- types of point layout and configuration
- switch and stock rails
- methods of point operation
- mechanical points
- the combined point machine
- the non-combined point machine
- the electro-pneumatic point machine
- rail point clamp locks
- facing point lock and detection tests
- insulation methods
- back drives

Before attempting the Self Test Questions which follow make sure that you fully understand what you have read. Take your time and check your answers to the Self-Assessment Questions again. When you feel confident begin the Self Test Questions.

# 14 Self Test

Attempt the following Self Test Questions. Take your time and think about the answers carefully. When you have finished the Self Test check your answers with those given in the Answers Section. If you experienced any difficulties return to the relevant part of the text and re-read the material. Do not move onto the next module until you feel confident in your understanding of this work.

## Self Test Question 1

State the function of the stretcher bars.

## Self Test Question 2

State the factors which you think will influence the design of a point layout.

## Self Test Question 3

Correctly identify the point configuration shown in figure 65.

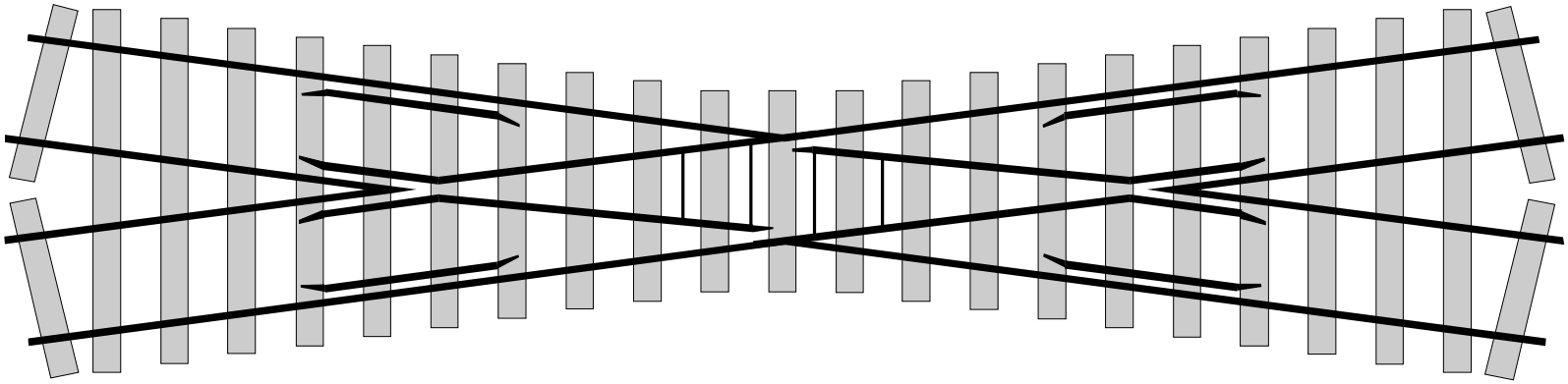


Figure 65

## Self Test Question 4

State two types of switch rail which are available.

## Self Test Question 5

State the two types of motor generally used in combined point machines. Also state their advantages and disadvantages.

## Self Test Question 6

State how locking and detection are achieved with a non-combined point machine.

## Self Test Question 7

State how an electro-pneumatic point machine is powered and how locking and detection are achieved.

## Self Test Question 8

State two of the advantages of the rail point clamp lock. State the cause of switch creep.

## Self Test Question 9

State the parts which require insulation on points operated by a combined point machine.

## Self Test Question 10

State one purpose of the basic back drive.

# 15 Answers

## 15.1 Introduction

This section provides the answers to both the

- Self-Assessment Questions
- Self Test Questions

Check your answers carefully. If you experience any difficulties, re-read the relevant parts of the text.

## 15.2 Answers to Self-Assessment Questions

### Self-Assessment Question 1

The Department of Transport requirements concerning the numbers of stretcher bars to be used on points states

All points to have two or more stretcher bars

## Self-Assessment Question 2

The other Department of Transport requirements applying to facing points on passenger lines are

- a bolt lock through a third stretcher bar
- a stock rail gauge tie (sole plate)
- switch and lock detection

## Self-Assessment Question 3

Figure 27 illustrates a single slip configuration.

## Self-Assessment Question 4

The stock and switch rails to be identified are flat bottom vertical stock rail with U.I.C. 54 switch rail and inclined bull head stock rail with undercut switch blade.

## Self-Assessment Question 5

The correctly completed version of Table 1 is as follows.

| Method of Point Operation | Points moved manually | Points moved by electric motor | Points moved by compressed air supply | Points moved by hydraulic fluid | Points locked by separate lock | Points locked by a lock in the machine | Mechanical Point detection | Electrical Point detection |
|---|---|---|---|---|---|---|---|---|
| Mechanical | ✓ | | | | ✓ | | ✓ | |
| Combined Electrical | | ✓ | | | | ✓ | | ✓ |
| Non-combined Electrical | | ✓ | | | | ✓ | | ✓ |
| Electro-pneumatic | | | ✓ | | | ✓ | | ✓ |
| Hydraulic | | | | ✓ | | ✓ | | ✓ |

## Self-Assessment Question 6

The two most commonly used combined electrical point machines are

- GEC HW 2000
- Westinghouse Style 63

The purpose of the clutch mechanism is to prevent the electric motor burning out if the points jam and to reduce the stress on the mechanism when the points have completed their travel.

## Self-Assessment Question 7

The principal differences between the non-combined and the combined electrically operated point machines are

- the combined machines incorporate the facing point lock and the detection
- the non-combined machines require separate facing point locks and detection facilities

## Self-Assessment Question 8

The main component parts of an SGE EP Point Machine are

- air control valve unit
- motor
- escapement mechanism
- type HV point detector

## Self-Assessment Question 9

An answer of the following form would be acceptable.

“Manual operation of an SGE EP Point Machine has 2 alternatives

- power operation
- wholly manual operation

Power operation is applied by removing the connector which carries the electrical inputs to the air control valve unit and then inserting a special manual key to operate the air valves.

Wholly manual operation is applied by inserting a suitable lever into the motor coupling and levering the points across. Wholly manual operation is dangerous if the electrical inputs are not disconnected or the air supply is still on.”

## Self-Assessment Question 10

Any two of the following advantages of the rail point clamp lock would be acceptable

- they are easily transportable in small units
- relatively low cost
- no need for extended timbers
- ability to cope with road spread up to a certain amount without failure
- ability to cope with switch creep

## Self-Assessment Question 11

A description along the lines of the following would be acceptable

“From the mid travel point the right hand actuator continues pushing the right hand drive lock slide to the right. The central bar of the lock slide is lodged against the right hand lock arm pushing it to the right and thereby pulling the right hand switch rail towards the stock rail. When the switch rail is nearly in contact with the stock rail the lock arm starts to slide past the edge of the stock rail lower web and begins to rise. Eventually the lock arm rises to the point where it is hooked onto the stock rail web and the central bar of the lock slide passes under the nose of the lock arm to lock it in position. The right hand actuator reaches the end of its travel before the rear arm of the lock slide strikes the face of the lower extension of the lock arm. A small space is left between them. It finally forces the switch rail against the stock rail and limits the travel of the right hand actuator. During this operation the coupling bar has been pulling the left hand actuator towards its closed position, which causes the hydraulic fluid in that actuator to return to the reservoir in the hydraulic power pack. As the right hand switch rail is pushed to the right, the stretcher bars pull the left hand switch rail in the same direction, so that the left hand lock arm follows the movement of the left hand lock slide.”

## Self-Assessment Question 12

The first step of the procedure laid down in Service A of the Signalling Maintenance Specification PA 21 states

1.1 Isolate the machine and disconnect the KR lines to the signal box.

Note: You will learn about the specific isolation methods when you receive practical instruction on the point machines. The KR lines are connections from the location box which indicate the state of the points to the interlocking circuits.

## Self-Assessment Question 13

The last operation for all services is to test by operation from the signal box and to observe that the apparatus functions correctly.

## Self-Assessment Question 14

The parts of a mechanical point layout which will require insulation are

- sole plates
- four foot stretcher bars
- facing point lock stretcher bar
- detector rods
- point rodding
- facing point lock rodding

## Self-Assessment Question 15

The two types of back drive are

- mechanical
- hydraulic

## Self-Assessment Question 16

A statement along the following lines would be acceptable

“Back drives are used to ensure that the full length of the switch rail moves in conjunction with the toe. When the switch rail is closed against the stock rail at the toe, the back drive(s) ensure that the whole of the planed area of the switch rail is correctly held against the stock rail. When the switch is open at the toe, the back drives ensure that there is sufficient clearance for the wheel flanges to easily pass between the switch rail and the stock rail over the full length of the planed section.”

# 15.3 Answers to Self Test Questions

## Answer to Self Test Question 1

The function of the stretcher bars is to tie both switch blades together. They ensure

- correct point opening
- that the switch rails move together as one unit

## Answer to Self Test Question 2

The factors which will influence point layout are

- the number of lines involved
- the direction of travel
- the switching to be done
- the line space available

## Answer to Self Test Question 3

Figure 65 illustrates a switch diamond configuration.

## Answer to Self Test Question 4

Any two from the following list would be acceptable

- undercut
- straight cut
- chamfered
- diamond
- U.I.C. 54

## Answer to Self Test Question 5

The two types of motor generally used in combined point machines are

- split field (also known as series field)
- permanent magnet

The comparative advantages and disadvantages of the two types of motor are

- the split field motor is smaller, provides a much higher starting torque and is much cheaper
- the permanent magnet motor is immune to AC currents and is always used in AC traction areas

## Answer to Self Test Question 6

In the non-combined point machine

- locking is achieved by crank and rods connection from the machine to a separate locking mechanism
- detection is achieved via a separate detector

## Answer to Self Test Question 7

An electro-pneumatic point is powered by compressed air. Locking is achieved in the escapement mechanism and detection in the point detector.

## Answer to Self Test Question 8

The rail point clamp lock has the following advantages any two of which would be acceptable

- ease of transport in small units
- low cost
- no need for extended timbers
- ability to cope with road spread up to a certain amount without failure
- ability to cope with switch creep

Switch creep is movement of the toe of the switch rail caused by temperature variations.

## Answer to Self Test Question 9

The parts of a combined point machine arrangement which require insulation are

- sole plate
- four foot stretcher bar
- facing point lock stretcher bar
- lock rod
- detector rods
- drive rod

## Answer to Self Test Question 10

Either of the following statements would be an acceptable answer

Back drives are used to ensure that the full length of the planed area of the switch rail is correctly held against the stock rail.

Back drives ensure that the full length of the switch rail moves in conjunction with the toe.

# 16 Glossary

Within the railway system there are a number of regional variations and individual conventions which mean that alternative terms may be used for

- equipment
- operations
- features

Module Four has used terms which normally correspond with those found in general use. However, it may be useful for you to familiarise yourself with the alternative terminology which you may encounter in your work. This glossary provides a number of variations which may occur

| | |
|---|---|
| ■ Stretcher Bars | - four foot stretcher bars |
| | - permanent way stretcher bars |
| | - points stretcher bars |
| | - switch stretcher bars |
| ■ Switch | - points |

| | |
|---|---|
| ■ Switch rail | - switch blade<br>- points blade |
| ■ Turn out | -switch<br>- points |